Winners and Losers: Competition, Creative Destruction, and Labor Income Risk*

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Abstract

Using U.S. administrative data, we find that technology-driven creative destruction in the product market passes through to worker earnings. The passthrough to incumbent worker earnings is both asymmetric and concentrated: profit drops from rival innovations lead to proportionally greater earnings declines and changes in the likelihood of job destruction than profit gains from their own firm's innovations, while top workers are significantly more exposed than the average worker. We develop an endogenous-growth model with monopsonistic labor markets and worker heterogeneity that replicates this asymmetry and the distribution of earnings risk. In the model, creative destruction exposes high-income workers to concentrated downside risk while increasing upward mobility for lower-income workers, shaping the welfare consequences of innovation policy.

Keywords: Creative destruction, innovation, labor income risk, monopsony, endogenous growth, earnings inequality, business dynamism, displacement risk.
JEL codes: E24, J31, J42, O31, O33, O40.

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The process of economic growth is inseparable from creative destruction, in which new entrants displace incumbent firms (Aghion and Howitt, 1992). In canonical macroeconomic models, creative destruction is largely benign: investors diversify firm-specific risks, and workers move costlessly from declining firms to expanding ones. In practice, much of a worker's human capital is specific to a particular job. Because such jobs are imperfect substitutes, the worker is partly bound to her employer and shares in its rents and its losses. We model this co-movement as arising from the monopsony power firms enjoy because of imperfect substitutability, though bargaining or implicit contracts could generate the same link. In this paper, we document the extent to which workers share in the gains and losses from creative destruction and interpret these facts through the lens of a structural model. A central insight is that creative destruction is a powerful force for earnings redistribution: it increases upward mobility for lower-paid workers while exposing higher-paid workers to substantial downside risk.

We begin by showing that increases in a firm's own innovation raise future profits and the total pay of its workers, while innovation by competing firms lowers both. At the firm level, these responses appear largely symmetric: firms that successfully innovate expand employment and their wage bill, while firms whose competitors innovate contract. However, this symmetry breaks down when we shift the focus from firm aggregates to individual workers. Using the panel structure of administrative employer–employee matched data from the United States, we estimate how individual earnings respond to innovation by the worker's employer and its competitors. Workers in innovating firms experience earnings gains, while workers in firms whose competitors innovate suffer losses—but these responses are strikingly asymmetric. The passthrough from profits to earnings is about 0.14 for own-firm innovations, consistent with existing estimates of profit sharing. For competitor innovations, the implied passthrough is 0.29—more than twice as large—even though the corresponding decline in firm profits is similar in magnitude.

This asymmetry in passthrough highlights the unique role of innovation in driving firm profits and worker earnings. When a firm innovates and expands, it raises wages and hires additional workers; part of the increase in the wage bill therefore accrues to new hires, not incumbents. Incumbent workers are already employed and do not share in the gains from employment growth. By contrast, when a competitor innovates and the firm loses market share, both the wage decline and the associated job loss fall squarely on incumbents. Incumbent workers thus bear a disproportionately large downside exposure to creative destruction but capture only a fraction of its benefits. We provide three additional

pieces of evidence consistent with this interpretation. First, we find that increases in competitor innovation significantly raise the probability of job destruction for incumbent workers—measured using a proxy for involuntary job loss based on whether a worker enters nonemployment or switches employers while simultaneously experiencing a sharp decline in earnings. Second, estimates from quantile regressions reveal that own-firm innovation produces positively skewed earnings growth—concentrating gains among a small subset of workers—whereas competitor innovation yields negatively skewed earnings changes, with most workers experiencing modest declines and a minority suffering large losses. Last, we show that innovations that are more distinct relative to the firm's existing technological base generate large increases in profits, payroll, and employment but only modest earnings gains for incumbent workers relative to equally valuable, less novel innovations.

Both the gains and the losses from innovation are disproportionately concentrated at the top of the earnings distribution. Workers in the top 5% of their industry's earnings distribution see earnings gains of 2.6 percentage points following a one-standard-deviation increase in own-firm innovation, compared to a 4 percentage point earnings decline from an equivalent increase in competitor innovation. The implied passthrough coefficients for these top earners are about three times larger than for the average worker, consistent with higher-skilled workers facing less elastic labor supply. This pattern also varies systematically across industries: the passthrough of innovation to worker earnings is significantly larger in industries with high rates of innovative activity (and potentially more scope for creative destruction) and in industries in which firms likely face less elastic labor supply curves: those with greater within-industry income dispersion as well as those with higher average wages. Once again, these exposures are highest among the highest-paid workers in these industries.

We next interpret these estimates through the lens of a model of creative destruction and labor market frictions. The model combines several key elements from the literature. Firms compete in producing different varieties, as in Aghion and Howitt (1992). As in Peters (2020), firms can engage either in internal or external innovation; internal innovation increases the productivity of existing product lines while external innovation allows the firm to steal business from a competitor; a competitive fringe of potential entrants provides an additional source of creative destruction. Unlike Peters (2020), the rate of industry innovation is stochastic: the model features innovation booms in the spirit of Berlingieri, Ridder, Lashkari, and Rigo (2025). The leading firm produces output using two complementary labor inputs: laborers and managers. The key distinction between laborers and managers in production is that the optimal ratio of managers to laborers is increasing in firm size.

Workers, either laborers or managers, optimally choose their preferred job out of a finite set; jobs differ in their match-specific productivity as in Roy (1951). Firms internalize their labor market power, since jobs are imperfect substitutes; hence they pay workers below their marginal product of labor as in Seegmiller (2024). Laborers have a more elastic labor supply than managers; since the more productive firms need to hire more workers but also a higher ratio of managers to laborers, they face more inelastic labor supply curves. Consequently, the more productive firms have higher profit margins as they optimally mark down wages further below the marginal product of labor.

In sum, incumbent workers in the model face two forms of income risk linked to innovation. The first operates through wages in continuing jobs: internal innovation raises a firm's relative productivity and increases labor demand and wages, while competitor innovation has the opposite effect. This channel generates roughly symmetric responses to own-firm and competitor shocks. The second source of risk—creative destruction—drives the asymmetry in worker outcomes consistent with our empirical results. A worker's human capital is tied to the product line on which she is employed. When a competitor or entrant displaces the incumbent producer, the job disappears and the worker must draw a new match from her remaining options, all of which pay less. Because high-income workers sort into matches where their human capital is most valuable, displacement imposes disproportionately large losses on them. This mechanism also limits the degree to which incumbent workers benefit from their own firm's innovation. Workers gain when the product line they work on improves, but they do not share in gains that arise from the firm expanding into other lines, such as when it steals a product from a competitor. As a result, incumbent workers capture only a fraction of the upside from innovation but bear most of the downside.

The calibrated model quantitatively replicates the key relation between technological progress and earnings risk that we uncover. In addition, the model reproduces key empirical features of worker-level earnings risk that are not explicit calibration targets. As in Guvenen, Karahan, Ozkan, and Song (2021), high-income workers face disproportionately negative skewness, reflecting their exposure to creative destruction when high-match-quality jobs disappear.

We conclude by performing two quantitative experiments. First, we explore implications of the well-documented secular decline in business dynamism (Haltiwanger, 2012; Akcigit and Ates, 2019) through the lens of the model. We show a related fact: the value share of radical innovation—patents that are technologically distinct from a firm's existing base—fell from roughly 40 percent in the 1990s to 20 percent in the 2010s. Recalibrating the

model to match this shift, we find that the decline in creative destruction can simultaneously account for several prominent macroeconomic trends: a falling aggregate labor share, rising firm concentration, declining dispersion in firm growth rates, and—consistent with the "Great Micro Moderation" (Sabelhaus and Song, 2010; Salgado, Bloom, Guvenen, Pistaferri, and Sabelhaus, 2023)—a reduction in the cross-sectional dispersion of firm-level employment growth and individual earnings growth. A key implication is that declining creative destruction lowers the earnings risk that workers face but at the same time increases inequality, as reduced displacement allows productivity differences across firms and workers to persist and widen.

Last, we use the calibrated model to study the welfare consequences of subsidizing innovation, or, equivalently, of exogenous shifts in the cost of innovation. In contrast to canonical growth models in which innovation unambiguously raises welfare, our environment features incomplete markets and uninsurable labor-income risk. As a result, incumbent workers value innovation very differently depending on how it affects creative destruction. A temporary subsidy that raises internal innovation generates welfare gains close to the complete-markets benchmark. By contrast, a subsidy that increases external innovation and entry—thereby raising the rate of creative destruction—reduces welfare for most workers, especially those at the top of the earnings distribution most exposed to displacement risk. High-income workers would pay several percent of annual consumption to avoid such a policy, while lower-paid workers benefit from the increased chance of drawing a better match.

These results highlight how increased business dynamism can redistribute earnings even as it reduces welfare for some workers, echoing the intuition of Krebs (2007) that the welfare costs of business cycles are higher when incorporating job displacement risk. That said, these calculations adopt the ex-post perspective of current incumbent workers who already know current labor earnings, which is the relevant one when considering changes in policy along a transition path. Intriguingly, behind the veil of ignorance (i.e., before learning one's productivity), these perceived costs vanish and each agent's willingness to pay coincides with the complete markets benchmark. Hence, our results also point to an additional potential channel—income risk—through which resistance to technological change can arise, especially among higher-paid workers.

In sum, our work emphasizes creative destruction as a driver of worker earnings risk, combining insights from Schumpeterian endogenous growth models (Aghion and Howitt, 1992; Peters, 2020; Acemoglu, Akcigit, Alp, Bloom, and Kerr, 2018) with models of worker comparative advantage (Roy, 1951; Hsieh, Hurst, Jones, and Klenow, 2019). It also

contributes to a nascent literature that emphasizes the importance of creative destruction for firm growth in the data (Akcigit and Kerr, 2018; Argente, Lee, and Moreira, 2024; Berlingieri et al., 2025), but differs in placing individual workers—rather than firms—at the center of the analysis. Models with creative destruction typically assume frictionless labor markets, which imply that creative destruction does not affect worker earnings; two notable exceptions are Aghion, Akcigit, Bergeaud, Blundell, and Hemous (2018), who study the impact of innovation on income inequality, and Aghion, Akcigit, Deaton, and Roulet (2016), who show that the degree of creative destruction is negatively related to measures of subjective well-being.

Our work is also related to a sizeable literature in labor economics that estimates the passthrough of firm shocks to workers (Guiso, Pistaferri, and Schivardi, 2005; Friedrich, Laun, Meghir, and Pistaferri, 2019; Lamadon, Mogstad, and Setzler, 2022; Chan, Xu, and Salgado, 2023) and interprets passthrough estimates as informative about the degree of insurance provided by firms to workers. By contrast, in our model the passthrough arises because workers differ in their match-specific productivity, which provides firms with labor market power. Our estimates of average passthrough from firm profits to worker earnings are nonetheless in line with the literature (Card, Cardoso, Heining, and Kline, 2018; Lamadon et al., 2022), and the greater exposure of higher-paid workers to innovation by their employer echoes Friedrich et al. (2019), who finds that the passthrough of firm-specific permanent productivity shocks is mostly concentrated among high-skilled workers. A subset of these papers studies the impact of firm innovation on the earnings of individual workers (van Reenen, 1996; Aghion, Bergeaud, Blundell, and Griffith, 2017; Kline, Petkova, Williams, and Zidar, 2019; Akcigit, Grigsby, and Nicholas, 2017; Aghion, Akcigit, Hyytinen, and Toivanen, 2018), but none examines how worker earnings respond to innovation by their firm's competitors, and they are therefore silent on the link between the overall level of creative destruction and worker earnings risk.

1 A Motivating Framework

To motivate the empirical analysis, we begin with a stripped-down version of the model in Section 3. The goal is to isolate the core mechanisms without the algebraic detail. This simplified environment does two things. First, it illustrates how labor-market frictions transmit creative destruction in product markets into workers' earnings. Second, it delivers a transparent estimating equation that guides our empirical specification. The full derivation and additional structure are provided in Appendix A.2.

Aggregate output $Y(t)$ is produced by a perfectly competitive final-goods sector that

aggregates a continuum of industry outputs $Y(j,t)$ via a Cobb-Douglas aggregator across industries, so the elasticity of substitution across industries is one and spending is equalized across industries.

Within each industry j, output is a CES composite of a continuum of intermediate varieties indexed by v:

$$Y(j,t) = \left(\int_0^1 Y_j(v,t)^{\frac{\sigma-1}{\sigma}} dv \right)^{\frac{\sigma}{\sigma-1}}. \tag{1}$$

where $\sigma > 1$ is the elasticity of substitution across varieties within an industry.

For now each firm produces a single intermediate variety, so the index v labels both firms and varieties; the full model in Section 3 allows firms to supply multiple varieties within an industry. An intermediate-good producer combines its current technology level $Q_j(v,t)$ with labor to produce output $Y_j(v,t) = Q_j(v,t)\, l(w_j(v,t))$. Because firms compete monopolistically within each industry, each variety producer sets price as a constant markup over marginal cost, reflecting the elasticity of substitution σ that governs within-industry competition.

In addition to their market power on the product side, firms also possess monopsony power in the labor market. Following Lamadon et al. (2022), each intermediate-good producer faces an upward-sloping labor supply curve for worker services, $l^s(w) = w^{1/\varepsilon}$, where $1/\varepsilon$ is the elasticity of labor supply and w is the posted wage of each firm—relative to total output $Y(t)$. Hence, a worker's per-period earnings are equal to $Y(t)\, w_j(v,t)$, and the marginal cost of labor is $w\,(1+\varepsilon)$, so the ratio of marginal to average labor cost is simply $1+\varepsilon$. We interpret this labor supply curve as applying to the extensive margin, with each worker supplying one unit of efficiency labor; the full model allows for both intensive and extensive margins.

Because each firm is a monopolistic competitor in its variety and a monopsonist in its local labor market, it internalizes how its production decision affects both the price it charges and the wage it must pay. Combining these two margins, the firm's total revenue over average cost satisfies

$$\frac{P_j(v,t)\, Y_j(v,t)}{Y(t)\, w_j(v,t)\, l_j(v,t)} = \frac{\sigma}{\sigma-1} \times [1+\varepsilon]. \tag{2}$$

That is, profit margins depend on the standard CES markup term (which decreases with the elasticity of substitution across varieties σ) and a monopsony adjustment term (which increases with the inverse elasticity of labor supply ε). Given this wedge, the firm's labor demand satisfies

$$l_j(v,t) = \left(\frac{\bar{A}}{\hat{Q}_j(v,t)} \right)^{1-\sigma} \left[\frac{\sigma}{\sigma-1} w_j(v,t)(1+\varepsilon) \right]^{-\sigma}. \tag{3}$$

In equilibrium, wages adjust so that firm-level labor demand depends only on the firm's productivity relative to its industry average, $\hat{Q}_j(v,t) \equiv Q_j(v,t)/\bar{Q}_j(t)$, where $\bar{Q}_j(t) \equiv \int_0^1 Q_j(v,t)\,dv$. The constant $\bar{A}$ is derived from the CES demand system and captures the level of aggregate productivity.

The key implication of monopsony power is that changes in a firm's relative productivity $\hat{Q}_j(v,t)$ now affect both wages and employment. Under perfect competition ($\varepsilon \downarrow 0$), a productivity shift alters only the quantity of labor demanded; the wage is pinned down by the common market wage. With monopsony power ($\varepsilon > 0$), a more productive firm raises both its wage and employment. As labor supply becomes more inelastic (larger ε), the adjustment occurs increasingly through wages rather than quantities.

A central focus of our analysis is how changes in firm productivity shape the wage dynamics of incumbent workers. Following an innovation shock that raises productivity for a subset of varieties in industry j, firm profits evolve according to

$$\mathrm{d}\log \pi_j(v,t) = \left(1 + \frac{1}{\varepsilon}\right) \frac{\sigma - 1}{\sigma + \frac{1}{\varepsilon}} \left(\mathrm{d}\log Q_j(v,t) - \mathrm{d}\log \bar{Q}_j(t)\right). \tag{4}$$

This expression highlights that future profits depend on the change in the firm's technology *relative* to its competitors, $\hat{Q}_j(v,t)$. Innovations that disproportionately benefit a subset of firms raise their relative productivity, while simultaneously lowering the relative position of competitors whose productivities do not change. In this sense, when the gains from innovation are concentrated, industry-level technological progress produces both winners and losers.

Under monopsonistic labor markets, these relative-productivity shocks affect not only profits but also the earnings of incumbent workers who remain at the firm. The equilibrium evolution of their wages is

$$\mathrm{d}\log w_j(v,t) = \frac{\sigma - 1}{\sigma + \frac{1}{\varepsilon}} \left(\mathrm{d}\log Q_j(v,t) - \mathrm{d}\log \bar{Q}_j(t)\right) \quad = \left(1 + \frac{1}{\varepsilon}\right)^{-1} \mathrm{d}\log \pi_j(v,t). \tag{5}$$

The elasticity of worker earnings with respect to innovation-driven changes in profits is decreasing in the labor supply elasticity ε^{-1}: wages respond more strongly when labor supply is less elastic.

Importantly, worker earnings are sensitive not only to improvements in their own firm's technology (consistent with the evidence in van Reenen, 1996; Kline et al., 2019) but also to innovation by competitors: holding own-firm fundamentals constant, an increase in the average productivity of rival firms reduces market share and depresses labor demand. This is the diffuse form of creative destruction that is a hallmark of Schumpeterian models, and it delivers two testable predictions. First, earnings respond to the employer's productivity

relative to competitors, so own-firm innovation raises earnings while competitor innovation lowers them. Second, the passthrough coefficient $(1 + 1/\varepsilon)^{-1}$ increases when labor supply is more inelastic, so workers with less elastic labor supply should exhibit greater exposure.

2 Creative Destruction, Firm Profits, and Worker Wages

We now document that innovation-driven creative destruction in the product market passes through to incumbent workers' earnings. The passthrough is both asymmetric and concentrated: profit losses from rival innovations translate into proportionally larger earnings declines than the gains from own-firm innovation, and these effects fall disproportionately on high-income workers. Our empirical strategy proceeds in two steps. First, we construct direct proxies for shifts in a firm's own technology and in the technology of its competitors, guided by equations (4) and (5). Second, we estimate how these innovation shocks affect firm profits and the extent to which these effects pass through to the wage earnings of incumbent workers.

2.1 Data and Measurement

We briefly summarize the data used in our analysis; full details are in Appendix A.1.

Firm Innovation

Our first step is to build empirical measures of innovation at the firm and competitor level. We follow Kogan, Papanikolaou, Seru, and Stoffman (2017) (henceforth KPSS), who estimate the market value of a patent from the stock price reaction of the innovating firm at the time of grant. Their approach delivers a direct, forward-looking measure of patent value, but it is available only for public firms because it relies on observed stock returns. We refer to this as the market value of a patent.

For each firm f in year t, we define:

$$A_{f,t} = \frac{\sum_{j \in P_{f,t}} KPSS_j}{K_{ft}}. \tag{6}$$

Here, $KPSS_j$ is the KPSS market value of patent j, and $P_{f,t}$ is the set of patents filed by firm f in year t. The numerator is the total market value of a firm's patents filed in year t. The denominator scales this by firm size, using the firm's book assets K_{ft} as our baseline. This adjustment accounts for the fact that larger firms file more patents. Results are robust if we instead scale by enterprise value. In the context of our model, $A_{f,t}$ proxies for changes in the quality of firm f's production technology, $Q_j(v, t)$.

We also construct an industry-level measure of competitor innovation:

$$A_{I\setminus f,t} = \frac{\sum_{f' \in I\setminus f} \sum_{j \in P_{f',t}} KPSS_j}{\sum_{f' \in I\setminus f} K_{f't}}. \tag{7}$$

This is the ratio of the total patent value of other firms in the same 3-digit SIC industry (excluding firm f) to the sum of their book assets.

The two measures differ in timing: for own-firm innovation ($A_{f,t}$), we date patents by their application year to avoid capturing anticipatory wage adjustments before the grant date; for competitors' innovation ($A_{I\setminus f,t}$), we use the grant year. This choice is not the main driver of our findings, as results are qualitatively similar if we also date the firm's patents as of their grant date. Further details on the timing convention are provided in Appendix A.1.

Firm Wage Bill and Employment

We measure firm wages and employment from the Longitudinal Business Database (LBD), linked to Compustat through Employer Identification Numbers (EINs). The raw sample runs from 1976 to 2019. Our analysis requires long-run growth rates of profits, employment, and wage bill, so we require that the firm has a five-year history. This means our sample of firm-year observations ends in 2014, and covers 12,500 firms with 132,000 firm-year observations. These are large, publicly traded firms. The average wage bill is $282 million, compared to a median of $27 million. The average firm-year employs 8,200 workers, while the median employs 930. Both pay and employment are highly skewed.

Worker earnings

We measure worker earnings using the CPS merged with Social Security's Detailed Earnings Record (DER). The CPS/DER panel tracks individuals across time and across firms from 1981 to 2019. These data report W-2 earnings and employer identifiers, which we use to link workers to Compustat firms by Employer Identification Numbers (EINs). We focus on prime-age workers (25–55), exclude the self-employed, and drop individuals earning less than a threshold equal to 20 hours per week at the federal minimum wage for 13 weeks (Guvenen, Ozkan, and Song, 2014). Our analysis requires five-year growth rates of wages, which means that, as with the firm data, our regression sample ends in 2014. This gives us a worker panel with 9,700 firms, 780,000 people, and 5.6 million person-years; after additionally requiring sufficient income history and firm controls, 2.9 million person-years remain, alongside 63,000 firm-years. This drop is almost entirely due to requiring firm R&D.

Our main outcome is the cumulative growth rate of worker earnings over h years:

$$g_{i,t:t+h} \equiv w^i_{t+1,t+h} - w^i_{t-2,t}. \tag{8}$$

For the bulk of our analysis we will focus on five-year horizons, $h = 5$. To remove life-cycle effects, we follow Autor, Dorn, Hanson, and Song (2014) and compute adjusted average earnings:

$$w^i_{t,t+h} \equiv \log\left(\frac{\sum_{j=0}^{h} \text{W-2 earnings}_{i,t+j}}{\sum_{j=0}^{h} D(\text{age}_{i,t+j})}\right). \tag{9}$$

Here, W-2 earnings are summed across all jobs for person i in year t, and $D(\text{age})$ adjusts for the mean life-cycle profile in earnings (Guvenen et al., 2014).

Two points are worth noting. First, averaging income growth over multiple years, as in (8)–(9), emphasizes persistent changes and smooths over transitory shocks or unemployment spells. Our baseline compares forward five-year earnings with the prior three years; other horizons give similar patterns. Second, the panel allows us to follow workers even if they leave a Compustat firm. A worker enters the sample in year t if employed by a matched firm, but subsequent earnings include income from any new employer. This ensures that $g_{i,t:t+h}$ captures the worker's true realized earnings growth, not just the growth in earnings during her tenure with a given firm.

2.2 Innovation, Profits, and Wage Earnings

We now estimate the empirical counterparts of equations (4) and (5), which relate profit and wage growth to firm and competitor innovation.

Firms

We begin by estimating the link between innovation, profits and wage earnings at the firm level. We measure cumulative growth in profits and average wages over horizon h.

$$g_{f,t:t+h} \equiv \log\left[\frac{1}{|h|}\sum_{\tau=1}^{h} X_{f,t+\tau}\right] - \log X_{f,t}. \tag{10}$$

We then estimate the following specification,

$$g_{f,t:t+h} = a\, A_{f,t} + b\, A_{I\setminus f,t} + c\, \mathbf{Z}_{ft} + u_{ft}. \tag{11}$$

The coefficients a and b capture how firm and competitor innovation affect profitability and average wages at the firm level. The controls $\mathbf{Z}_{ft}$ include firm size (log market capitalization and log book assets), R&D intensity, the KPSS value of the firm's patent portfolio (assuming 20% depreciation), the firm's idiosyncratic volatility, lagged gross profitability, and SIC

3-digit industry and year fixed effects. Standard errors are clustered by industry and year. To compare magnitudes, we normalize both $A_{f,t}$ and $A_{I\setminus f,t}$ to unit standard deviation. Appendix A.1 contains further details.

The top two rows of Figure 1 plot the responses of profits and average wages to firm and competitor innovation. The pattern is clear. When firms innovate, both profits and wages rise. A one-standard-deviation increase in firm innovation raises profits by 5.7 percent and average wages by 2.0 percent over the next five years. Competitor innovation works in the opposite direction. A one-standard-deviation increase in competitor innovation reduces focal firm profits by 4.1 percent, with only a small and statistically insignificant effect on worker pay.

At first glance, the flat response of average wages to competitor innovation might suggest that competitor innovation does not pass through to workers. That conclusion is misleading. It ignores the extensive margin: firms adjust employment as well as wages. The bottom two rows of Figure 1 plot the responses of employment and the total wage bill. We see that firm employment strongly responds to innovation: innovation by the firm leads to a 2.9 percent increase in employment and a 4.9 percent increase in the wage bill over the same period. Similarly, innovation by the firm's competitors is followed by a 2.7 percent decline in employment and a 3.4 percent decline in the wage bill over the next five years.

A final point is the absence of pre-trends in these firm-level responses. Figure 1 plots the estimated coefficients a and b across horizons $h = -5 \ldots 10$. For $h = -1$ to $h = -5$, both coefficients are small and statistically insignificant. In other words, past changes in profits, wages, or employment show no correlation with either A_f or $A_{I\setminus f}$. This lack of pre-existing trends reduces concerns that our innovation measures simply capture unobserved heterogeneity driving firm outcomes.

Individual Workers

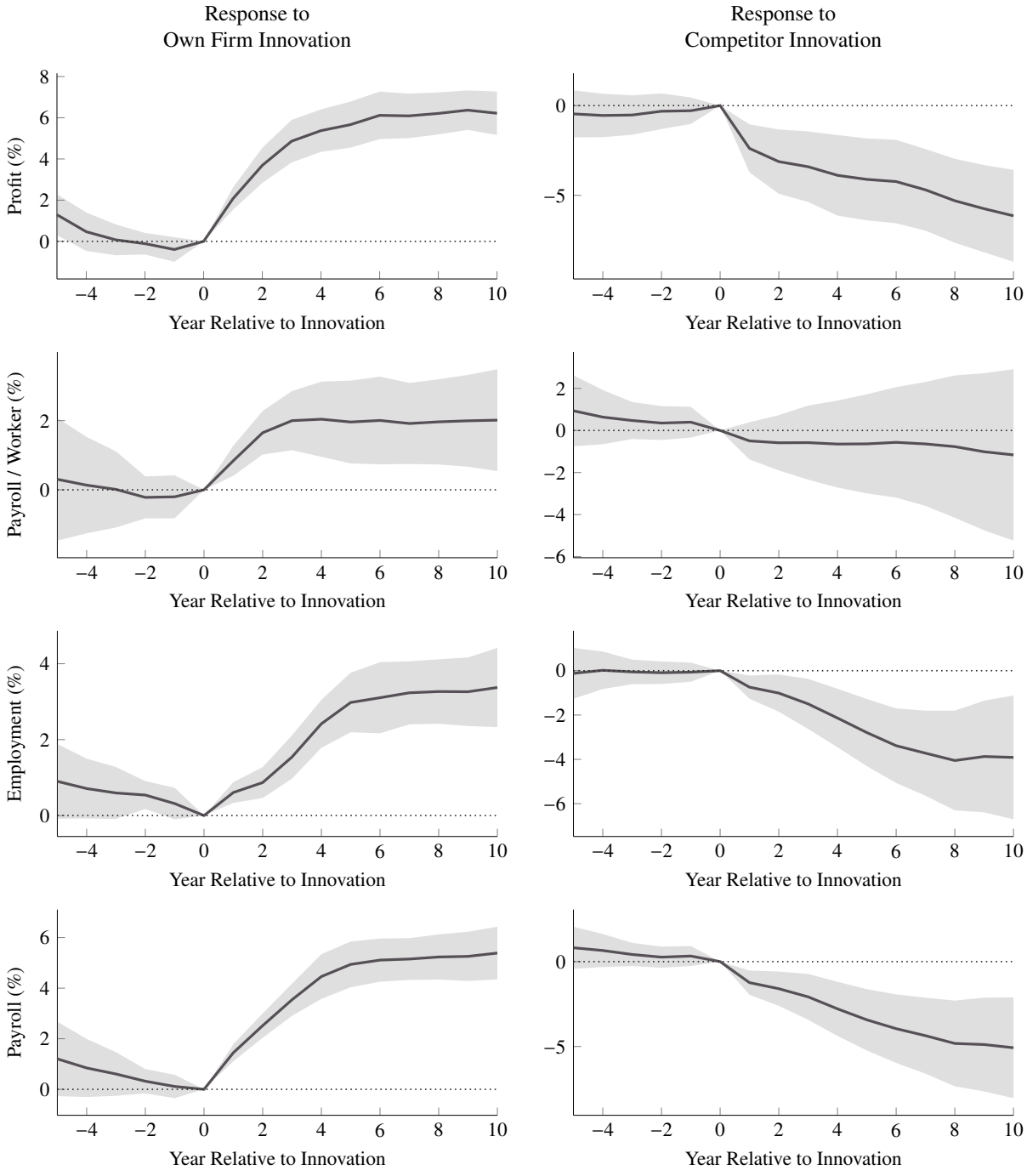
A limitation of the firm-level analysis for studying worker outcomes is that it conditions on continuing employment. For example, individual workers can still experience earnings losses while average wages rise at the firm level if firms selectively lay off lower-paid employees. We therefore move from firm aggregates to individual earnings to see how innovation shocks transmit directly to workers.

We estimate:

$$g_{i,t:t+h} = a\, A_{f,t} + b\, A_{I\setminus f,t} + c\, \mathbf{Z}_{ft} + d\, \mathbf{X}_{i,t} + \varepsilon_{i,t}, \tag{12}$$

where $g_{i,t:t+h}$ is the cumulative growth of worker i's earnings. A positive a means workers share in their firm's innovation rents, while a negative b means competitor innovation erodes

Figure 1: Cumulative Changes in Firm Employment, Pay, and Profit Following Innovation



Note: Firm outcomes following a one–standard deviation innovation shock at $t = 0$, shown as cumulative percentage changes. Pay and employment are from the LBD; profit is from Compustat. Regressions control for lagged market capitalization, stock return volatility, assets, patent stock, and industry and year fixed effects. Own-firm innovation is the stock-market surprise to a patent filing; competitor innovation is the sum of such surprises for other firms in the same industry.

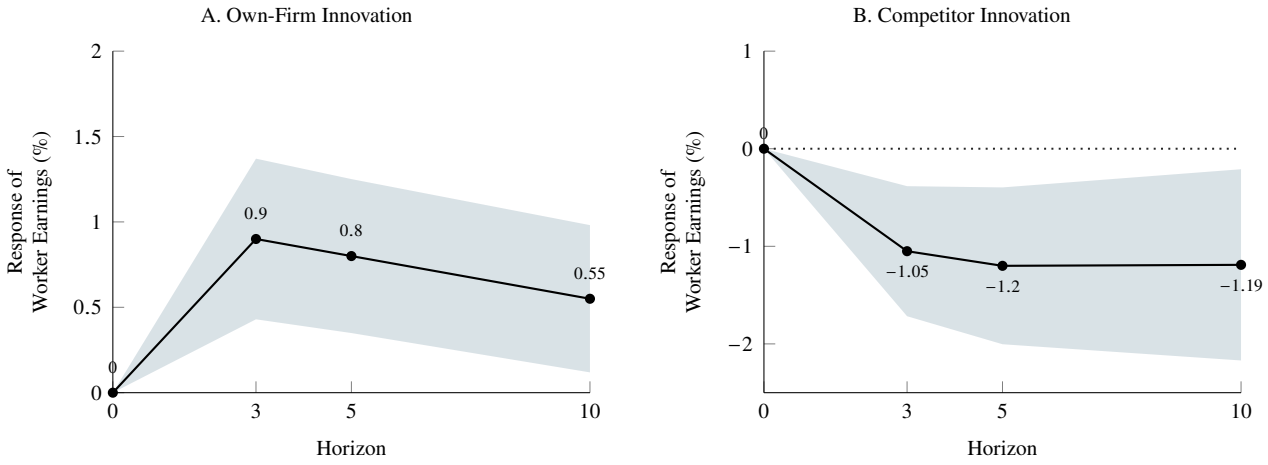
their earnings. These coefficients are identified by comparing the future earnings path for workers employed in firms that differ in their degree of innovation exposure but are otherwise similar in terms of age and current and past level of wage earnings. We control for the same set of firm-level variables $\mathbf{Z}_{ft}$ as in equation (11), plus a vector of worker controls $\mathbf{X}_{i,t}$ that includes: non-parametric controls for worker age, past earnings level, and recent earnings growth rates. We also include worker income × industry and worker income × year fixed effects, with workers partitioned into five bins of the within-industry, within-year earnings distribution. We weight observations by the inverse of the number of workers in each firm-year and cluster standard errors at the industry-year level. Appendix A.1 contains further details.

Figure 2 reports the estimated a and b coefficients for incumbent workers over horizons of three to ten years. The results confirm that innovation shocks pass through to individual workers. Firm innovation raises worker earnings, while competitor innovation lowers them—and the latter effect is larger and more persistent. At the five-year horizon, a one-standard-deviation increase in firm innovation (A_f) raises cumulative earnings growth by 0.8 percentage points. The same increase in competitor innovation ($A_{I\setminus f}$) reduces cumulative earnings growth by 1.2 percentage points. Looking across horizons, competitor innovation continues to depress earnings long after the shock. A one-standard-deviation increase cuts earnings by 1.1 percent at three years and 1.2 percent at ten years. By contrast, the effect of own-firm innovation fades: earnings gains are 0.9 percent at three years and only 0.6 percent at ten years. Thus, workers benefit modestly from their own firm's innovations, but competitor innovations impose larger and more durable earnings losses.

Contrasting the estimates from the worker-level to the firm-level regressions yields two insights. First, the ratio of changes in worker earnings to changes in firm profits gives the passthrough coefficient—equation (5) in the simple model. Our estimates of a from equations (12) and (11) imply a passthrough of about $0.8/5.7 \approx 0.14$, which is close to values in the literature (e.g., Card et al., 2018; Lamadon et al., 2022). By contrast, the estimates of b imply a much larger worker earnings decline relative to firm profits: $1.2/4.1 \approx 0.29$. Put differently, for every 1 percent fall in firm profits from competitor innovation, worker earnings drop by about 0.29 percent. The passthrough of innovation shocks is thus asymmetric: smaller for profit gains from own innovation, larger for losses from competitor innovation.

Second, incumbent workers bear a disproportionate share of the costs of creative destruction. Comparing Figure 2 (earnings of incumbent workers) with the second row of Figure 1 (average pay for all continuing employees) reveals that incumbents' earnings fall significantly more than the firm-wide average wage following competitor innovation.

Figure 2: Worker Earnings Responses to Own-Firm and Competitor Innovation



Note: Panels report worker earnings responses at 3-, 5-, and 10-year horizons to own-firm and competitor innovation. Labels above points show estimated effects from the plotted data. Estimation details (controls, weighting, and clustering) follow Section 2.2 and Appendix A.1.

The gap reflects worker turnover: when a firm expands after innovating, part of the surplus accrues to new hires; when a firm contracts after competitors innovate, incumbents absorb the full burden. Incumbents thus capture fewer of the gains from own-firm innovation and bear more of the losses from competitor innovation.

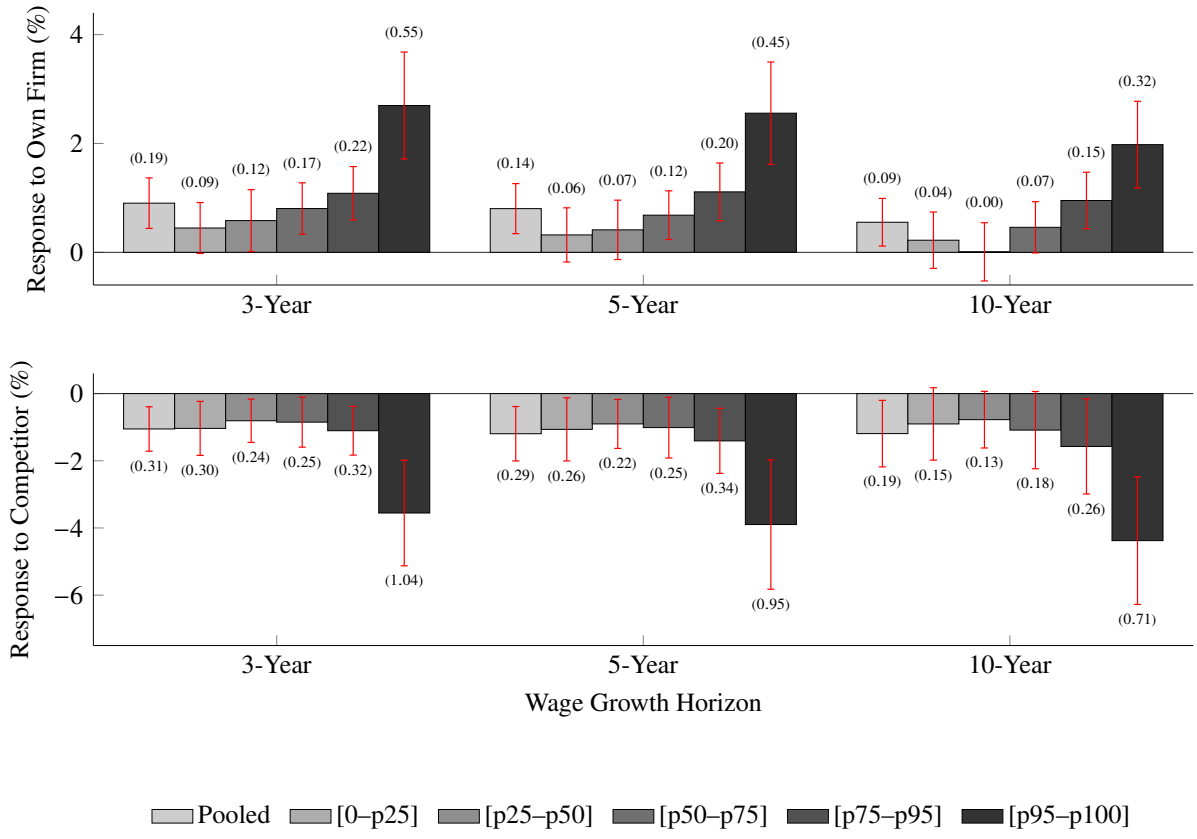
Worker Heterogeneity

We next ask which workers gain most from their own firm's innovation and which are most exposed to competitor-driven losses. The simple model in Section 1 predicts that the passthrough of profit shocks depends on the elasticity of labor supply, ε. A natural starting point is that higher-skilled workers face lower labor supply elasticities than the average worker. To test this, we re-estimate equation (12), allowing the coefficients a and b to vary with a worker's prior income relative to others in the same industry and year. Workers are grouped into five bins of the within-industry, within-year earnings distribution: bottom quartile (0–25th percentile), second quartile (25–50th), third quartile (50–75th), the rest of the top quartile (75–95th), and top ventile (95–100th). The coefficients on the control variables are also allowed to vary by income group.

Figure 3 reports the resulting a and b estimates for each group across horizons of three to ten years, together with the implied profit-sharing elasticities. The gradient is steep: higher-paid workers are much more exposed to innovation shocks than lower-paid workers. At the five-year horizon, a one-standard-deviation increase in firm innovation raises earnings growth by 2.6 percentage points for workers in the top 5% of the earnings distribution. By

contrast, a one-standard-deviation increase in competitor innovation reduces their earnings growth by 4 percentage points. The implied passthrough coefficients—shown in parentheses in the figure—are about three times larger than those for the average worker. In short, both the gains and the losses from innovation are disproportionately concentrated at the top of the earnings distribution.

Figure 3: Heterogeneous Responses of Wage Growth to Innovation by Prior Worker Earnings



Note: Heterogeneous worker-earnings responses by prior income group; parenthetical passthrough values scale wage-growth responses by the corresponding 5-year firm-profit responses. See Section 2.2 and Appendix A.1 for full specification details.

Industry Heterogeneity

To investigate which workers are more or less exposed to innovation-related income risk, we allow coefficients to vary by industry characteristics. First, we expect the creative destruction channel to be more relevant in firms belonging to industries with high rates of innovation. To illustrate this effect, we re-estimate equations (11) and (12) separately depending on whether the level of industry innovation (the industry analogue of (6)) is above or below the median. As we see in Panel A of Table 1, the magnitudes of the estimated coefficients a and

b are significantly larger in industries with high rates of innovative activity. In particular, the responses of firm profits, total worker pay, and employment to competitor innovation are larger by an order of magnitude in industries in which the rate of innovation is high relative to firms in the other industries. The last two columns show that the response of worker earnings for top workers (those in the top quartile) is also significantly larger in industries with high levels of innovation.

Table 1: Firm Outcomes and Wage Growth for Industry Subgroups

		5 Year Firm Outcomes			5 Year Wage Growth	
		Profit	Pay	Employ.	P50–75	P95–100
Panel A. Industry Innovation Rate						
Own Firm	High	8.790 (0.775)	7.310 (0.652)	4.430 (0.739)	1.230 (0.304)	3.780 (0.732)
	Low	2.510 (0.714)	2.180 (0.580)	1.480 (0.612)	-0.099 (0.124)	0.235 (0.267)
Competitor	High	-4.560 (1.570)	-3.970 (1.190)	-2.470 (1.040)	-0.823 (0.621)	-4.270 (1.300)
	Low	-0.183 (0.432)	-0.200 (0.439)	-0.079 (0.399)	-0.375 (0.145)	-0.438 (0.203)
Panel B. Industry Income Inequality						
Own Firm	High	7.190 (0.613)	6.030 (0.546)	3.640 (0.613)	0.869 (0.261)	2.930 (0.588)
	Low	3.800 (0.797)	2.240 (0.580)	1.670 (0.625)	0.043 (0.140)	0.271 (0.230)
Competitor	High	-4.310 (1.340)	-4.000 (1.040)	-2.670 (0.900)	-1.030 (0.585)	-4.440 (1.220)
	Low	0.380 (0.520)	-0.137 (0.628)	-0.265 (0.641)	-0.026 (0.230)	-0.062 (0.344)
Panel C. Average Industry Wage						
Own Firm	High	7.540 (0.652)	6.380 (0.577)	3.810 (0.624)	1.030 (0.271)	3.440 (0.579)
	Low	2.980 (0.707)	1.760 (0.512)	1.750 (0.725)	0.269 (0.180)	-0.646 (0.492)
Competitor	High	-4.240 (1.410)	-4.010 (1.070)	-2.700 (0.942)	-0.784 (0.527)	-4.810 (1.040)
	Low	-0.656 (0.456)	-0.764 (0.531)	-0.711 (0.580)	0.448 (0.447)	-0.605 (0.849)

Note: This table contains responses to own firm and competitor innovation, where industries are grouped into high or low relative to the median industry. All coefficients are multiplied by 100, with standard errors reported in parentheses next to each estimate. Industry income inequality is based on the cross-sectional variance within an industry in a given year. Industry innovation rate is measured by the rate of an industry's patenting activity. Industry wages are measured using the CPS-DER linked sample.

Second, we consider two indirect proxies for the labor supply elasticity in our model. First, the within-industry dispersion of earnings: in our quantitative model below, the elasticity of labor supply is endogenously negatively related to the degree of dispersion in workers' productivity across different potential jobs they consider. Second, the average level of earnings: industries with higher average wages are more likely to employ high-skill workers whose labor supply is less elastic. The results in panels B and C of the table are consistent with our intuition. Panel B shows that both own firm and competitor innovation are only significantly related to worker earnings within the subset of industries with high levels of income dispersion. This result obtains despite the fact that own firm innovation is a

significant predictor of firm outcomes within the low income dispersion industries. Panel C shows that similar results obtain when comparing industries with higher average wages to those with lower average wages. Together, these results show that innovation shocks transmit unevenly, with the strongest effects in industries where labor supply is likely to be less elastic.

2.3 What Drives the Passthrough of Innovation to Worker Earnings?

The results so far reveal a striking asymmetry: the passthrough of competitor innovation to worker earnings is roughly twice as large as the passthrough of own-firm innovation. The simple model of Section 1 cannot generate this, because it features only incremental innovation and abstracts from employment transitions. In practice, innovation involves both improving existing products and taking market share from rivals. We now examine three channels that can generate the asymmetry: ex-post heterogeneity in worker outcomes, job destruction, and differences between radical and incremental innovations.

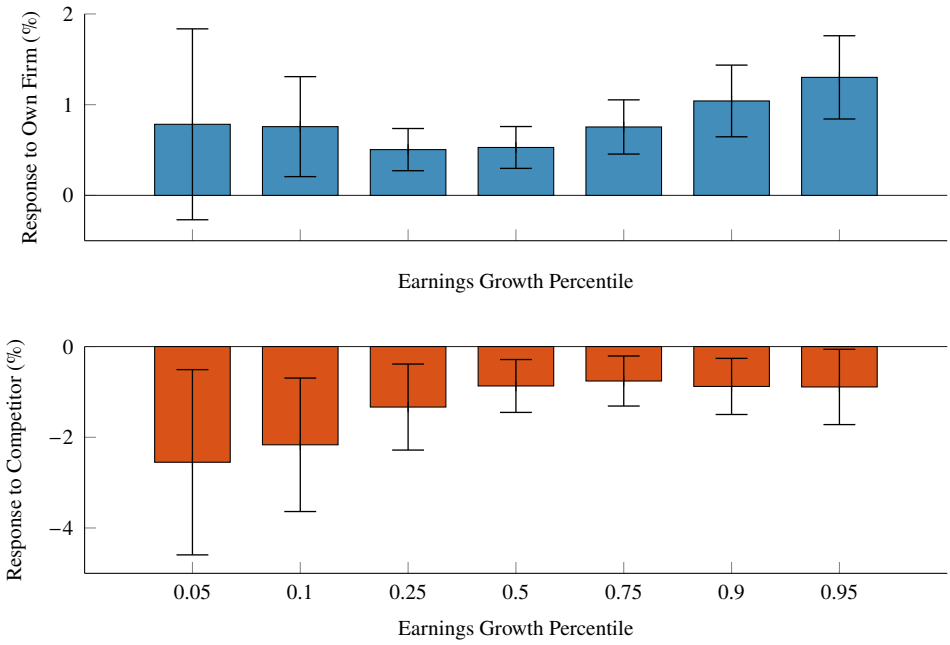
Ex-post Heterogeneity in Worker Outcomes

So far we have focused on average effects. Yet averages mask substantial ex-post heterogeneity across incumbent workers. To uncover this, we estimate equation (12) using the quantile regression approach of Schmidt and Zhu (2025), which characterizes how the entire distribution of earnings growth shifts in response to firm or competitor innovation.

Figure 4 plots the marginal effects—analogous to coefficients a and b—for different quantiles of the earnings growth distribution. Competitor innovation is associated with more negatively skewed subsequent earnings growth for workers. A one-standard-deviation increase in competitor innovation lowers the 5th percentile of earnings growth by 2.6 percentage points, compared to 0.9 points at the median. By contrast, higher innovation by the worker's own firm is associated with more positively skewed earnings growth—a 1.3 percentage point increase in the 95th percentile compared to a 0.5 percentage point increase in the median earnings growth. Because gains and losses are concentrated, the welfare implications for risk-averse workers are substantially worse than the mean suggests; we quantify this in Section 4.3.

We next examine how these effects vary by worker income. Appendix Figure SA.1 shows quantile regression estimates of equation (12) where coefficients a and b vary with the worker's prior income rank. The income gradient in risk is even more pronounced in the tails: top earners face both greater passthrough and greater income risk than the average worker. For example, when competitors innovate, a one-standard-deviation increase reduces

Figure 4: Innovation and the Distribution of Wage Growth: Quantile Regression Estimates



Note: Average marginal effects from quantile regressions (following Schmidt and Zhu 2025) for own-firm and competitor innovation across the earnings-growth distribution. Standard errors use the weighted bootstrap. Variable definitions and full regression specifications are in Section 2.2 and Appendix A.1.

the 5th percentile of earnings growth for top workers by 7.1 percentage points—far larger than the 2.6 point decline at the median. Conversely, when their own firm innovates, top workers see a 3.5 point increase at the 95th percentile, compared to 1.8 points at the median. Thus, high earners not only capture more of the upside from their own firm's innovation but also bear much more downside risk from competitor innovation.

Job Destruction

A significant component of the passthrough of innovation to incumbent workers' earnings risk arises from the possibility of involuntary job loss, which typically entails large and persistent income costs. To test this channel, we re-estimate equation (12) replacing the dependent variable with an indicator equal to one if the worker experiences an involuntary exit within the next five years. Because the data do not record the reason for separations, we define involuntary exit as a change of employer coinciding with earnings growth below the 20th percentile.

Table 2 reveals an important reason behind the asymmetric passthrough. Competitor innovation raises the likelihood of involuntary exit: a one-standard-deviation increase in competitor innovation increases the probability by 0.81 percentage points. By contrast, a similar increase in own-firm innovation lowers the probability of involuntary exit by 0.33

percentage points. The effects are especially pronounced for high-income workers. A one-standard-deviation increase in competitor innovation raises their likelihood of involuntary exit by 1.87 percentage points, compared to only 0.70 points for lower-paid workers. These results confirm that job destruction is an important mechanism through which innovation shocks translate into earnings risk, particularly for top earners.

Table 2: Effects of Innovation on Likelihood of Involuntary Exit

Pr(Involuntary Exit)	Pooled	By earnings group				
		P0–25	P25–50	P50–75	P75–95	P95–100
Own Firm Innovation	-0.33%	-0.17%	-0.22%	-0.31%	-0.42%	-0.79%
	(0.12%)	(0.14%)	(0.18%)	(0.12%)	(0.09%)	(0.22%)
Competitor Innovation	0.81%	0.70%	0.74%	0.75%	0.89%	1.87%
	(0.29%)	(0.32%)	(0.33%)	(0.33%)	(0.30%)	(0.41%)

Note: Involuntary exit is defined as leaving a firm and experiencing 5 year wage growth in the bottom 20% of the sample for that given year. Columns show probability of involuntary exit in response to a one standard deviation change in innovation based on the estimates from a linear probability model. Controls are identical to the other worker-level regressions in the text, and standard errors (shown in parentheses) are clustered by industry and year.

Radical vs Incremental Innovations

The discussion above suggests that incremental innovations—improvements to a firm's existing product portfolio—should be associated with higher passthrough to incumbent workers than radical innovations, which are more likely to involve stealing market share from competitors. The logic is that part of a worker's human capital is tied to a specific product; when a firm acquires a new product line, some gains accrue to newly hired workers rather than incumbents, reducing the benefits passed through to the latter. This implies an important asymmetry: radical innovations generate larger gains for the firm overall, but a smaller share of those gains reaches incumbent workers. We now examine this prediction in the data.

We classify patents as radical (novel) or incremental using the text-based measure of Kelly, Papanikolaou, Seru, and Taddy (2021), which uses the pairwise similarity between patents issued to the same firm. A patent i is called novel if it is sufficiently distinct from the firm's previous five years of patents:

$$N_i = 1 \Leftrightarrow \max_{j \in P(f)} \rho_{i,j} \leq 0.5, \tag{13}$$

where $\rho_{i,j}$ is the textual similarity between patents i and j, and $P(f)$ is the set of patents filed by firm f in the past five years. If a firm lacks five years of history, patents are compared to all their prior patents. A firm's first patent is always novel. Roughly 35% of patents meet this criterion.

At the firm level, we then decompose innovation into radical and incremental components:

$$A^r_{f,t} = \frac{\sum_{j \in P_{f,t}} KPSS_j \, N_j}{K_{ft}}, \qquad A^i_{f,t} = A_{f,t} - A^r_{f,t}. \tag{14}$$

The correlation between A^i_f and A^r_f is approximately 22%.

We then estimate a modified version of equations (11) and (12), decomposing own-firm innovation into radical and incremental parts:

$$g_{i,t:t+h} = a \, A_{I \setminus f,t} + b^r \, A^r_{f,t} + b^i \, A^i_{f,t} + c \, \mathbf{Z}_{ft} + d \, \mathbf{X}_{i,t} + \varepsilon_{i,t}. \tag{15}$$

where i indexes either firms or workers, and the controls are the same as in earlier specifications. For comparability, coefficients b^r and b^i are scaled by the standard deviation of own-firm innovation.

Table 3 presents the results. Panel A shows that radical innovations generate larger increases in firm profits, total labor compensation, and employment than incremental innovations. More importantly, Panel B shows that this pattern does not extend to workers: radical innovations raise profits much more than incremental ones (9.3 vs. 3.9 percentage points), but increase pay for top workers by less (2.3 vs. 2.8 percentage points). The implied passthrough coefficients are 0.12 vs. 0.17 among all workers, and 0.25 vs. 0.73 for workers in the top 5th percentile.

Table 3: Firm vs Worker Responses to Radical and Incremental Innovation

	Own Firm Innovation		
	Radical	Incremental	Competitor
A. Firm Outcomes			
Profit	9.25% (0.99)	3.88% (1.01)	-4.24% (1.20)
Total Payroll	7.70% (0.65)	3.36% (0.48)	-3.98% (1.04)
Employment	5.59% (0.71)	1.21% (0.54)	-3.42% (0.87)
B. Worker Wage Growth			
Pooled	1.09% (0.52) [11.79]	0.66% (0.32) [17.08]	-1.20% (0.41) [28.19]
[0%, 25%)	1.29% (0.74) [13.99]	-0.26% (0.50) [-6.72]	-1.07% (0.48) [25.12]
[25%, 50%)	0.12% (0.56) [1.29]	0.73% (0.46) [18.77]	-0.89% (0.37) [21.10]
[50%, 75%)	0.96% (0.68) [10.43]	0.56% (0.46) [14.42]	-1.01% (0.46) [23.83]
[75%, 95%)	1.50% (0.77) [16.18]	0.81% (0.39) [21.00]	-1.42% (0.49) [33.50]
[95%, 100%]	2.31% (1.17) [24.95]	2.82% (0.66) [72.59]	-3.91% (1.00) [92.12]

Note: Radical (novel) patents are measured based on textual similarity to prior patents. The table compares firm and worker outcomes at a five-year horizon across the three innovation shocks. Standard errors, also in percentage points, are in parentheses; the implied passthrough from profits to wages is in brackets.

2.4 Robustness and Additional Results

Alternative Measures

Our results are robust to three alternative measurement choices, all reported in Tables SA.1 and SA.2. First, we replace 3-digit SIC industries with the product-market similarity measure of Hoberg and Phillips (2016) (TNIC3), which likely captures a firm's actual competitors more accurately; its main drawback is that it is available only from 1989 onward, which substantially reduces the sample. Second, we scale the innovation measures in (6) and (7) by enterprise value—market capitalization plus book assets minus the value of common equity—rather than by book assets. Third, our W-2 earnings measure omits Incentive Stock Options (ISOs), which are untaxed on exercise and therefore absent from W-2 data; because 97% of ISO income accrues to workers in the top decile of earnings (Eisfeldt, Falato, and Xiaolan, 2022), this omission biases our estimates of passthrough for top workers towards zero. We therefore replicate the analysis using adjusted gross income (AGI) and wage and salary income (WSI) from individual tax returns (Form 1040), which include realized gains from stock option exercise (Panel E; see also Appendix A.1). Because the 1040-based records are available continuously only after 1999, we use them for robustness rather than as our main analysis. In all three cases the estimates are quantitatively and economically similar to our baseline.

Pre-trends

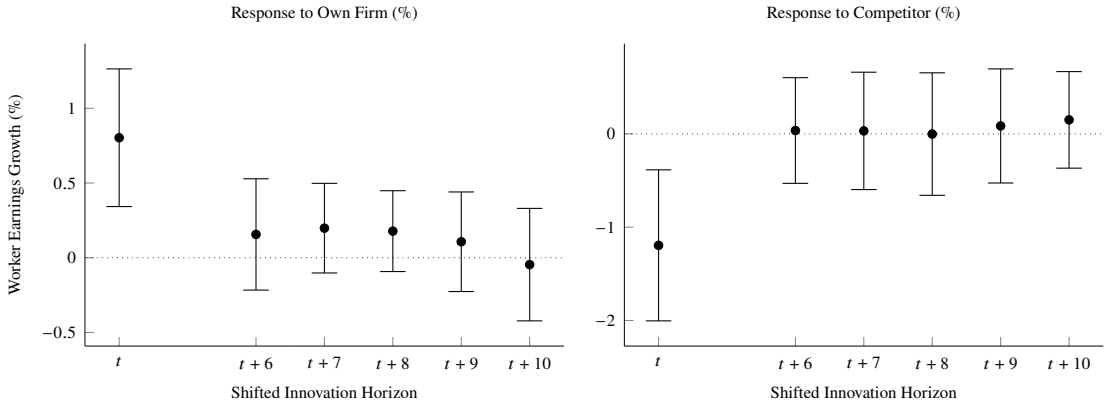
Our identification relies on cross-sectional variation in KPSS patent values as innovation shocks, conditional on an extensive set of firm and worker controls. The key identifying assumption is that, conditional on observables—which include measures of past patenting and R&D investment—the arrival of a highly-valued patent (one that elicits a significant increase in firm value) represents an unanticipated shift in a firm's technology; the absence of pre-trends in Figure 1, noted above, is consistent with this interpretation. Still, persistently innovative firms may employ workers with steeper earnings profiles, while firms lagging in innovation may be on slower growth paths and employ workers with flatter income trajectories. If our controls for firm R&D, past patenting, and worker earnings history do not fully capture these differences, our estimates could reflect such selection rather than the innovation shock itself.

To address this concern, we ask whether current worker earnings growth predicts future innovation by the firm or its competitors, re-estimating equation (12) with the measures of own-firm and competitor innovation, A_f and $A_{I\setminus f}$, shifted forward,

$$g_{i,t:t+5} = a\, A_{f,t+h} + b\, A_{I\setminus f,t+h} + c\, \mathbf{Z}_{ft+h} + d\, \mathbf{X}_{i,t} + \varepsilon_{i,t}. \tag{16}$$

Our choice of $h = 6 \ldots 10$ avoids any overlap with the five-year period used to measure wage growth, and we shift the timing of the firm-level controls $\mathbf{Z}$ in parallel to remove any predictable component in innovation. Figure 5 shows no economically or statistically significant relation between wage growth and subsequent innovation: high wage growth among a firm's incumbent workers does not forecast future firm innovation. It is therefore difficult to reconcile our findings with any omitted factor that jointly drives both innovation and worker earnings trajectories.

Figure 5: 5-Year Wage Growth and Future Innovation: A Placebo Test



Note: Placebo tests relate past wage growth to future innovation (from lead 6 onward), probing pre-trends and reverse timing. The dependent variable is cumulative worker earnings growth from t to $t + 5$. Specifications follow Section 2.2 and Appendix A.1.

Alternative Mechanisms

Our interpretation is that the results reflect passthrough of product-market creative destruction to worker earnings, in which case the effect of innovation should operate through the focal firm's profits. If instead shifts in industry innovation were driven by changes in product demand at competing firms, we would expect increases in competitor innovation to be directly associated with lower worker earnings at the focal firm. Because profits are themselves an outcome of innovation, we cannot include both profits and innovation in the same regression; we therefore re-estimate equation (12), replacing $A_{I\setminus f}$ and A_f with profit growth of the competitor and the focal firm, constructed analogously to equations (6) and (7). Panels A and B of Appendix Figure SA.2 show that only profits of the worker's own firm are systematically related to wage growth: once we control for own-firm profits, changes in competitor profitability have no economically meaningful effect. Replacing profit growth with stock returns yields the same conclusion; if anything, stock returns of competing firms are positively correlated with worker earnings in the focal firm, a pattern consistent with

other common shocks to industries (e.g., demand shocks) that drive both overall industry stock returns and workers' outside options. In short, the evidence is consistent with our interpretation that the passthrough of innovation shocks to worker earnings operates through firm profits.

3 The Model

The empirical evidence in Section 2 documents that the passthrough of creative destruction to worker earnings is asymmetric and concentrated—a pattern the simple model in Section 1 cannot generate. We now build a richer model that reproduces all three features of the data: asymmetric passthrough, concentration of risk among high-income workers, and skewed earnings dynamics.

We extend the simple model along two key dimensions. First, we endogenize labor supply and introduce rich worker heterogeneity. Firms employ both low-skilled laborers and high-skilled managers, and technology is skill biased so that more productive firms require a higher manager-to-worker ratio. Both types make flexible labor supply decisions along extensive and intensive margins. Following Roy (1951), workers select the best job from a distribution of offers and choose their effort level when employed, generating intensive-margin variation in hours or efficiency units supplied. These features jointly produce realistic heterogeneity across firms and workers.

Second, we incorporate business stealing by allowing firms to invest in two types of innovation. The first is *internal* innovation, which improves the profitability of a firm's existing products, as in Section 1. The second is *external* innovation, undertaken by incumbents and potential entrants, which improves upon a competitor's product and thereby steals its product line, as in Aghion and Howitt (1992). The loss of a product line affects not only firm profitability but also the evolution of workers' human capital: business-stealing events can destroy valuable, high-productivity matches for some workers while creating new, potentially superior, opportunities for others. Section 3.5 provides intuition for how these ingredients jointly generate asymmetric passthrough.

3.1 Static Equilibrium

We begin by outlining the static equilibrium block of the model, in which firms post wages and prices and workers supply labor. We retain the notation of Section 1 unless noted otherwise, and highlight only the new elements.

Production

Firms in each industry compete in producing different varieties as in (1). Following Acemoglu et al. (2018), Bertrand competition among potential producers, combined with a small fixed cost of entry, ensures that only the most efficient firm in each variety produces in equilibrium and sets the monopoly price (see Appendix A.2 for details).

A firm with productivity $Q_j(v,t)$ that is active in industry j and variety v produces output using a constant-returns-to-scale technology that combines two types of labor, laborers and managers,

$$Y_j(v,t) = \left((1-B) \left(Q_j(v,t) l_j(v,t) \right)^{\frac{\psi-1}{\psi}} + B \left(\bar{Q}_j(t)^{\alpha} Q_j(v,t)^{1-\alpha} m_j(v,t) \right)^{\frac{\psi-1}{\psi}} \right)^{\frac{\psi}{\psi-1}}, \tag{17}$$

Here, $l_j(v,t)$ and $m_j(v,t)$ denote efficiency units of laborers and managers, who are complements in production ($\psi \in (0,1)$), so a more productive firm needs both more laborers and more managers. The parameter $\alpha \in [0,1]$ governs skill bias: because managerial productivity is not fully variety-specific, more productive firms demand a disproportionately higher manager-to-laborer ratio, generating strong sorting of high-skilled workers into large firms—similar to Burstein and Vogel (2017). The leading producer in each variety posts a price $P_j(v,t)$ and piece rates $w_j(v,t)$ and $s_j(v,t)$ per efficiency unit of laborers and managers, expressed relative to aggregate output $Y(t)$; thus, a laborer's per-period earnings equal the number of efficiency units she supplies times $w_j(v,t)\,Y(t)$.

Labor Supply

Workers supply labor along both an extensive margin of job choice and an intensive margin of effort. A worker's labor income equals the product of the number of efficiency units she supplies per unit of effort z, the amount of effort e she chooses, and the posted wage $w_j(v,t)\,Y(t)$ per efficiency unit. Because effort costs are symmetric across all jobs, we can treat the problem sequentially: each worker first chooses the job that offers the highest wage per unit of effort, and then chooses her optimal level of effort.

Workers differ in their productivity across a finite set of potential jobs, and select the job that maximizes earnings, as in the Roy (1951) model. Upon entering the job market, each worker i is randomly assigned a type $\tau(i) \in$ (Laborer, Manager) with probabilities $(\omega, 1-\omega)$, and observes her match-specific productivity in K jobs (industry–variety pairs)—a tractable stand-in for the information frictions that prevent workers from knowing their productivity in all potential jobs. This device is analogous to the simultaneous-search models used in

industrial organization, where a finite consideration set captures frictions in consumer choice (see Honka, Hortaçsu, and Wildenbeest, 2019, for a survey); our formulation is closest to Bils, Kaymak, and Wu (2025). Let $j(k,i,t)$ and $v(k,i,t)$ denote the industry and variety associated with her k^{th} job, respectively; worker i's match-specific productivity in that job follows a Fréchet distribution,

$$z_{j(k,i,t)}(i, v(k,i,t), t) \sim \text{Fréchet}(T, \theta_{\tau(i)}), \tag{18}$$

where the scale parameter T is the same for laborers and managers, while the dispersion parameters $\theta_{\tau(i)}$ are different. We assume $\theta_l > \theta_m$, so that the productivity of managers is more dispersed than the productivity of laborers; this assumption implies that the labor supply of managers is less elastic on the extensive margin than that of laborers. In addition to these K options, each worker also has a fixed outside option (home production) in which she earns $R\,Y(t)$ per unit effort and has productivity $z^*_{i,t} = 1$. We defer the discussion of how worker productivity evolves to Section 3.2.

Each worker chooses the option that maximizes her wage per unit of effort e. We let $z^*_{i,t}$ and $w^*_{i,t}$ denote the productivity and wage, respectively, associated with worker i's optimal choice. Given the Fréchet distribution of productivity z, the probability that a type-$\tau(i)$ worker selects job k from the set of posted wages $\{w_j\}_{j=1}^K$ is

$$P\left[\text{option } k \text{ selected}|\tau(i), \{w_j\}_{j=1}^K\right] = \frac{T w_k^{\theta_{\tau(i)}}}{\sum_{\kappa=1}^K T w_\kappa^{\theta_{\tau(i)}}} \left(1 - \exp\left(-\left[T \sum_{\kappa \in K} w_\kappa^{\theta_{\tau(i)}}\right] R_{\tau(i)}^{-\theta_{\tau(i)}}\right)\right). \tag{19}$$

Appendix A.2 derives the implied measure of laborers and managers who supply labor to each firm after integrating over the distribution of wages posted by other firms.

We next discuss workers' labor supply decisions along the intensive margin. Workers are hand-to-mouth and have Greenwood, Hercowitz, and Huffman (1988) preferences over consumption c and work effort e,

$$U(c,e) = u\left(c - Y(t)\frac{e^{1+\varphi}}{1+\varphi}\right), \tag{20}$$

where $u(\cdot)$ is a monotonically increasing and concave function satisfying the Inada conditions, and $\varphi > 0$ captures the disutility of effort. The assumption that the disutility of labor is proportional to output $Y(t)$ preserves balanced growth.

Each worker chooses their level of effort to maximize (20). Because each worker consumes her wage earnings in each period, the optimal choice of effort is

$$e^*_{i,t} = (w^*_{i,t} z^*_{i,t})^{\frac{1}{\varphi}}. \tag{21}$$

Workers in better-matched, higher-paying jobs therefore supply more effort. Given (21), the total earnings for worker i in period t can be written as

$$e^*_{i,t}\, w^*_{i,t}\, z^*_{i,t}\, Y_t = (w^*_{i,t}\, z^*_{i,t})^{1+\frac{1}{\varphi}}\, Y_t = c^*_{i,t}. \tag{22}$$

Given the extensive- and intensive-margin decisions described above, the labor supply curves faced by a firm are deterministic functions of its posted wages, derived in Appendix A.2. In contrast to the simple model of Section 1, the quantity of labor supplied per worker is now flexible. By introducing effort on the intensive margin, we add a second channel through which labor supply responds to posted wages. As a result, there are three distinct channels through which labor supply responds to a firm's posted wage. The first is the *intensive margin*: workers in higher-wage jobs supply more effort. The second is *reallocation across employers*: a higher wage draws workers away from competing firms. The third channel operates through *non-employment*: higher wages draw workers out of home production. When a firm needs to attract a very large number of workers, the latter two channels become less important, leaving only the intensive-margin response. Consequently, the total labor-supply elasticity faced by an individual firm declines with the wage and asymptotes to $1/\varphi$. This declining elasticity is the source of endogenous monopsony power in the model.

Static Equilibrium

Equilibrium in each period requires that (i) firms choose wages and prices to maximize flow profits, taking other firms' behavior as given, and (ii) firms hold correct beliefs about their competitors' actions. Firms are atomistic and take aggregate quantities, price indices, and the distribution of competing wage offers as given, while internalizing how their own posted wages affect the labor they attract. Because each worker considers only a finite set of jobs, a firm can attract a substantial share of the workers who consider it while remaining negligible relative to the economy.

Panel A of Appendix Figure SA.3 shows that more productive firms employ a disproportionately larger share of managers. Although laborers and managers are complements ($\psi < 1$), managerial productivity does not scale one-for-one with firm productivity; as firms expand, they therefore require a higher manager-to-laborer ratio. Panel B shows that the labor supply elasticity for managers is lower than that of laborers on the extensive margin. For a given worker type, more productive firms face a less elastic labor supply curve because they already attract a larger share of the workers who consider them. A higher posted wage raises labor supply both by inducing greater effort under GHH preferences and by increasing the probability that the firm offers the highest wage per unit of effort. The latter response weakens as the firm's

wage rises relative to competing offers, because its own wage also enters the denominator of the job-choice probability. Intuitively, the workers who remain with other employers have increasingly favorable matches there relative to the expanding firm, so further wage increases generate smaller proportional gains in employment. In addition, more productive firms employ relatively more managers than laborers. Both of these mechanisms imply that more productive firms face steeper labor supply curves. As a result, we see in Panel C that firm profit margins increase with the firm's productivity advantage relative to competitors. Last, Panel D shows that average wage earnings paid by the firm increase in firm productivity, and that this increase is driven by managers. Because labor supply is upward sloping and becomes steeper when the firm already employs many workers, more productive firms must pay more to attract the managers they need; earnings per laborer are close to flat, so the rise in the firm-wide average also reflects the shift in composition toward managers. Each firm therefore trades off product-market power against the cost of attracting labor from competing employers, setting a constant markup over marginal cost while posting wages below the marginal product of labor (see Appendix A.2 for the formal profit equation and a characterization of the static equilibrium).

3.2 Innovation and Worker Productivity Dynamics

We now describe the dynamic block of the model and delegate all details to Appendix A.2. Appendix Figure SA.4 presents the timeline of events within each instant.

Our empirical results exploit time-series variation in the arrival rate of innovation within industries, which indicates that innovation tends to cluster over time. To capture this clustering in a tractable model with a continuum of product lines, we introduce an industry-specific source of risk that produces common shocks to innovation outcomes: an industry-level stochastic clock $\Gamma_j(t)$, a compound Poisson process with non-negative increments:

$$\mathrm{d}\Gamma_j(t) = \mu \, \mathrm{d}N_j(t), \qquad N_j(t+\Delta) - N_j(t) \sim \text{Poisson}\left(\frac{\Delta}{\mu}\right), \tag{23}$$

where $\mathrm{d}N_j(t)$ is a Poisson process with intensity $1/\mu$. Thus, $\Gamma_j(t)$ acts as a subordinator for technological progress: it increases in random increments whose expected size satisfies $E[\mathrm{d}\Gamma_j(t)] = \mathrm{d}t$. The parameter μ governs the lumpiness of innovation: larger μ produces more volatile industry-level growth, while aggregate quantities evolve deterministically because shocks are independent across the continuum of industries.

Innovation by Incumbent Firms

Similar to Akcigit and Kerr (2018), we allow for two different dimensions of innovation: a firm can innovate within a variety in which it is the leading producer (internal innovation) or in a random variety it does not currently produce (external innovation). To streamline the exposition, we first take firms' innovation investment decisions as given. As in Peters (2020), our assumptions on the cost of innovation will imply that firm investment rates are constant across varieties, which allows us to describe innovation dynamics at the variety level.

Internal innovation. The leading producer in each variety chooses its investment in internal innovation, denoted I, which determines the likelihood of achieving a successful improvement in productivity. An investment level I at time t^- generates a stochastic intensity $I\, d\Gamma_j(t)$ of internal innovation over the next instant. Given the definition of $\Gamma(t)$ in (23), the expected arrival rate of successful innovation equals $I\, dt$.

When a firm succeeds in innovating, its productivity increases by a random number of steps. Each step raises productivity by a factor $\lambda > 1$, and the number of steps reflects the amount of business time that elapses during the innovation burst governed by (23). Thus, conditional on not being displaced by a concurrent external innovation, a firm that invests I experiences a stochastic productivity change in variety v given by

$$\mathrm{d}q_j(v,t) \equiv \mathrm{d}\log Q_j(v,t) = \log\lambda\, N_j^I(v,t) - \delta\,\mathrm{d}t, \tag{24}$$

where δ captures the (smooth) rate of depreciation of the firm's productivity and

$$N_j^I(v,t) \sim \text{Poisson}\big(I_j(v,t^-)\, d\Gamma_j(t)\big)\,, \tag{25}$$

is a variable that counts the number of successful internal innovation steps; it can exceed one because a positive jump in $d\Gamma_j(t)$ represents a burst of business time, allowing multiple innovation steps to occur within an instant.

Therefore, conditional on the incumbent not being displaced by an external innovator, the firm's relative productivity evolves as

$$\mathrm{d}\hat{q}_j(v,t) \equiv \mathrm{d}\log Q_j(v,t) - \mathrm{d}\log \bar{Q}_j(t) = N_j^I(v,t)\log\lambda - \tilde{g}\,\mathrm{d}\Gamma_j(t). \tag{26}$$

That is, a firm's competitive position erodes whenever the rest of its industry innovates faster than it does, and improves only when the firm itself successfully innovates. Internal innovations generate upward jumps proportional to the number of successful steps $N_j^I(v,t)$, while industry-wide innovation bursts raise average productivity $\bar{Q}_j(t)$, creating a downward drift in relative productivity at rate $\tilde{g}\, d\Gamma_j(t)$.

Appendix Figure SA.5 illustrates how the subordinator $\Gamma_j(t)$ maps calendar time into

business time, producing the lumpy, industry-correlated shocks that our empirical strategy uses.

Firms choose the level of internal innovation I to equate costs with marginal benefits. The cost of generating new arrivals with intensity I is equal to

$$Y(t)\, c_I\left(I; \hat{q}\right) = \frac{Y(t)}{\varphi_I}\, \Phi\left(\hat{q}\right)\, I^{\zeta}. \tag{27}$$

Following Peters (2020), the function $\Phi\left(\hat{q}\right)$ is chosen so that the equilibrium rate of internal innovation $\bar{I}$ is constant across firms, ensuring tractability by making the innovation problem separable across product lines.

External innovation. The second form of innovation—*external* or *business-stealing* innovation—allows a firm to displace a competitor and take over its product line. This is the source of creative destruction in the model: when a product line is displaced by a competitor, incumbent workers lose their firm-specific match and must draw a new one, generating the concentrated downside risk that drives asymmetric passthrough. External innovation is undirected: the firm draws target varieties at random and improves their productivity by a factor λ. For each product line in which the firm is the leading producer, it can generate business-stealing improvements according to

$$N_j^x(v,t) \sim \text{Poisson}\big(x_j(v,t^-)\, \mathrm{d}\Gamma_j(t)\big)\,, \tag{28}$$

where the firm chooses the intensity $x_j(v,t)$ at cost

$$Y(t)\, c_x(x_j) = \frac{Y(t)}{\varphi_x}\, x_j^{\zeta}. \tag{29}$$

per product line the firm currently operates. These assumptions ensure that the aggregate rate of creative destruction $\bar{x}$ is constant across firms and time. Following Peters (2020), we assume that a small fraction of business-stealing innovations reset the target line's productivity, which keeps the distribution of relative productivities stationary. The mechanism is also economically sensible: sometimes firms create new products that completely displace an existing one, as opposed to merely producing the same product as a rival at lower marginal cost.

Innovation by New Entrants

A competitive fringe of potential entrants provides a second channel of creative destruction alongside incumbents' external innovation. Each potential entrant pays a flow cost $Y(t)/\varphi_e$ to attempt a business-stealing innovation; free entry ensures that the expected gains from entry equal this cost, pinning down the equilibrium entry rate $\bar{e}$. Each successful entrant arrives with a single product line.

Worker Productivity

Each worker is infinitely lived and begins the model with K potential jobs, with initial match-specific productivity drawn from the Fréchet distribution as in equation (18). Between innovation events, each worker's match quality evolves according to an Ornstein–Uhlenbeck process,

$$\mathrm{d}\, z^*(i,k,t) = -\rho\, z^*(i,k,t)\,\mathrm{d}t + \sqrt{2\rho}\,\mathrm{d}W_{i,k,t}, \tag{30}$$

which gradually pulls match quality toward the mean, capturing depreciation of firm-specific human capital and generating realistic voluntary turnover even absent creative destruction. When an external innovation displaces job k—that is, when the associated product line is lost to creative destruction—the worker receives a new potential job with a fresh productivity draw,

$$z^*(i,k,t) \sim N(0,1), \tag{31}$$

and new industry and variety indices drawn from the product lines that changed their active producer during the innovation burst. Because the worker had previously selected the best-paying job among her K options, this reset typically reduces her best available match quality—the key mechanism generating concentrated downside earnings risk.

3.3 Equilibrium: Dynamics

We begin with a discussion of aggregate dynamics and then turn to the dynamics of firms and product lines; the implications for worker earnings are taken up in Section 3.5.

Aggregate Output

The model features a balanced growth path (BGP) in which the distribution of *relative* productivities $\hat{q}$ is constant across time and across industries. This property follows from the absence of aggregate shocks and the Cobb–Douglas aggregator across industries in equation (17).

In particular, productivity in industry j evolves according to

$$\mathrm{d}\log \bar{Q}_j(t) = \tilde{g}\,\mathrm{d}\Gamma_j(t) - \delta\,\mathrm{d}t, \tag{32}$$

with growth rate

$$\tilde{g} \equiv (\bar{I} + \bar{x} + \bar{e})(\lambda - 1), \tag{33}$$

which reflects the equilibrium rates of internal innovation $\bar{I}$, external innovation by incumbents $\bar{x}$, and external innovation by entrants $\bar{e}$. Our assumptions on the cost of innovation imply that these equilibrium rates are constant across firms and over time. Aggregate output grows deterministically at approximately $\tilde{g}-\delta$ (see equation 76 for a derivation and exact expression).

Firms and Product Lines

The value of a firm is the sum of the values of all product lines for which it is currently the leading producer:

$$V_{f,t} = Y(t) \sum_{n=1}^{N_{f,t}} V\big(\hat{q}_{j(f)}(n, f, t)\big)\,. \tag{34}$$

Here, $V(\hat{q})$ denotes the value of a single product line with relative productivity $\hat{q}$, expressed relative to aggregate output. Because our innovation cost functions make internal and external investment independent across varieties, the firm's dynamic problem is separable across product lines.

In choosing innovation investment, each firm takes aggregate growth, the distribution of relative productivities, and the rates of creative destruction and entry as given. These objects are determined jointly in equilibrium by firms' optimal innovation choices and the free-entry condition.

The value of a single product line satisfies the Hamilton–Jacobi–Bellman equation

$$\begin{aligned}(r - g)\, V(\hat{q}) \;= \max_{I,x} \Bigg\{ & \pi\big(\hat{q}\big) - c_I(I; \hat{q}) - c_x(x) + x\, \mathcal{G}_x\, V \\ & + \frac{1}{\mu}\Bigg[e^{-(I+\bar{x}+\bar{e})\mu} \sum_{k=0}^{\infty} \frac{(I\,\mu)^k}{k!}\; V\Big(\hat{q} + k \log \lambda - \tilde{g}\,\mu\Big) - V(\hat{q}) \Bigg] \Bigg\},\end{aligned} \tag{35}$$

The value of a product line incorporates four forces: current flow profits, the option value of improving the line through internal innovation, the possibility of stealing a competitor's line through external innovation, and the risk of losing the line to creative destruction. The required return on the left-hand side is the interest rate minus the aggregate growth rate, because values are expressed relative to output. A detailed economic interpretation of each term and the full derivation of the first-order conditions are provided in Appendix A.2.

Firms invest in each type of innovation up to the point where marginal cost equals the expected gain in product-line value, so the equilibrium rates satisfy

$$c_I'(\bar{I}; \hat{q}) = \mathcal{G}_I V \qquad \text{and} \quad c_x'(\bar{x}) = \mathcal{G}_x\, V, \tag{36}$$

where the operators $\mathcal{G}_I$ and $\mathcal{G}_x$ are defined in equations (89) and (83) in Appendix A.2.

Similarly, the equilibrium rate of new firm entry $\bar{e}$ is pinned down by the free entry condition, which states that entrants cannot make positive profits,

$$\mathcal{G}_e\, V - \frac{1}{\varphi_e} \leq 0, \tag{37}$$

where $\mathcal{G}_e V$ captures the incremental value of a successful external innovation by an entrant—equation (85) in Appendix A.2. Equation (37) holds with equality if entry is positive $\bar{e} > 0$.

3.4 Calibration

We first show how to map the key explanatory variables in Section 2—the KPSS value of innovation—into model counterparts, and then describe our parameter choices.

Mapping to Observable Measures of Innovation

To run the same regressions in the model as in the data, we construct model counterparts of the KPSS value of innovation. For firm f, this is the sum of market-value gains across its product lines following an innovation event—which increases through both internal innovation (raising the value of existing lines) and external innovation (adding new lines). At the industry level, the total KPSS value is proportional to $d\Gamma_{j,t}$, the realization of the industry innovation process. This proportionality means that our empirical regressions of worker earnings on competitor innovation reflect the effect of the industry-level subordinator process described above.

Parameter Choice

We calibrate the model in two steps: we first choose a set of parameters based on a priori information, and then calibrate the remaining parameters to match key features of the data.
Parameters calibrated a priori. We first calibrate a subset of parameters to a priori information. The steepness of the cost of internal innovation ζ is set to 2, following Acemoglu et al. (2018) and Peters (2020). We calibrate the degree of innovation uncertainty $\mu = 1/2$ so that, on average, there are two $d\Gamma$ shocks per year, and when they occur innovation progresses as if 6 months of the discrete time process has occurred. We set the number K of job offers that a worker has at any time to ten.
Parameters calibrated to firm and worker data. We calibrate the remaining 16 parameters to approximately match 37 moments, listed with their sources in Table SA.3. *Aggregate targets*: a growth rate of output of 2% along the balanced growth path and an aggregate labor share of 0.6. *Worker-level targets*: average income risk (the standard deviation of 5-year earnings growth), worker mobility (the probability a worker leaves the firm over the next three years, both on average and by income group), and inequality (each income group's share of aggregate earnings). *Regression moments*: the 5-year responses of firm profits, pay per worker, and employment to own-firm and competitor innovation (6 moments, Figure 1), together with the corresponding responses of individual worker earnings, both pooled (Figure 2) and by worker earnings level (Figure 3); we work with these latter 12 moments

in terms of passthrough from firm profits to worker earnings. *Labor market structure and innovation composition*: labor supply elasticities by productivity quartile from Seegmiller (2024), and the dollar share of total innovation stemming from x, set to 0.35 to match the average dollar share of novel patents under the text-similarity measure of Section 2.3. Last, we calibrate the function $\Phi(\hat{q})$ capturing the cost of internal innovation so that equilibrium internal innovation $\bar{I}$ is independent of firm productivity $\hat{q}$; Appendix Figure SA.6 plots the resulting cost function together with the ergodic distribution of $\hat{q}$.

Parameters. Appendix Table SA.4 reports our calibrated parameters. Two features are most noteworthy. First, creative destruction is relatively rare: it occurs at rate $\bar{x} + \bar{e} = 0.084$, implying an average product-line lifespan of about 12 years, with just over $\frac{3}{4}$ of this activity coming from incumbents' external innovation rather than from entry. Second, managers exhibit far greater skill dispersion than laborers ($\theta_m = 2.13$ vs. $\theta_l = 6.81$), which helps generate heterogeneous passthrough across the income distribution.

Model Fit

Appendix Table SA.3 summarizes the model's fit to the data. The model closely reproduces the firm-level responses of profits, employment, and the wage bill to competitor innovation, though it overstates the corresponding responses to own-firm innovation by roughly a factor of two (Panel A). It reproduces the heterogeneous passthrough of innovation to worker earnings across income groups, including the steeper gradient and asymmetry for high earners (Panel B; Appendix Figure SA.7 shows that the model also matches earnings responses at other horizons). It broadly matches the empirical earnings shares by income group (Panel C) and the decline in job switching with income, although it overstates both the level of switching and the steepness of that gradient (Panel D), and qualitatively matches the relationship between firm productivity and labor supply elasticity from Seegmiller (2024) (Panel E). The model somewhat understates the five-year standard deviation of earnings growth (0.28 vs. 0.57 in the data), while the labor share is quite close to the empirical target (Panel F).

3.5 Model Mechanisms

Two mechanisms jointly generate asymmetric and concentrated passthrough consistent with the empirical patterns established in Section 2.2.

Sources of Earnings Risk

Incumbent workers face two sources of income risk. The first is the passthrough of incremental innovation, the focus of Section 1: equation (49), the generalization of (3), implies that

earnings respond to a firm's relative productivity $\hat{q}_j(v,t)$ through its labor-demand schedule, so internal and competitor innovation move wages—and the probability of job switching—in opposite directions but by broadly symmetric amounts. The asymmetry documented in Section 2 comes from the second source, creative destruction. Because a worker's human capital z_{ijt} is specific to her product line and she has already selected the best-paying of her K options, losing that line forces a draw that pays less; by the same token, she does not share in gains from innovations that expand the firm along *other* lines, which accrue partly to new hires. Panel B of Appendix Table SA.3 reflects the resulting asymmetry: among the top 5% of earners, passthrough is around 43% for positive shocks but closer to 100% for negative ones.

Concentration of Earnings Risk

These two sources of risk both fall disproportionately on higher-paid workers for two key reasons. First, higher-income workers are more likely to be managers, whose fatter-tailed distribution of firm-specific human capital implies less elastic labor supply and hence greater wage sensitivity to changes in $\hat{q}$ than laborers face (Appendix Figure SA.8 shows the distribution of skill and earnings for both groups).

The second reason is that top earners, either managers or laborers, typically have high idiosyncratic productivity and are employed in high-$\hat{q}$ product lines. These higher-paid workers are particularly vulnerable to job destruction because their current job is typically far better than their next-best option; passthrough rises sharply in the upper tail of the gap distribution (Appendix Figures SA.9 and SA.10).

4 Model Implications

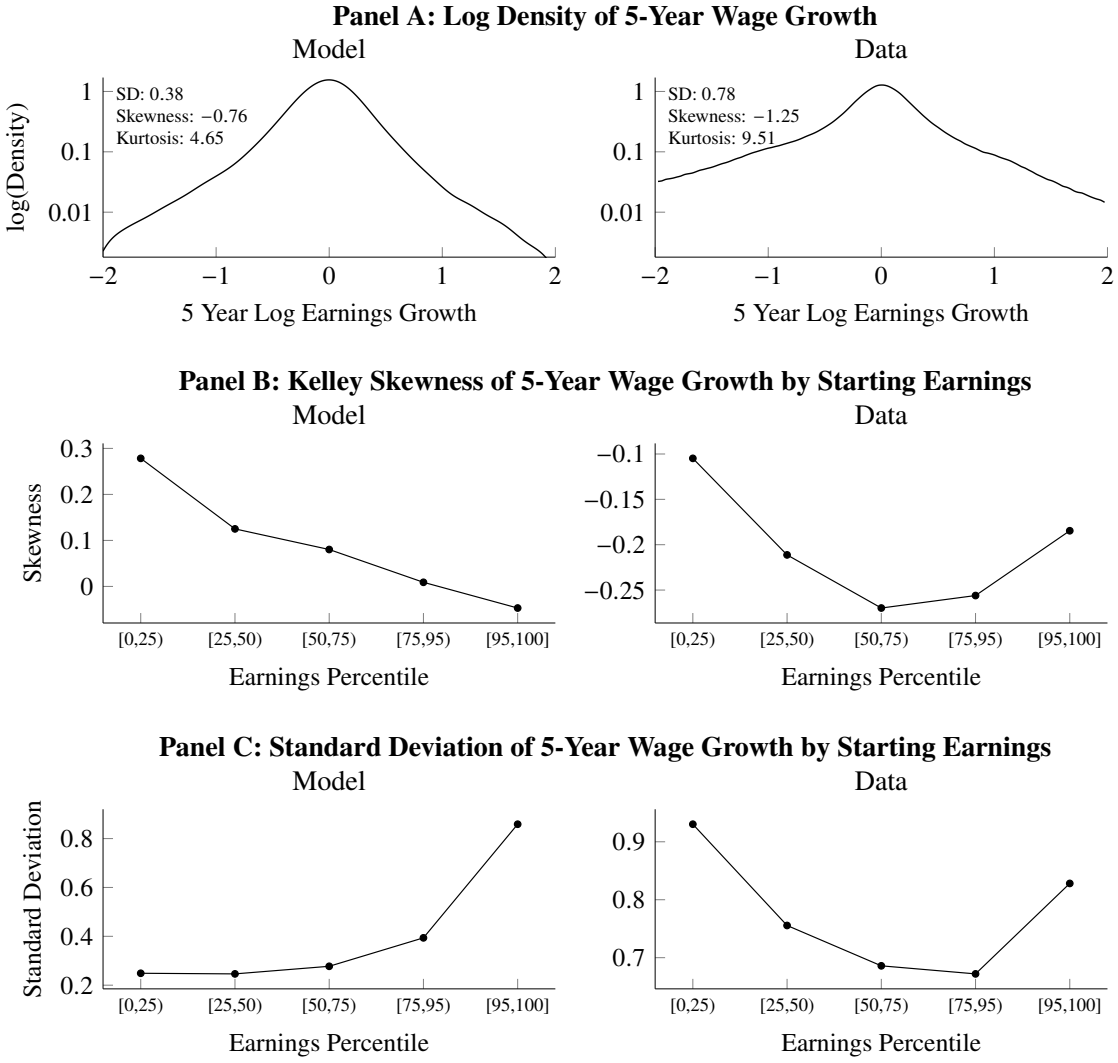
We now turn to the quantitative implications of the model: its fit to moments that are not targeted in the calibration, the consequences of the secular decline in radical innovation, and the welfare effects of innovation policy when labor-income risk is uninsurable.

4.1 Non-targeted Moments

Worker Mobility and Earnings Risk. Using U.S. administrative data, Guvenen et al. (2021) document that earnings growth is highly skewed—high-income workers experience large, concentrated downside shocks—and that both low- and high-income workers exhibit greater earnings-growth volatility. Figure 6 shows that the model reproduces most of these patterns: excess kurtosis (Panel A) through occasional large displacements, more negative skewness for high-income workers (Panel B) because creative destruction destroys their unusually

productive matches, and an increasing earnings-growth variance for top earners (Panel C). The model does not reproduce the U-shape at the bottom of the distribution. The model also makes empirically plausible predictions for the response of involuntary exit to innovation shocks across income groups (Figure 7), despite none of these moments being targeted.

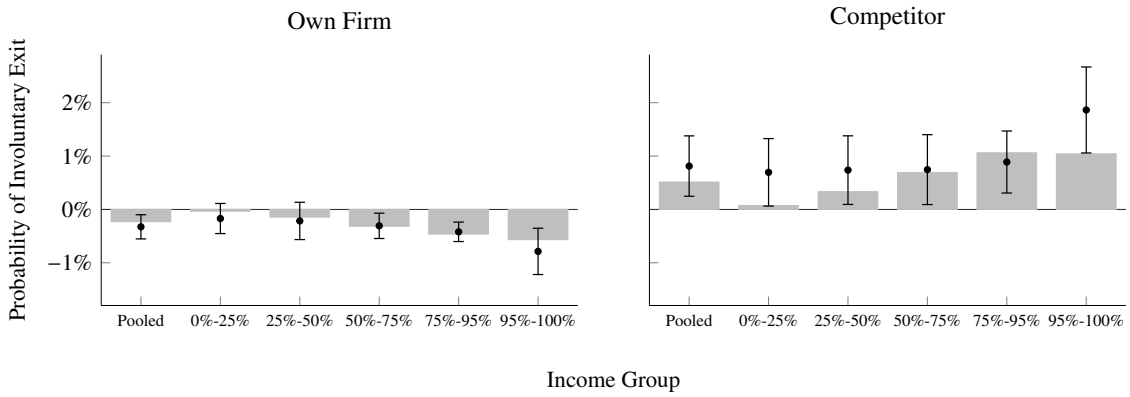
Figure 6: Comparing the Income Process to Guvenen et al. (2021)



Note: Panel A presents the log density of 5 year log earnings growth compared with the same numbers as reported by Guvenen et al. (2021), who provide summary statistics by earnings percentile. Panel B shows the Kelly Skewness, computed as the (P90 - P50) - (P50 - P10), and Panel C shows the standard deviation.

Labor Share and Productivity. Seegmiller (2024) shows that firms at the top of the productivity distribution have lower labor shares, reflecting their reliance on workers with relatively inelastic labor supply. The model reproduces this pattern, which arises here from

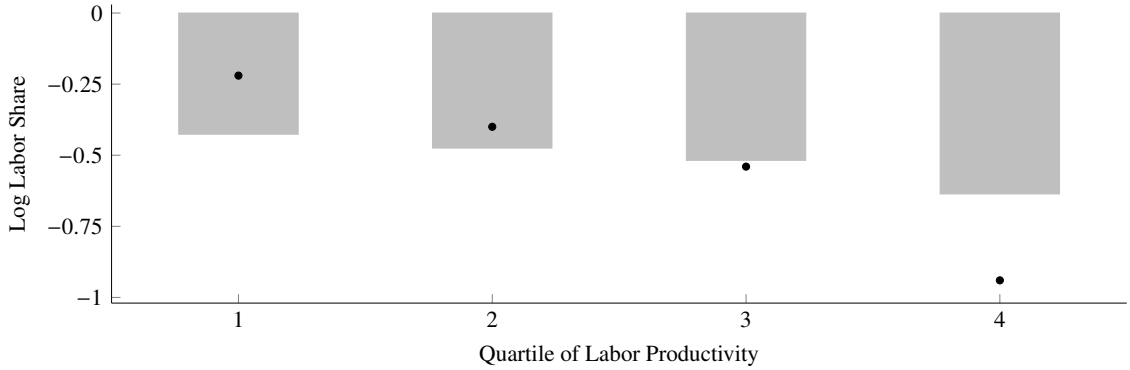
Figure 7: Probability of Involuntary Exit Following Shock



Note: The figure shows the probability of an involuntary exit — exiting the firm and experiencing wage growth in the bottom 20 percent of the sample — in the model and the data. The x axis sorts responses into income-bin interactions and the y axis reports linear probability model estimates in the model (columns) and the data (points with error bars). Model-implied coefficients are smaller in magnitude than their empirical counterparts but share their sign and usually fall inside the 95 percent confidence interval.

the larger markdowns set by more productive firms; Figure 8 shows the fit across productivity quartiles.

Figure 8: Log Labor Share by Productivity Quartile: Model vs. Seegmiller (2024)



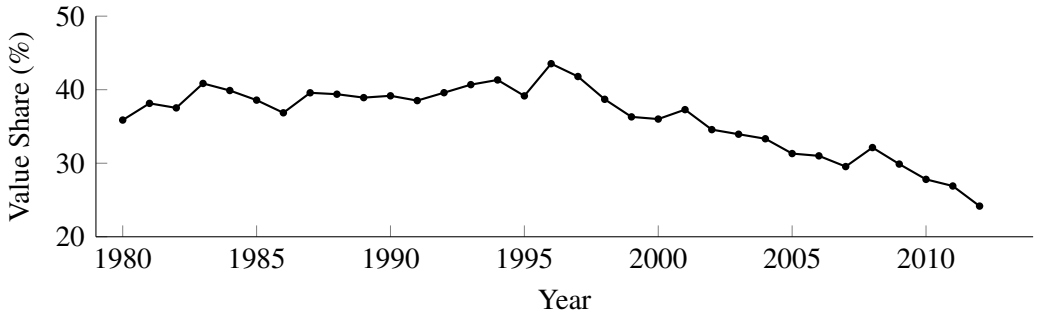
Note: Figure shows the log labor share over the same quartiles. Point estimates in the data are taken from Table 3 of Seegmiller (2024).

4.2 Declining Business Dynamism and Income Risk

A well-documented feature of the U.S. economy over recent decades is the decline in business dynamism, particularly in firm entry rates (Haltiwanger, 2012; Akcigit and Ates, 2019). Figure 9 reveals a complementary trend among incumbent firms: the value share of innovation accounted for by radical patents—those that are technologically distinct from the firm's existing base, as defined in equation (14)—has fallen from roughly 40 percent in the 1990s to 20 percent in the 2010s. While the existing literature focuses on the decline in entry, our measure captures a decline in business stealing by incumbents, which corresponds

to external innovation x in our model. Together, these trends point to a broad reduction in creative destruction operating through multiple margins.

Figure 9: Value Share of Radical Innovation Over Time



Note: Time series of the relative value (KPSS) of patents that correspond to radical innovation compared to the total value of patents issued in each year. The classification of patents as radical innovation is based on the textual similarity to prior patents issued to the same each firm—see equation (14) in the main text.

We next explore what a reduction in the rate of creative destruction implies for firms and workers in our model. We recalibrate the model's share of external innovation—the rate of creative destruction x compared to the total—to match the average radical-innovation share in the 1990s (40 percent) and the 2010s (20 percent). Motivated by the evidence in the literature on declining entry, we also reduce the rate of entry by the same proportion. We hold all other parameters at their baseline values.

In the model, a decrease in the rate of creative destruction lowers the risk of firm displacement and hence lowers earnings risk faced by workers. In addition, however, it increases the dispersion in productivity across product lines—since it effectively lowers the rate at which the relative productivity of product lines resets to one.

Labor Share. The observed decrease in the rate of creative destruction implies a decline in the aggregate labor share from 58.8% to 55.3% in our model. As in the data, this decline is driven by large and highly-productive firms: as the risk of creative destruction falls, firms are more likely to grow larger and more productive. The log labor share for firms in the top productivity quartile declines by approximately 0.093 log points, compared to a 0.08 decline in the data (Seegmiller, 2024), while that of firms in the bottom productivity quartile declines by just 0.01, compared to 0.02 in the data. This result concords with evidence that the decline in the labor share is driven by the most productive, "superstar" firms (Autor, Dorn, Katz, Patterson, and Van Reenen, 2020).

Dispersion in Firm Growth Rates. A decline in the risk of creative destruction also compresses the dispersion of firm growth rates: our calibrated model implies that the dispersion in 5-

year arc employment growth (p90-p10 spread) declined by approximately 0.4 between the 1990s and 2010s, compared to approximately 0.25 in the data (Salgado et al., 2023), and that this decrease is primarily driven by the right tail, a pattern that is consistent with the data (Decker, Haltiwanger, Jarmin, and Miranda, 2016).

Concentration and Between-firm Inequality. The increased dispersion in firm productivity leads to increased concentration: in our calibrated model, the revenue share of the top 0.1 percent of firms increases by approximately 15 log points between 1990 and 2010, consistent with the approximately 14 log points that Kwon, Ma, and Zimmermann (2024) report over the same period. Greater productivity dispersion also generates more heterogeneity in labor demand, and hence in posted wages, across firms, which raises income inequality: our calibration implies that the between-firm component accounts for more than half (55%) of the increase in the variance of worker log earnings levels, compared to 67% in the data (Barth, Bryson, Davis, and Freeman, 2016; Song, Price, Guvenen, Bloom, and Von Wachter, 2019).

Worker Earnings Risk. Moving from the 1990s to the 2010s parameterization reduces the 90 / 10 spread of five-year earnings growth from 0.54 to 0.47—a decline of roughly 7 percentage points. This prediction aligns with the "Great Micro Moderation" documented by Sabelhaus and Song (2010); Salgado et al. (2023), who show that the cross-sectional dispersion of individual earnings growth has declined in an almost secular fashion since 1980, a pattern that is pervasive across demographic groups. In particular, they find a decline in the 90 / 10 spread of men's five-year earnings growth from around 1.52 to around 1.42 by 2010—a comparable decline in income risk, though the level is quite different between the data and the model. Our model provides a structural mechanism for this fact: the secular decline in creative destruction—operating through reduced radical innovation by incumbents—lowers the rate at which workers are displaced from productive matches, thereby reducing the non-diversifiable component of earnings risk.

In brief, a decline in the rate of creative destruction lowers the degree of uncertainty facing workers and firms (it reduces dispersion in growth rates) yet at the same time it crystallizes differences across firms and workers by increasing inequality (it increases dispersion in levels).

4.3 Welfare

A central implication of standard endogenous growth models (e.g., Romer, 1990) is that innovation generates knowledge spillovers and therefore should be subsidized to raise welfare. That conclusion does not necessarily apply in our setting. Markets are incomplete, and higher innovation increases workers' exposure to labor-income risk. With risk-averse workers and

uninsurable earnings fluctuations, the private willingness to pay for additional innovation may be significantly lower than the complete-markets benchmark.

To study this question, we introduce additional structure on worker preferences. Workers have constant-relative-risk-aversion (CRRA) utility, so equation (20) becomes

$$u(C_{it}) = \frac{C_{it}^{1-\gamma}}{1-\gamma}, \tag{38}$$

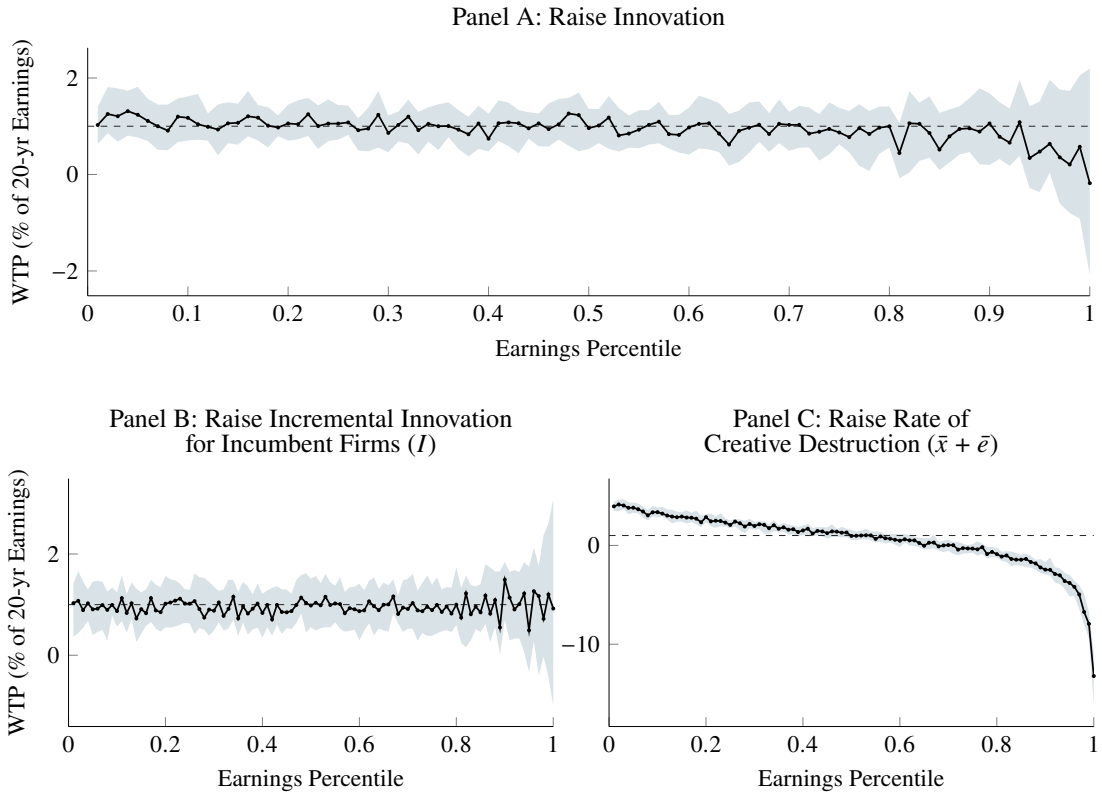
where the consumption index $C_{i,t}$ combines goods consumption and the disutility of effort. Workers are hand-to-mouth and consume their wage income. We evaluate each worker's willingness to pay—measured as a fraction a_i of consumption over the next $T = 20$ years—for a one-time increase in the innovation rate that raises aggregate growth by $\Delta = 1\%$ for a single year. We set $\beta = 0.04$ and $\gamma = 1$ (log utility); we choose a relatively low risk aversion because of our hand-to-mouth assumption. See Appendix A.2 for details.

We consider three experiments, each calibrated to deliver this same one-year increase in aggregate growth: (i) an increase in all forms of innovation—internal I, external x, and entry e; (ii) an increase in internal innovation only; and (iii) an increase in external innovation and entry only. Each can be interpreted as an economy-wide reduction in the cost of innovation, arising either from a subsidy or from technological progress (an investment-specific shock). For simplicity, we shift p_r such that the distribution of wages and firm productivity remains unchanged across the three experiments; costs are adjusted so that x and e change proportionally when external innovation changes, and so that I, x, and e remain proportional when we change the aggregate rate.

Figure 10 plots workers' willingness to pay for each policy as a function of their initial earnings percentile; the dashed line marks the complete-markets benchmark. Because workers are heterogeneous in productivity and current job offers, there is significant dispersion in willingness to pay. We therefore plot the median together with the 25th and 75th percentiles for each earnings bin.

Panel A shows that workers' willingness to pay falls below the complete-markets benchmark, especially for those at the top of the earnings distribution. Workers from the bottom through the 80th percentile are willing to pay 1% percent of their consumption for a one-year, 1% increase in the growth rate. For workers above the 80th percentile, willingness to pay declines to 0.77 percent. Averaged across workers, the mean willingness to pay is 0.96 percent of earnings over the next 20 years. Panels B and C reveal that this pattern is driven almost entirely by workers' aversion to creative destruction. Panel B shows that when subsidies apply only to internal innovation, workers' willingness to pay tracks the complete-

Figure 10: Willingness to Pay for a Temporary Increase in Innovation



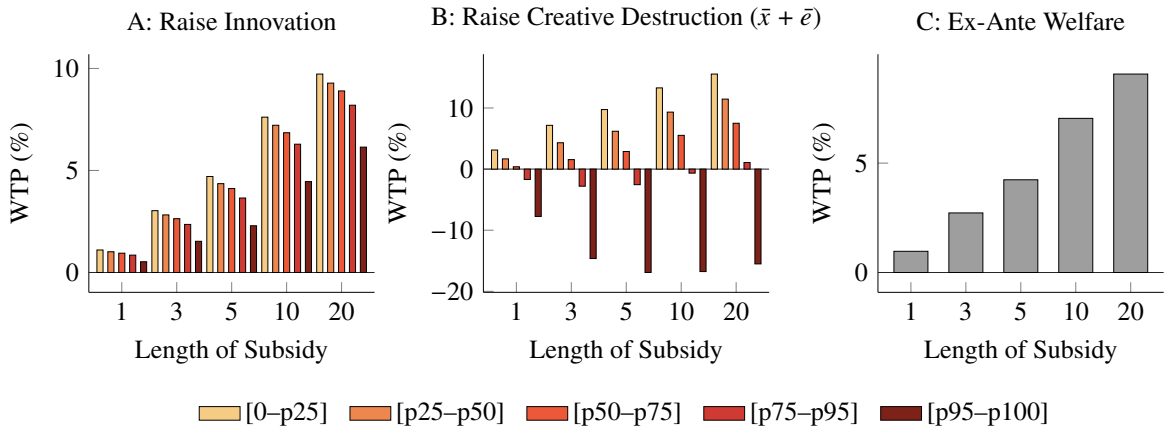
Note: Willingness-to-pay (WTP) by starting earnings percentile for temporary innovation-policy shocks for employed workers. Panel A raises both incremental and creative-destruction innovation rates; Panel B raises incremental innovation only; Panel C raises creative destruction only. Lines show medians and shaded bands show interquartile ranges; the dashed line is the complete-markets benchmark. Specification details follow Section 4.3 and Appendix A.1.

markets benchmark quite closely—the average is approximately 0.97 percent of earnings. Any dispersion around that benchmark simply reflects differences in match quality with the employer.

Panel C paints a very different picture when we raise the rate of creative destruction, $\bar{x} + \bar{e}$. Workers above the 68th percentile of the earnings distribution exhibit *negative* willingness to pay for additional creative destruction or business dynamism. Those at the 80th percentile would pay roughly 0.88 percent of their consumption *to avoid* subsidizing creative destruction and higher entry, and an average worker in the top decile would pay 5.3 percent. For these workers, creative destruction poses a first-order earnings risk by displacing their firm-specific human capital. By contrast, workers below the 53rd percentile gain from increased business dynamism, since displacement raises the odds of drawing a better match than their current one. Averaged across all workers, the mean willingness to pay is 0.55 percent of earnings over 20 years.

Figure 11 extends this analysis by varying the duration of the innovation subsidy from one to twenty years. Panel A shows that when all forms of innovation are subsidized, willingness to pay increases with the length of the subsidy for workers across the earnings distribution, but the dispersion across income groups widens substantially. Willingness to pay is uniformly decreasing in income given the duration of the subsidy. Panel B performs an analogous calculation but for subsidies which only raise the rate of creative destruction (analogous to Figure 10 Panel C). Here, willingness to pay grows monotonically with the duration of the subsidy for workers in the bottom half of the income distribution, and workers above the median would uniformly prefer to avoid such a policy due to the elevated exposure to displacement risk across all subsidy lengths. Panel C shows the welfare for either subsidy ex ante—the willingness to pay for the subsidy before a worker knows their offer set. In this setting, workers uniformly want to pay for the subsidy since under the veil of ignorance they care about aggregate growth of the economy, which benefits from innovation.

Figure 11: Willingness to Pay for Subsidies of Different Lengths



Note: Panel A raises both incremental and creative-destruction innovation rates proportionally, as in the prior experiment's panel A; Panel B raises creative destruction only, as in the prior experiment's panel C. Both show the average WTP conditional on knowing your own human capital draws, both pooled and within income percentiles. Panel C shows the willingness to pay for the innovation subsidy under the veil of ignorance, where a worker does not yet know their starting offer set. This is the same across both counterfactuals ex ante. The x axis shows how welfare relates to the length of the subsidy, ranging from 1 year out of a 20 year income path to a full 20 year subsidy. Panels A and B who are employed at the time of the subsidy, while the ex-ante welfare number also includes unemployed workers.

In sum, uninsurable labor-income risk sharply reduces the welfare gains from innovation relative to the complete-markets benchmark. The key force is creative destruction: because worker human capital is non-diversifiable, displacement from a high-quality match imposes losses that cannot be insured away. Not all types of innovation are equivalent from a worker welfare perspective—policies that expand the technological frontier without displacing

workers are broadly beneficial, while those that rely on business stealing generate concentrated losses for the workers least able to bear them.

At the same time, these welfare losses are fundamentally *ex post*—conditional on a worker's current match quality. Behind the veil of ignorance, before workers learn their position in the productivity distribution, willingness to pay for additional innovation coincides with the complete-markets benchmark: higher creative destruction raises the level and compresses the spread of lifetime earnings, so all agents benefit equally in expectation. These ex-ante valuations are reported in Panel C of Figure 11.

5 Conclusion

We document that creative destruction in the product market passes through to worker earnings in an asymmetric and concentrated fashion, and develop a structural model that rationalizes these patterns. The key insight is that worker human capital is tied to individual product lines: incumbent workers have disproportionately large downside exposure to creative destruction but capture only a fraction of its benefits. This asymmetry reshapes the welfare evaluation of innovation: because displacement from a high-quality match cannot be insured away, most workers—especially high earners with the most to lose—would forgo faster creative destruction even though it raises aggregate growth. At the same time, from an ex-ante perspective—before workers learn their position in the match-quality distribution—innovation is unambiguously beneficial, since creative destruction compresses the dispersion of lifetime earnings.

We close by drawing out several broader implications of our analysis.

The design of innovation policy. In contrast to the canonical view that innovation should be subsidized to internalize knowledge spillovers (Romer, 1990), our analysis implies that the *composition* of innovation matters as much as its level. This distinction bears directly on a prominent strand of the growth literature that emphasizes radical innovation by new entrants—rather than incremental improvement by incumbents—as the primary engine of long-run growth, and stresses the benefits of the reallocation and creative destruction that follow from new entry (Akcigit and Kerr, 2018; Acemoglu et al., 2018). Our work shows that subsidies that raise internal innovation—improvements to existing products that do not displace incumbents—deliver welfare gains close to the complete-markets benchmark, whereas subsidies that accelerate entry and business stealing concentrate losses on the workers least able to insure against them. The very margin that is most valuable for growth

is thus the one that carries the steepest (ex-post) welfare costs—a tradeoff that is absent from growth models with frictionless labor markets.

Declining business dynamism. Our framework implies that the secular decline in creative destruction has rich implications for both firms and workers. Lower displacement reduces the non-diversifiable earnings risk that workers cannot insure away; at the same time, it allows productivity differences across firms and workers to persist and widen, raising the labor share of the most productive firms, firm concentration, and between-firm earnings inequality. The model thus supplies a common mechanism for several trends often studied in isolation—the falling aggregate labor share (Autor et al., 2020), rising concentration, and the compression of earnings-growth dispersion (Salgado et al., 2023)—and identifies a partial silver lining of declining dynamism: less risk at the cost of more inequality.

Asset prices and precautionary saving. Our work shows that creative destruction concentrates earnings risk among high earners, who hold a disproportionate share of aggregate wealth. Firms with high growth opportunities stand to benefit from creative destruction, and their value rises precisely in high-marginal-utility states. This pattern offers an additional microfoundation for the apparent overvaluation of high-growth companies, as workers are willing to accept lower returns on these firms in exchange for partial insurance against their loss in human capital (Kogan, Papanikolaou, and Stoffman, 2020). In addition, the same uninsurable risk likely strengthens the precautionary saving motive of high earners, connecting our mechanism to recent work on rising risk among the rich (Braxton, Herkenhoff, Huang, Nattinger, Rothbaum, and Schmidt, 2025).

The political economy of technological change. The workers most exposed to creative destruction—high earners with the most valuable matches—are also those who command the largest share of current output. Their willingness to pay to avoid faster creative destruction is significant. Our results thus connect to a literature on the political economy of resistance to welfare-improving change. Fernandez and Rodrik (1991) show that uncertainty over who will gain generates a status-quo bias against reforms that a majority would support ex post, while Acemoglu and Robinson (2000) emphasize that whether such resistance succeeds turns on the political power of those who stand to lose. In our setting these two ingredients coincide: the concentrated income losses from faster creative destruction fall on high earners, who also command disproportionate political influence—giving them both the motive and the means to resist technological change that raises aggregate welfare.

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Supplemental Appendix to
"Winners and Losers: Competition, Creative Destruction, and Labor Income Risk"

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A Supplementary Material

A.1 Data Appendix

Data Construction

Our data on worker earnings combine individual records from the Current Population Survey (CPS) with employer-employee matched administrative records from the Social Security Administration's Detailed Earnings Record (DER). The CPS/DER panel covers 1981–2019 and allows us to track individuals over time and across firms. We match it to Compustat firms using Employer Identification Numbers (EIN), yielding approximately 9,700 firms and 61.6 million worker-year observations. Because we analyze five-year growth rates throughout, the last year in which we measure a firm innovation shock is 2014, with the associated growth rate ending in 2019.

We restrict the sample to prime-age workers, ages 25 to 55, and exclude the self-employed and individuals earning less than the equivalent of 20 hours per week for 13 weeks at the federal minimum wage, following Guvenen et al. (2014). This leaves 12,500 firms and 132,000 firm-years in the firm panel, and 9,700 firms, 780,000 people, and 5.6 million person-years in the worker panel. Requiring sufficient income history and firm controls — almost entirely the lagged R&D requirement — reduces these to 63,000 firm-years and 2,849,000 person-years. Table SA.5 reports summary statistics.

The firm wage bill and employment come from the Longitudinal Business Database (LBD), an establishment-level dataset of non-farm firms whose wage and employment records are sourced from tax filings (Jarmin and Miranda, 2002; Chow, Fort, Goetz, Goldschlag, Lawrence, Perlman, Stinson, and White, 2021). We link these to Compustat, which we use for firm control variables and profits.

To proxy for pay to incumbent employees, we take individual records from the linked DER-CPS sample and define the focal firm as the employer at $t = 0$. We then track wages paid to that fixed set of employees by the focal firm at $t \in \{-5, \ldots, 10\}$ and aggregate by firm,

so the variable omits compensation of workers who subsequently join or leave. Coverage is intermittent between 1990 and 2000 and becomes a continuous panel thereafter; when a year is missing we compute average earnings from the income that is available.

Worker level regressions We construct flexible controls for worker age and lagged earnings by linearly interpolating between third-degree Chebyshev polynomials in workers' lagged income quantiles within industry-age bins at 10-year age intervals. To absorb mean reversion following large transitory shocks, we add third-degree Chebyshev polynomials in lagged income growth percentiles, with coefficients that differ across five bins formed on a worker's income rank within their firm's industry. So that our point estimates are comparable to the firm-year analysis in Section 2.2, we weight observations by the inverse of the number of workers in each firm-year. Standard errors are clustered at the industry-year level; for quantile regressions we block-resample at the SIC3 level to allow for persistence within industries.

Income percentiles are measured from DER wages relative to the worker's SIC3 industry in the year observed, and income bins are indicators for the [0%, 25%), [25%, 50%), [50%, 75%), [75%, 95%), and [95%, 100%] ranges.

The vector $\mathbf{Z}$ includes one lagged value of the dependent variable together with the logs of book assets and market capitalization, since firm size may induce a mechanical correlation between the dependent variable and our innovation measure. We also control for the depreciated stock of prior patents — depreciating past innovations at 20 percent per year — and for R&D spending scaled by assets, because firms with valuable past innovations and high R&D may persistently innovate at a different rate. Finally, we control for firm idiosyncratic volatility σ_{ft}, which may mechanically affect our innovation measure and is likely correlated with growth opportunities and earnings risk, and for firm profitability, defined as sales minus cost of goods sold scaled by assets.

Firm and Industry Splits For the exercises in Table 1 the industry innovation rate is the value of the patents an industry puts out each year, inequality is the annual cross-sectional standard deviation of average firm wages in a given year, and average industry wages are based on the worker-level CPS / DER linked sample. An industry is high if it is above median and low otherwise.

Robustness Measures We report three alternative measurement choices in Table SA.1. First, we replace SIC3 industries with the text-based definition of Hoberg and Phillips (2016), which measures product market competition from the language similarity of 10-K filings, and re-run equations 8 and 11. Second, to address potential mismeasurement of firm size, we normalize by enterprise value — market capitalization plus book assets less common equity

— rather than market value in equation 6. Last, we construct a "residualized" measure of innovation that nets out predictability in the KPSS measure, estimating a Poisson regression on the firm-level controls together with industry and year fixed effects and using the difference between the realized and predicted KPSS value.

A.2 Model Appendix

We describe the setup and equilibrium conditions of the model in greater detail, reproducing key equations from the main text where needed. Throughout, we require $\theta_l \geq \theta_m$ and $\theta_m > 1$: the first makes type l workers more substitutable, the second delivers finite moments. We restrict $\alpha \in (0, 1)$, so that technology shocks cannot raise managerial productivity faster than that of laborers. The remaining restrictions are standard: $\delta > 0$, $B \in (0, 1)$ so that production requires both inputs, $\sigma > 0$, $\varphi > 0$, and I, x, and e all positive.

Static Equilibrium Conditions

Relative Prices. There is a single consumption good $Y(t)$ produced by a perfectly competitive final goods sector that is a nested CES aggregate of intermediate goods, $Y_j(v,t)$, indexed by variety v and industry j:

$$Y(t) = \exp\left(\int_0^1 \log\left(\int_0^1 Y_j(v,t)^{\frac{\sigma-1}{\sigma}} dv\right)^{\frac{\sigma}{\sigma-1}} dj\right), \tag{39}$$

where $\sigma > 1$ is the elasticity of substitution between product varieties within an industry. The production function for intermediate good in industry j and variety v is produced only by the most efficient producer according to

$$Y_j(v,t) = Q_j(v,t)\left((1-B)\left[Q_j(v,t)\,l_j(v,t)\right]^{\frac{\psi-1}{\psi}} + B\left[\bar{Q}_j(t)^{\alpha}\,Q_j(v,t)^{-\alpha}\,m_j(v,t)\right]^{\frac{\psi-1}{\psi}}\right)^{\frac{\psi}{\psi-1}}. \tag{40}$$

where $l_j(v,t)$ and $m_j(v,t)$ captures the number of efficiency units of laborers and managers, respectively, and ψ is the elasticity of substitution between skill types. We let

$$\bar{Q}_j(t) \equiv \int_0^1 Q_j(v,t)dv \quad \text{and} \quad \hat{Q}_j(v,t) \equiv \frac{Q_j(v,t)}{\bar{Q}_j(t)} \tag{41}$$

denote the average productivity across all varieties within industry j and the relative productivity in variety v, respectively.

Using the final goods producer's input demand from the outer nest of the production function (39), we get that the demand for industry j output $Y_j(t) = \left(\int_0^1 Y_j(v,t)^{\frac{\sigma-1}{\sigma}} dv\right)^{\frac{\sigma}{\sigma-1}}$, where $Y_j(v,t)$, satisfies

$$Y_j(t) = \frac{Y(t)}{P_j(t)}, \tag{42}$$

an expression which holds for all industries $j \in [0,1]$. Within industries, the demand for intermediate good $Y_j(v,t)$ is

$$Y_j(v,t) = Y_j(t)\left[\frac{p_j(v,t)}{P_j(t)}\right]^{-\sigma} = \frac{Y(t)}{P_j(t)}\left(\frac{P_j(v,t)}{P_j(t)}\right)^{-\sigma}, \tag{43}$$

an expression which holds for all varieties, v, and industries $j \in [0,1]$, where the price index for industry j output, $P_j(t)$, is given by:

$$P_j(t) = \left(\int_0^1 P_j(v,t)^{1-\sigma} dv\right)^{\frac{1}{1-\sigma}}. \tag{44}$$

Further notice that revenue only depends on relative prices:

$$p_j(v,t)Y_j(v,t) = Y(t)\left(\frac{P_j(v,t)}{P_j(t)}\right)^{1-\sigma} \equiv Y(t)\left[\hat{P}_j(v,t)\right]^{1-\sigma},$$

a property which will help immensely along the balanced growth path.

Rearranging (42) and equating it for two different industries yields the following condition

$$P_j(t)Y_j(t) \equiv P_j(t)\bar{Q}_j(t)\left(\int_0^1\left[\frac{Y_j(v,t)}{\bar{Q}_j(t)}\right]^{\frac{\sigma-1}{\sigma}} dv\right)^{\frac{\sigma-1}{\sigma}} = P_k(t)\bar{Q}_k(t)\left(\int_0^1\left[\frac{Y_k(v,t)}{\bar{Q}_k(t)}\right]^{\frac{\sigma-1}{\sigma}} dv\right)^{\frac{\sigma-1}{\sigma}}. \tag{45}$$

Along the balanced growth path we consider, the distribution of $\hat{p}_j(v,t)$ and $\hat{Q}_j(v,t)$ across varieties will be constant over time, hence the integral over varieties is the same on both sides of (45), which simplifies to

$$\frac{P_j(t)}{P_k(t)} = \frac{\bar{Q}_k(t)}{\bar{Q}_j(t)} \tag{46}$$

and aggregate output satisfies

$$Y(t) = \int_0^1 P_j(t)Y_j(t)dj = \int_0^1 \bar{Q}_j(t)\underbrace{\left(\int_0^1\left[\frac{Y_j(v,t)}{\bar{Q}_j(t)}\right]^{\frac{\sigma-1}{\sigma}} dv\right)^{\frac{\sigma-1}{\sigma}}}_{\equiv \bar{A}} dj = \bar{A}\bar{Q}(t). \tag{47}$$

Labor Markets and Equilibrium Wages. Workers of type $\tau \in \{l,m\}$ observe K job offers and for each job $k \in K$ have firm-specific match quality $z_{ik} \sim$ Fréchet(T_τ, θ_τ) Each of those jobs pays wages per efficiency unit $\{w_k\}_{k=1}^K$. Workers face an effort cost $c(e) = \frac{e^{1+\phi}}{1+\phi}$, where $\phi > 0$. This implies optimal effort $e_{ik}^\star = (w_k z_{ik})^{1/\phi}$. Letting $\beta \equiv 1 + \frac{1}{\phi}$, total earnings per unit of output are $y_{ik} = w_k e_{ik}^\star = (w_k z_{ik})^\beta = w_k^\beta z_{ik}^\beta$.

A power function of a Fréchet random variable is again Fréchet, and so is the maximum of a set of Fréchet draws, so the distribution of take-home earnings is available in closed form:

$$z_{ik}^\beta \sim \text{Fréchet}(T_\tau, \tilde{\theta}_\tau), \quad y_{ik} \sim \text{Fréchet}(S_k, \tilde{\theta}_\tau), \qquad S_k = T_\tau w_k^{\theta_\tau} \quad (\beta\tilde{\theta}_\tau = \theta_\tau). \tag{48}$$

So adding the effort choice only impacts the shape parameter but leaves the scale parameter unchanged.

Let $\Lambda_\tau = T_\tau \sum_{k=1}^{K} w_k^{\theta_\tau}$. Fréchet is max-stable, so $y_i^* = \max_{k \le K} y_{ik} \sim \text{Fréchet}(\Lambda_\tau, \tilde{\theta}_\tau)$.

However, all workers also face a common outside option which pays wages at rate R. All workers have a single efficiency unit in that option, giving a post-effort wage rate R^β. Their probability of participating in the market is given by $\pi_\tau = 1 - \exp(-\Lambda_\tau R^{-\theta_\tau})$, which follows from the CDF of the Fréchet distribution. This gives us a final probability that the worker chooses a given job: $P\{\text{choose k}\} = \pi_\tau \left(w_k^{\theta_\tau} / \sum_{\kappa=1}^{K} w_\kappa^{\theta_\tau} \right)$.

Aggregate labor supply of laborers and managers faced by a firm are deterministic functions of posted wages w (for laborers) and s (for managers):

$$l(w,t) = \omega K T_l^{[1+1/\varphi]/\theta_l} w^{\frac{1}{\varphi}} \times \mathbb{E}_{\mathcal{W}_{i,t}} \left[\left(\frac{w^{\theta_l}}{w^{\theta_l} + \mathcal{W}_i^{\theta_l}} \right)^{[1+1/\varphi]/\theta_l - 1} \gamma\left(1 - \tfrac{1+1/\varphi}{\theta_l}, -\left[T_l(w^{\theta_l} + \mathcal{W}_{i,t}^{\theta_l})\right] R^{-\theta_l}\right) \right] \tag{49}$$

$$m(s,t) = (1-\omega) K T_m^{[1+1/\varphi]/\theta_m} s^{\frac{1}{\varphi}} \times \mathbb{E}_{\mathcal{S}_{i,t}} \left[\left(\frac{s^{\theta_m}}{s^{\theta_m} + \mathcal{S}_{i,t}^{\theta_m}} \right)^{[1+1/\varphi]/\theta_m - 1} \gamma\left(1 - \tfrac{1+1/\varphi}{\theta_m}, -[T_m(s^{\theta_m} + \mathcal{S}_{i,t}^{\theta_m})] R^{-\theta_m}\right) \right] \tag{50}$$

where $\gamma(\cdot,\cdot)$ denotes the lower incomplete gamma function, which arises because the distribution of the maximum of truncated Fréchet draws appears in the derivation (see Nadarajah (2009) for details) and

$$\mathcal{W}_{i,t} \equiv \left\{ \sum_{\kappa=1}^{K-1} [w_{j(\kappa,i,t)}(v(\kappa,i,t),t)]^{\theta_l} \right\}^{\frac{1}{\theta_l}} \quad \text{and} \quad \mathcal{S}_{i,t} \equiv \left\{ \sum_{\kappa=1}^{K-1} [s_{j(\kappa,i,t)}(v(\kappa,i,t),t)]^{\theta_m} \right\}^{\frac{1}{\theta_m}}. \tag{51}$$

Along a balanced growth path these functions are time-invariant. Three channels link a firm's posted wage to the quantity of labor it attracts: (i) the *intensive margin* ($w^{1/\varphi}$), reflecting workers' effort choices; (ii) *reallocation across employers* (the term inside the expectation), capturing how a higher wage draws workers away from competing firms; and (iii) *non-employment* (the incomplete gamma function), governing the inflow from home production.

Firm Profit Maximization Problem. We assume a two stage pricing game in which firms first pay a small fixed cost $\varepsilon > 0$ to enter, taking the limit as $\varepsilon \downarrow 0$ as in Acemoglu, Gancia, and Zilibotti (2012); Acemoglu et al. (2018). In a second stage, entrants compete a la Bertrand by simultaneously posting prices. In equilibrium, the most efficient producer posts a price equal to the standard CES markup over marginal cost.

The firm profit maximization problem, expressed relative to aggregate output, is:

$$\Pi_j(v,t) = \max_{\hat{P}} \left[\hat{P}^{1-\sigma} - \left\{ w\left(\frac{\bar{A}}{\hat{Q}_j(v,t)} \hat{P}^{-\sigma} \right) + \hat{Q}_j(v,t)^{\alpha} s\left(\frac{\bar{A}}{B\hat{Q}_j(v,t)^{1-\alpha}} \hat{P}^{-\sigma} \right) \right\} \frac{\bar{A}}{\hat{Q}_j(v,t)} \hat{P}^{-\sigma} \right]. \quad (52)$$

The firm trades off its product-market power (setting a high price) against the cost of attracting labor (paying higher wages). The functions $w(\cdot)$ and $s(\cdot)$ are the inverses of the labor supply curves in equation (49). We can re-write this as

$$\Pi_j(v,t) = \max_{P,l,m} Y(t) \left[\left(\frac{P}{P_j(t)} \right)^{1-\sigma} - w(l)l - s(m)m \right], \quad (53)$$

$$\text{s.t. } Q_j(v,t) \cdot \left[(1-B)l^{\frac{\psi-1}{\psi}} + B[\hat{Q}_j(v,t)^{-\alpha} m]^{\frac{\psi-1}{\psi}} \right]^{\frac{\psi}{\psi-1}} = Y_j(t) \left(\frac{P}{P_j(t)} \right)^{-\sigma} = \frac{Y(t)}{P_j(t)} \left(\frac{P}{P_j(t)} \right)^{-\sigma}.$$

Incorporating equations (45) and (47), yields the following identity

$$\frac{Y(t)}{P_j(t)Q_j(v,t)} = \frac{\bar{A}\bar{Q}(t)}{P_j(t)Q_j(v,t)} = \frac{\bar{A}\bar{Q}(t)}{Q_j(v,t)} \frac{\bar{Q}_j(t)}{\bar{Q}(t)} = \frac{\bar{A}}{\hat{Q}_j(v,t)}, \quad (54)$$

where $\hat{Q}_j(v,t) \equiv \frac{Q_j(v,t)}{\bar{Q}_j(t)}$. Hence, we can write the constraint as

$$\begin{aligned} \left[(1-B)l^{\frac{\psi-1}{\psi}} + B[\hat{Q}_j(v,t)^{-\alpha} m]^{\frac{\psi-1}{\psi}} \right]^{\frac{\psi}{\psi-1}} &= \frac{Y(t)}{Q_j(v,t)P_j(t)} \left(\frac{P}{P_j(t)} \right)^{-\sigma} \\ &= \frac{\bar{A}}{\hat{Q}_j(v,t)} \left(\frac{P}{P_j(t)} \right)^{-\sigma} = \frac{\bar{A}\hat{Y}_j(v,t)}{\hat{Q}_j(v,t)}, \end{aligned} \quad (55)$$

where the first equality uses 54 and the second equality uses 43.

Next, defining $\hat{Y}_j(v,t) \equiv \frac{Y_j(v,t)}{Y_j(t)}$, we derive the labor requirement functions:

$$\chi(\hat{Q}_j(v,t), \hat{Y}_j(v,t)) \equiv \frac{m(\hat{Q}_j(v,t), \hat{Y}_j(v,t))}{l(\hat{Q}_j(v,t), \hat{Y}_j(v,t))} = \frac{B^{\psi} \hat{Q}_j(v,t)^{\alpha(1-\psi)}}{(1-B)^{\psi}} \frac{mc_l(\hat{Q}_j(v,t), \hat{Y}_j(v,t))^{\psi}}{mc_m(\hat{Q}_j(v,t), \hat{Y}_j(v,t))^{\psi}} \quad (56)$$

$$l(\hat{Q}_j(v,t), \hat{Y}_j(v,t)) = \frac{\bar{A}\hat{Y}_j(v,t)}{\hat{Q}_j(v,t)} \left[(1-B) + B[\hat{Q}_j(v,t)^{-\alpha} \chi(\hat{Q}_j(v,t), \hat{Y}_j(v,t))]^{\frac{\psi-1}{\psi}} \right]^{\frac{-\psi}{\psi-1}} \quad (57)$$

$$mc_l(\hat{Q}_j(v,t), \hat{Y}_j(v,t)) = w(l(\hat{Q}_j(v,t), \hat{Y}_j(v,t)))(1 + \tfrac{d \log w(l)}{d \log l}|_{l=l(\hat{Q}_j(v,t), \hat{Y}_j(v,t))}), \quad (58)$$

$$mc_m(\hat{Q}_j(v,t), \hat{Y}_j(v,t)) = s(m(\hat{Q}_j(v,t), \hat{Y}_j(v,t)))(1 + \tfrac{d \log s(m)}{d \log m}|_{m=m(\hat{Q}_j(v,t), \hat{Y}_j(v,t))}), \quad (59)$$

$$\chi_j(v,t) \equiv \frac{m_j(v,t)}{l_j(v,t)} = \frac{B^{\psi} \hat{Q}_j(v,t)^{\alpha(1-\psi)}}{(1-B)^{\psi}} \frac{[w_j(v,t)(1+\varepsilon_j^w(v,t))]^{\psi}}{[s_j(v,t)(1+\varepsilon_j^s(v,t))]^{\psi}} \quad (60)$$

$$l_j(v,t) = \frac{\bar{A}\hat{Y}_j(v,t)}{\hat{Q}_j(v,t)} \left[(1-B) + B[\hat{Q}_j(v,t)^{-\alpha} \chi_j(v,t)]^{\frac{\psi-1}{\psi}} \right]^{\frac{-\psi}{\psi-1}}, \quad (61)$$

where $m_j(v,t)$ is immediate from $\chi_j(v,t)$ and $\varepsilon_j^w(v,t)$ and $\varepsilon_j^s(v,t)$ are the inverse labor supply elasticities for w and s, respectively.

Let $\pi_j(v,t) = \frac{\Pi_j(v,t)}{Y(t)}$ be the profit per unit of output. Combining (54) with the firm's labor requirement functions yields

$$\pi_j(v,t) = \max_{\hat{P}} \left[\hat{P}^{1-\sigma} - \left\{ w \left(\frac{\bar{A}}{\hat{Q}_j(v,t)} \hat{P}^{-\sigma} \right) + \hat{Q}_j(v,t)^{\alpha} s \left(\frac{\bar{A}}{B\hat{Q}_j(v,t)^{1-\alpha}} \hat{P}^{-\sigma} \right) \right\} \frac{\bar{A}}{\hat{Q}_j(v,t)} \hat{P}^{-\sigma} \right]. \tag{62}$$

We can rewrite the FOC for the optimal price in terms of elasticities, yielding

$$\hat{P}(\hat{Q}) = \underbrace{\frac{\sigma}{\sigma-1}}_{\text{CES markup}} \frac{\bar{A}}{\hat{Q}_j(v,t)} \underbrace{\left[(1-B)^{\psi} [w_j(v,t)(1+\varepsilon_j^w(v,t))]^{1-\psi} + B^{\psi} \hat{Q}_j(v,t)^{\alpha(1-\psi)} [s_j(v,t)(1+\varepsilon_j^s(v,t))]^{1-\psi} \right]^{\frac{1}{1-\psi}}}_{\text{monopsonist's marginal cost}} \tag{63}$$

where $w(l)$ and $s(m)$ are the inverse labor supply curves in (49), and

$$\varepsilon_j^w(v,t) \equiv \frac{\partial w(l(\hat{Q}_j(v,t)))}{\partial l(\hat{Q}_j(v,t))} \frac{l(\hat{Q}_j(v,t))}{w(l(\hat{Q}_j(v,t)))}, \; \varepsilon_j^s(\hat{Q}_j(v,t);s) \equiv \frac{\partial s(m(\hat{Q}_j(v,t)))}{\partial m(\hat{Q}_j(v,t))} \frac{m(\hat{Q}_j(v,t))}{s(m(\hat{Q}_j(v,t)))} \tag{64}$$

are the elasticity of the inverse labor supply curve.

To calculate the elasticity of labor supply we also need to compute the expected number of efficiency units that a firm would get given a posted wage w. Workers leave their outside option if the take home pay is greater than $R^{1+\frac{1}{\varphi}}$. The expected number of efficiency units for job k is

$$N_k = \frac{\tilde{\theta}_\tau S_k}{w_k} \int_{R_\tau^\beta}^{\infty} y^{-\tilde{\theta}_\tau} e^{-\Lambda_\tau y^{-\tilde{\theta}_\tau}} dy = \frac{S_k}{w_k} \Lambda_\tau^{1/\tilde{\theta}_\tau - 1} \gamma\left(1 - \tfrac{1}{\tilde{\theta}_\tau}, \Lambda_\tau R_\tau^{-\theta_\tau}\right) \tag{65}$$

which gives aggregate labor supply as a function of wages,

$$l(w) = \omega K \frac{S_k}{w_k} \Lambda_l^{1/\tilde{\theta}_l - 1} \gamma\left(1 - \tfrac{1}{\tilde{\theta}_l}, \Lambda_l R_l^{-\theta_l}\right) \tag{66}$$

where the expression for type m is analogous.

We can take logs and differentiate this with respect to w to compute the elasticities:

$$\varepsilon_j^\tau = (\theta_\tau - 1) + \frac{T_\tau \theta_\tau w_k^{\theta_\tau} \mathbb{E}_W \left[\Lambda_\tau^{\frac{1}{\tilde{\theta}_\tau}-2} \left(\left(\frac{1}{\tilde{\theta}_\tau} - 1 \right) \gamma(1 - \frac{1}{\tilde{\theta}_\tau}, \Lambda_\tau R_\tau^{-\theta_\tau}) + \left(\Lambda_\tau R_\tau^{-\theta_\tau} \right)^{1-\frac{1}{\tilde{\theta}_\tau}} e^{-\Lambda_\tau R_\tau^{-\theta_\tau}} \right) \right]}{\mathbb{E}_W \left[\Lambda_\tau^{\frac{1}{\tilde{\theta}_\tau}-1} \gamma(1 - \frac{1}{\tilde{\theta}_\tau}, \Lambda_\tau R_\tau^{-\theta_\tau}) \right]} \tag{67}$$

Finally, we need $\bar{A}$. Combining the production function with the CES demand function,

$$\frac{Y_j(v,t)}{\bar{Q}_j(t)} = \hat{Q}_j(v,t) l_j(v,t) = \frac{Y(t)}{P_j(t)\bar{Q}_j(t)} \hat{P}_j(v,t)^{-\sigma} = \frac{\bar{A}\bar{Q}(t)}{\bar{Q}_j(t)} \frac{\bar{Q}_j(t)}{\bar{Q}(t)} \hat{P}_j(v,t)^{-\sigma} = \bar{A} \hat{P}_j(v,t)^{-\sigma}, \tag{68}$$

which we can rearrange and integrate to get

$$\begin{aligned}\bar{A} &= \frac{\int_0^1 \int_0^1 \hat{Y}_j(v,t)\mathrm{d}j\,\mathrm{d}v}{\int_0^1 \int_0^1 \hat{P}_j(v,t)^{-\sigma}\mathrm{d}j\,\mathrm{d}v} \\ &= \int_0^1 \int_0^1 \hat{Q}_j(v,t)\left[(1-B)l_j(v,t)^{\frac{\psi-1}{\psi}} + B[\hat{Q}_j(v,t)^{-\alpha}m_j(v,t)]^{\frac{\psi-1}{\psi}}\right]^{\frac{\psi}{\psi-1}}\mathrm{d}j\,\mathrm{d}v,\end{aligned} \tag{69}$$

where the last equality follows because $\int_0^1 \hat{P}_j(v,t)^{-\sigma}\mathrm{d}v = \int_0^1 \frac{Y_j(v,t)}{Y_j(t)}\mathrm{d}v = 1$. The last equality says that we can write (69) as a labor-weighted average of relative labor productivity $\hat{Q}_j(v,t)$ times the aggregate number of efficiency units of labor.

Dynamic model

Understanding the Business-Time Construction. Appendix Figure SA.5 shows how $\Gamma_j(t)$ maps calendar time into business time. Panel A shows that business time advances only when $\Gamma_j(t)$ jumps. Panel B shows that the relative productivity $\hat{Q}_j(v,t)$ is a standard Poisson process in business time, and Panel C shows this in calendar time. For each product line where the firm is producing, external improvements arrive according to

$$N_j^x(v,t) \sim \text{Poisson}\big(x_j(v,t^-)\,\mathrm{d}\Gamma_j(t)\big)\,. \tag{70}$$

These improvements are undirected: when $N_j^x(v,t) > 0$, the firm draws $N_j^x(v,t)$ varieties at random and improves their productivity by λ. A fraction p_r of these innovations reset productivity in the affected lines to $\lambda\,\bar{Q}_j(t)$. It is possible for $N_j^x > 0$ and $N_j^e > 0$ in instant of calendar time; we assume the gains from internal innovation accrue to the firm that ultimately acquires the product line.

The number of external innovations from entrants in a product line is

$$N_j^e(v,t) \sim \text{Poisson}\big(e_j(t^-)\,\mathrm{d}\Gamma_j(t)\big)\,. \tag{71}$$

where $e_j(t)$ is the mass of new entrants. Multiple firms may innovate at the same instant and product line. In this case a random successful external innovator becomes the new producer.

Worker productivity dynamics. We work with a transformed index

$$z^*(i,k,t) \equiv \Phi^{-1}\left[F_{\tau(i)}\left(z_{j(i,k,t)}(i,v(i,k,t),t)\right)\right], \tag{72}$$

which maps the Fréchet productivity draw z into a standard normal variable z^* that follows OU dynamics. When a product line is displaced by $\bar{x}$ or $\bar{e}$ we draw a new job from the product lines that changed their active producer during the innovation burst with match quality to $z^*(i,k,t) \sim N(0,1)$.

Dynamics of $\hat{q}$, $\bar{q}$, and aggregate growth. Let $q_j(v,t) \equiv \log Q_j(v,t)$. Under our assumptions we can write the dynamics of $q_j(v,t)$ as

$$\mathrm{d}q_j(v,t) \;:=\; \log\lambda\,\mathrm{d}N_j(v,t) - \underbrace{\big(1-\tilde{D}_j(v,t)\big)\hat{q}_j(v,t^-)}_{\text{reset to }\bar{q}_j(t)} - \delta dt\,, \tag{73}$$

where, given n jumps at the product line level, the probability that the product line is not 'reset' equals

$$\Pr\big(\tilde{D}_j(v,t)=1 | dN_j(v,t)=n\big) \;=\; \left[\frac{I[\hat{q}_j(v,t^-)]+\tau(1-p_r)}{I[\hat{q}_j(v,t^-)]+\tau}\right]^n \equiv \gamma_n(\hat{q}_j(v,t^-)). \tag{74}$$

Hence, we can calculate (using $\hat{q}$ as shorthand for $\hat{q}_j(v,t^-)$)

$$\begin{aligned} E\left[\frac{\mathrm{d}Q_j(v,t)}{Q_j(v,t^-)}\,\middle|\, d\Gamma_{j,t}=\mu\right] &= e^{-(I(\hat{q})+\tau)\mu} \\ &\quad\times\sum_{k=1}^{\infty}\frac{\big([I(\hat{q})+\tau]\,\mu\big)^k}{k!}\left[\gamma_k(\hat{q})(\lambda^k-1)+(1-\gamma_k(\hat{q}))\left(\frac{\lambda^k}{\hat{Q}_j(v,t^-)}-1\right)\right]. \end{aligned} \tag{75}$$

Let $\bar{Q}_j(t) \equiv \int_0^1 Q_j(v,t)dv$, where $\bar{q}_j(t) \equiv \log\bar{Q}_j(t)$. The growth rate of industry productivity is

$$\begin{aligned} \frac{\mathrm{d}\bar{Q}_j(t)}{\bar{Q}_j(t^-)} &= \int_0^1 \hat{Q}_j(v,t)\cdot\frac{\mathrm{d}Q_j(v,t)}{Q_j(v,t^-)}\mathrm{d}v = E\left[\hat{Q}_j(v,t)\frac{\mathrm{d}Q_j(v,t)}{Q_j(v,t^-)}\right] \\ &= \mathbf{1}(\mathrm{d}\Gamma_{jt}=\mu)\,E\left[\hat{Q}_j(v,t)\frac{\mathrm{d}Q_j(v,t)}{Q_j(v,t^-)}\,\middle|\,\Gamma_{jt}=\mu\right]. \end{aligned}$$

So it follows that

$$\begin{aligned} E\left[\hat{Q}_j(v,t)\tfrac{\mathrm{d}Q_j(v,t)}{Q_j(v,t^-)}\middle| d\Gamma_{j,t}=\mu\right] &= E\Bigg\{e^{-(I(\hat{q})+\tau)\mu}\sum_{k=1}^{\infty}\frac{\big([I(\hat{q})+\tau]\,\mu\big)^k}{k!} \\ &\qquad\times\left[\gamma_k(\hat{q})(\lambda^k-1)\hat{Q}_j(v,t^-)+(1-\gamma_k(\hat{q}))\left(\lambda^k-\hat{Q}_j(v,t^-)\right)\right]\Bigg\}. \\ (\text{if } I(\hat{q})=I) &= e^{-(I+\tau)\mu}\sum_{k=1}^{\infty}\frac{\big([I+\tau]\,\mu\big)^k}{k!} \\ &\qquad\times\left[\gamma_k(0)(\lambda^k-1)\underbrace{E\{\hat{Q}_j(v,t^-)\}}_{=1}+(1-\gamma_k(0))\left(\lambda^k-E\{\hat{Q}_j(v,t^-)\}\right)\right] \\ &= e^{-(I+\tau)\mu}\sum_{k=1}^{\infty}\frac{\big([I+\tau]\,\mu\big)^k}{k!}(\lambda^k-1) = \exp((\lambda-1)(I+\tau)\mu)-1. \end{aligned} \tag{76}$$

where the second line replaces $\gamma_k(\hat{q})$ with $\gamma_k(0)$ to reflect the fact that $\gamma_k(\hat{q})$ is constant when I is constant, which allows us to take the expectation inside the sum. If we work in logs, (76) implies that if I is constant,

$$\mathrm{d}\bar{q}_j(t) = (\lambda-1)(I+\tau)\mathrm{d}\Gamma_{j,t} - \delta dt \equiv \tilde{g}\,\mathrm{d}\Gamma_{j,t} - \delta dt. \tag{77}$$

When I is not constant, we need to instead define $\tilde{g}$ using the more complicated expression in the top line of (76). Finally, the aggregate growth rate equals

$$g\mathrm{d}t = \frac{\mathrm{d}\bar{Q}(t)}{\bar{Q}(t)} = E\left[\frac{\bar{Q}_j(t^-)}{\bar{Q}(t)}\frac{\mathrm{d}\bar{Q}_j(t)}{\bar{Q}_j(t^-)}\right] = \underbrace{E\left[\frac{\bar{Q}_j(t^-)}{\bar{Q}(t)}\right]}_{=1} E\left[\frac{\mathrm{d}\bar{Q}_j(t)}{\bar{Q}_j(t^-)}\right] = \frac{1}{\mu}[\exp((\tilde{g}-\delta)\mu)-1]\mathrm{d}t \approx (\tilde{g}-\delta)\mathrm{d}t, \tag{78}$$

where iid industry productivity increments allow us to go from the second to the third equality. While $g \neq \tilde{g}$ due to Jensen's inequality, the difference between the two is small when μ isn't large.

Combining (73) with (77) yields the law of motion for the $\hat{q}_j(v,t)$ of the most efficient producer:

$$\mathrm{d}\hat{q}_j(v,t) = \log\lambda\,\mathrm{d}N_j(v,t) - \left(1-\tilde{D}_j(v,t)\right)\hat{q}_j(v,t^-) - \tilde{g}\,\mathrm{d}\Gamma_{j,t}. \tag{79}$$

For brevity, we omit j and v subscripts, since $\hat{q}$ is iid across varieties and industries along the BGP.

HJB, Innovation costs $\Phi(\hat{q})$, and ODEs for product line valuations. The value of a product line is the present discounted value of profits. The flow payoff $Y(t)\pi\left(\hat{q}_j(v,t)\right)$ is discounted at rate r, with $\pi(0)=0$. If $\frac{\dot{Y}(t)}{Y(t)} = g\,\mathrm{d}t$, the value per unit of $Y(t)$ is

$$V(\hat{q}_j(v,t)) = \mathbb{E}\left[\int_0^\infty e^{-(r-g)\,t}\left[\pi\left(\hat{q}\right) - c_I\left(I;\hat{q}\right) - c_x\left(x;\hat{q}\right)\right]\mathrm{d}t\right], \tag{80}$$

where profits and costs include new product lines obtained through external innovation. We can write this valuation recursively as

$$\begin{aligned}(r-g)\,V(\hat{q}) \;=\; \max_{I(\hat{q}),x}\;& \pi\left(\hat{q}\right) - c(I,x;\hat{q}) + x\mathcal{G}_x V \\ &+\frac{1}{\mu}\left[e^{-(I(\hat{q})+\tau)\mu}\sum_{k=0}^{\infty}\frac{\left(I(\hat{q})\,\mu\right)^k}{k!}\,V\left(\hat{q}-\tilde{g}\,\mu+k\log\lambda\right) - V(\hat{q})\right],\end{aligned} \tag{81}$$

where the second line obtains from the following direct calculation

$$\begin{aligned}\left[E_t V(\hat{q}_t)\,|\,d\Gamma_t=\mu, N_t^x=0\right] \;=\;& e^{-(I(\hat{q}_{t^-})+\tau)\mu}\,V(\hat{q}_{t^-}\underbrace{-\tilde{g}\,\mu}_{\text{competitor price impact}}) \\ &+\sum_{k=1}^{\infty} e^{-(I(\hat{q}_{t^-})+\tau)\mu}\frac{\left((I(\hat{q}_{t^-})+\tau)\,\mu\right)^k}{k!}\underbrace{\left[\left[\frac{I(\hat{q}_{t^-})}{I(\hat{q}_{t^-})+\tau}\right]^k V(\hat{q}_{t^-}-\tilde{g}\,\mu+k\log\lambda)\right]}_{\text{improve product line}} \\ &+\sum_{k=1}^{\infty} e^{-(I(\hat{q}_{t^-})+\tau)\mu}\frac{\left((I(\hat{q}_{t^-})+\tau)\,\mu\right)^k}{k!}\underbrace{\left[(1-\left[\frac{I(\hat{q}_{t^-})}{I(\hat{q}_{t^-})+\tau}\right]^k)\,0\right]}_{\text{lose product line}} \\ =\;& e^{-(I(\hat{q}_{t^-})+\tau)\mu}\sum_{k=0}^{\infty}\frac{\left(I(\hat{q}_{t^-})\,\mu\right)^k}{k!}\,V(\hat{q}_{t^-}-\tilde{g}\,\mu+k\log\lambda).\end{aligned} \tag{82}$$

Notice that increases in τ decrease the continuation value through a higher likelihood that a competitor steals the line, which drops the value of profits from the existing line to zero.

Next, we need to work out $\mathcal{G}_x V$. By direct calculation and an application of the tie-breaking rules for multiple jumps discussed in the main text, we get

$$\mathcal{G}_x V = \mathbb{E}\left\{ \frac{\exp(-[I(\hat{q})+\tau]\mu)}{P_{x\geq 1}} \sum_{k=1}^{\infty} \frac{[I(\hat{q})+\tau]^k \mu^k}{k!} \left[\underbrace{\sum_{n=1}^{k} \sum_{j=1}^{n} \frac{k!\, p_x(\hat{q})^j p_e(\hat{q})^{n-j} p_I(\hat{q})^{k-n}}{j!(n-j)!(k-n)!}}_{\substack{\text{Prob of } n \text{ bus stealing jumps} \\ \text{and at least 1 } x \text{ jump}}} \frac{1}{n} \left[\begin{array}{l} (1-p_r^n) V(k\log\lambda + \hat{q} - \tilde{g}\mu) \\ +p_r^n V(k\log\lambda - \tilde{g}\mu) \end{array} \right] \right] \right\}, \tag{83}$$

where $P_{x\geq 1} = 1 - \exp(-x\mu)$ and

$$p_I(\hat{q}) = \frac{I(\hat{q})}{I(\hat{q})+\tau}, \qquad p_x(\hat{q}) = \frac{x}{I(\hat{q})+\tau}, \qquad p_e(\hat{q}) = \frac{e}{I(\hat{q})+\tau}. \tag{84}$$

The $1/n$ factor reflects what happens if two firms steal at the same time, capturing the probability that a firm has the last innovation.

Analogously, we can compute $\mathcal{G}_e V$, the same object for an entry-related jump.

$$\mathcal{G}_e V = \mathbb{E}\left\{ \frac{\exp(-[I(\hat{q})+\tau]\mu)}{P_{e\geq 1}} \sum_{k=1}^{\infty} \frac{[I(\hat{q})+\tau]^k \mu^k}{k!} \left[\underbrace{\sum_{n=1}^{k} \sum_{j=1}^{n} \frac{k!\, p_e(\hat{q})^j p_x(\hat{q})^{n-j} p_I(\hat{q})^{k-n}}{j!(n-j)!(k-n)!}}_{\substack{\text{Prob of } n \text{ bus stealing jumps} \\ \text{and at least 1 } e \text{ jump}}} \frac{1}{n} \left[\begin{array}{l} (1-p_r^n) V(k\log\lambda + \hat{q} - \tilde{g}\mu) \\ +p_r^n V(k\log\lambda - \tilde{g}\mu) \end{array} \right] \right] \right\}, \tag{85}$$

where $P_{e\geq 1} = 1 - \exp(-e\mu)$. A firm wins the product line given an x jump (a successful external innovation) with probability

$$P_{x \text{ success}} = \frac{\exp(-\tau\mu)}{1-\exp(-x\mu)} \sum_{k=1}^{\infty} \frac{(\tau\mu)^k}{k!} \frac{1}{k} \underbrace{\left[1 - \left(\frac{e}{\tau}\right)^k\right]}_{\text{Prob of at least 1 x jump}}, \tag{86}$$

along with its counterpart for an external innovation by a new entrant, which is

$$P_{e \text{ success}} = \frac{\exp(-\tau\mu)}{1-\exp(-e\mu)} \sum_{k=1}^{\infty} \frac{(\tau\mu)^k}{k!} \frac{1}{k} \left[1 - \left(\frac{x}{\tau}\right)^k\right]. \tag{87}$$

In practice, the probability of multiple jumps is very low in our calibrated model.

Following Peters (2020), we choose $\Phi(\hat{q})$ so that innovation intensities are constant across $\hat{q}$, defining it implicitly so that it can be recovered from the ODE that pins down firm valuations. The FOC with respect to $I(\hat{q})$ equals

$$\Phi(\hat{q}) \frac{\zeta I(\hat{q})^{\zeta-1}}{\varphi_I} = \mathcal{G}_I V, \tag{88}$$

where

$$\mathcal{G}_I V \equiv \underbrace{e^{-(I(\hat{q})+\tau)\mu} \left[\sum_{k=0}^{\infty} \left(\frac{[I(\hat{q})\mu]^k}{k!} \right) \left[V\left(\hat{q} - \tilde{g}\,\mu + (k+1)\log\lambda\right) - V\left(\hat{q} - \tilde{g}\,\mu + k\log\lambda\right) \right] \right]}_{\mathcal{G}_I V} \tag{89}$$

If $I(\hat{q})$ is constant, then the following choice of $\Phi(\hat{q})$ is consistent with a constant arrival rate of I:

$$\Phi(\hat{q}) = \frac{\varphi_I}{\zeta\,(I^*)^{\zeta-1}}\; e^{-(I+\tau)\mu}\left[\sum_{k=0}^{\infty}\left(\frac{[I\mu]^k}{k!}\right)\left[V\left(\hat{q}-\tilde{g}\,\mu+(k+1)\log\lambda\right)-V\left(\hat{q}-\tilde{g}\,\mu+k\log\lambda\right)\right]\right] \tag{90}$$

Analogously, we solve for th φ_x consistent with a value of radical innovation by incumbents, x:

$$\varphi_x = \frac{\zeta\, x^{\zeta-1}}{\mathcal{G}_x\, V}. \tag{91}$$

If $\Phi(\hat{q})$ satisfies 90 and we plug in the optimal I and x into the HJB (81), valuations follow

$$\begin{aligned}(r-g)\,V(\hat{q}) \quad &= \quad \pi\left(\hat{q}\right)+x\left(1-\frac{1}{\zeta}\right)\mathcal{G}_x\,V \\ &- \quad \frac{I}{\zeta}\; e^{-(I+\tau)\mu}\left[\sum_{k=0}^{\infty}\left(\frac{[I\mu]^k}{k!}\right)\left[V\left(\hat{q}-\tilde{g}\,\mu+(k+1)\log\lambda\right)-V\left(\hat{q}-\tilde{g}\,\mu+k\log\lambda\right)\right]\right] \\ &+ \quad \frac{1}{\mu}\left[e^{-(I+\tau)\mu}\sum_{k=0}^{\infty}\frac{(I\,\mu)^k}{k!}\;V\left(\hat{q}-\tilde{g}\,\mu+k\log\lambda\right)-V(\hat{q})\right],\end{aligned} \tag{92}$$

which, once we've discretized over a $\hat{q}$ grid, reduces to a linear system of equations.

Computing Certainty Equivalent Welfare

We derive the willingness to pay for a subsidy that raises g by one percentage point for a single year out of 20 years of earnings, setting g to 2 percent and the time discount factor β to 4 percent.

Worker preferences are such that given a wage rate they maximize effort:

$$\max_{e} u\left(e_{k,i,t}z_{j(k,i,t)}Y(t)w_{k,t}-Y(t)\frac{e^{1+\varphi}}{1+\varphi}\right).$$

At optimal effort e^*, $u(w,z,e^*) = Y_t^{1-\gamma}u\left(\frac{\varphi}{1+\varphi}[wz]^{1+\frac{1}{\varphi}}\right)$. Workers are hand-to-mouth, so $c_t = \frac{\varphi}{1+\varphi}[wz]^{1+\frac{1}{\varphi}}$. We want to find a, a multiple of consumption, such that workers are indifferent:

$$\begin{aligned}&E_0\left[\int_0^{20} e^{-(\beta-g(1-\gamma))t}u(a\times c_t)dt\right] \\ &\quad = E_0\left[\int_0^{1} e^{-(\beta-(g+\Delta)[1-\gamma])t}u(\tilde{c}_t)dt+\int_1^{20} e^{-(\beta-g[1-\gamma])t+\Delta(1-\gamma)}u(\tilde{c}_t)dt\right]\end{aligned} \tag{93}$$

where $\tilde{c}_t$ is the counterfactual consumption for the worker under the subsidy.

CRRA preferences mean that the utility function is homogenous of degree $1-\gamma$, implying

$$a_i = \left(\frac{E_0\left[\int_0^1 e^{-(\beta-(g+\Delta)[1-\gamma])t}u(\tilde{c}_t)dt+\int_1^{20} e^{-(\beta-g[1-\gamma])t+\Delta(1-\gamma)}u(\tilde{c}_t)dt\right]}{E_0\left[\int_0^{20} e^{-(\beta-g(1-\gamma))t}u(c_t)dt\right]}\right)^{\frac{1}{1-\gamma}}$$

which we evaluate by simulation. For $\gamma = 1$ utility is log, which gives

$$a_i = \exp\left(\frac{\Delta\left[\int_0^1 t\, e^{-\beta t}\, dt + \int_1^{20} e^{-\beta t}\, dt\right] + E_0\left[\int_0^{20} e^{-\beta t}\, \log(\tilde{c}_t/c_t)\, dt\right]}{\int_0^{20} e^{-\beta t}\, dt}\right)$$

All counterfactuals raise growth by the same amount. When we raise the aggregate innovation rate we do so proportionally, so that

$$\frac{I_{cf}}{I_{cf} + x_{cf} + e_{cf}} = \frac{I}{I + \bar{x} + \bar{e}} \qquad \text{and} \qquad I_{cf} + x_{cf} + e_{cf} = b[I + \bar{x} + \bar{e}]$$

where b is chosen so that $\Delta = 0.01$. When we raise $\bar{x} + \bar{e}$, we keep those proportional as well.

The role of risk aversion. Because a_i is a ratio of certainty equivalents, a counterfactual that scales consumption by a common factor across states and dates leaves a_i unchanged for every γ: risk aversion prices changes in the shape of a worker's consumption distribution, not its level. The experiments in Panels A and B move consumption close to proportionally, so willingness to pay there is nearly identical under $\gamma = 1$ and $\gamma = 2$ (Figure SA.11); γ has bite only in the creative-destruction experiment. There, raising γ does not simply scale losses up—it moves willingness to pay in opposite directions in different parts of the earnings distribution. The reason is that every worker can fall back on the common outside option paying rate R, which bounds consumption from below. Faster aggregate growth raises the value of that option along with everything else. As γ rises the certainty equivalent loads more heavily on the worst states in the support, which is precisely where the counterfactual is relatively favorable. For the workers with the most to lose from displacement—top earners, whose outside options are worst relative to their current match (Figure SA.9)—this channel partly offsets their greater exposure to displacement risk, which is why losses in the upper tail of Panel C are smaller under $\gamma = 2$ than under log utility even as losses through the ninetieth percentile are slightly larger.

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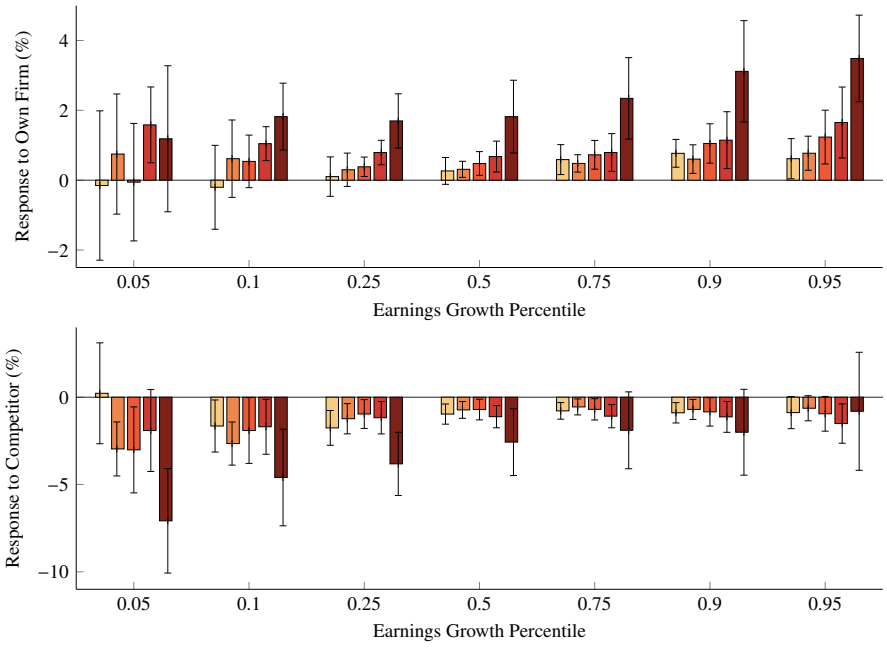
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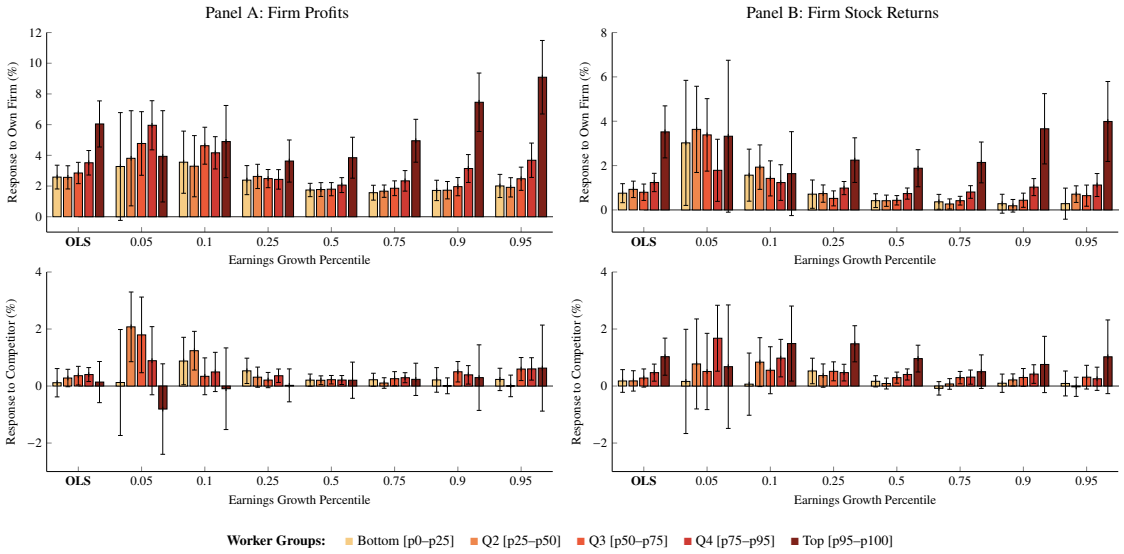
B Appendix Tables and Figures

Figure SA.1: Innovation and the Wage Growth Distribution by Income Group: Quantile Regression Estimates



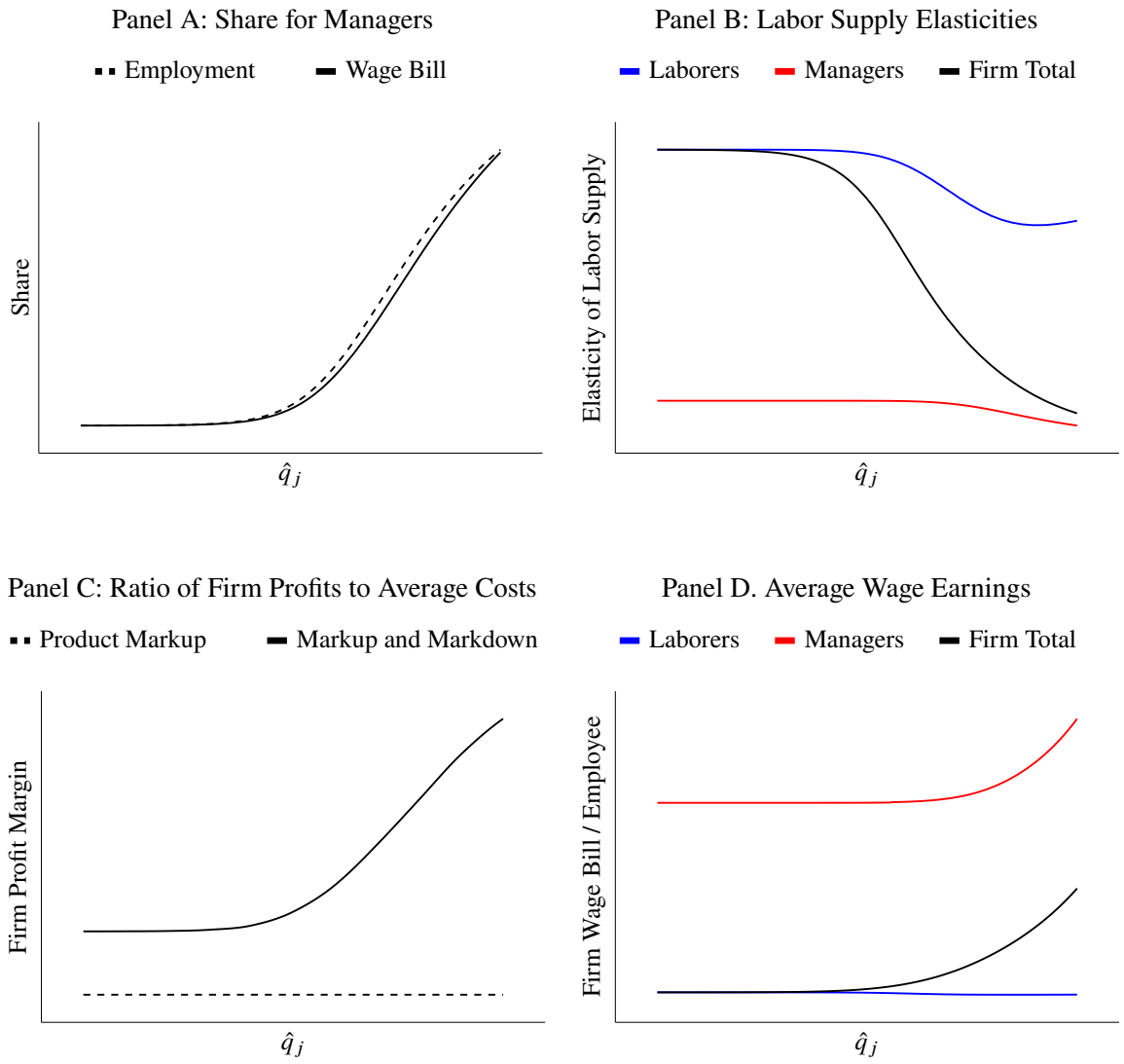
Note: The figures report average marginal effects from quantile regression, estimated using the method of Schmidt and Zhu (2025). Standard errors are estimated using the block weighted bootstrap. For details on controls and data construction see Appendix A.1.

Figure SA.2: Passthrough of Firm Profits and Stock Returns to Worker Wage Growth by Income Group



Note: The figure shows responses of five year wage earnings growth to shocks to own firm and competitor profits (Panel A) and stock returns (Panel B), for both OLS and quantile regressions. Firm-level controls are lagged market capitalization, stock return volatility, assets, and the lagged depreciated patent stock; worker-level controls are lifecycle wage effects approximated by Chebyshev polynomials and lagged wage growth, together with industry × income bin and year × income bin fixed effects. Income bins are indicators for the $[0\%, 25\%)$, $[25\%, 50\%)$, $[50\%, 75\%)$, $[75\%, 95\%)$, and $[95\%, 100\%]$ quantiles within an industry and year. Competitor innovation is the SIC3 industry aggregate less the own firm.

Figure SA.3: Static Equilibrium: Inverse Labor Supply Elasticities, Employment, and Wage Bill by Type



Note: Panel A shows the manager share of the wage bill and of employment across the $\hat{Q}_j(v,t)$ distribution. Panel B shows the elasticity of wages with respect to labor supply for both types, together with the wage-bill weighted average ("total"). Panel C compares firm profitability across the $\hat{Q}_j(v,t)$ distribution with the CES-implied markup.

Figure SA.4: Timeline of worker and firm actions within a period.

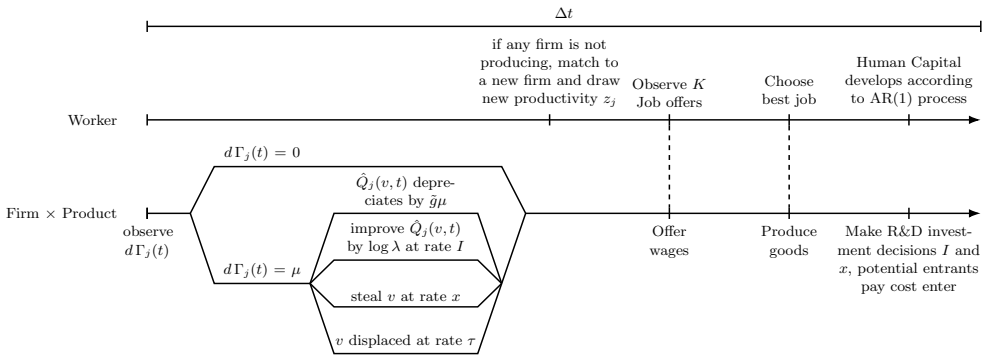
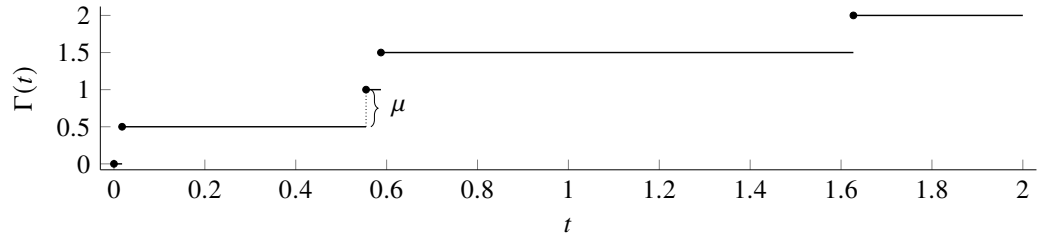
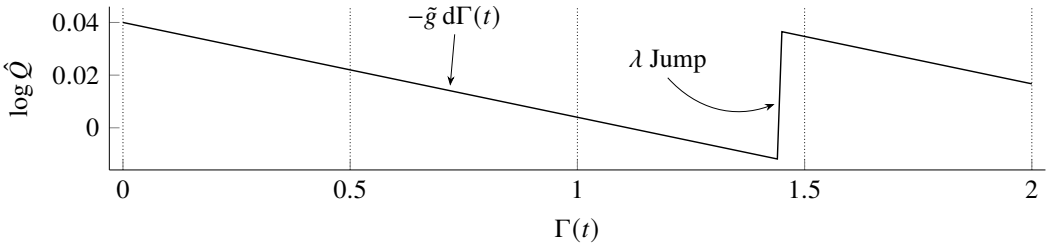


Figure SA.5: Time for the Innovation Process vs. Time for the Observer

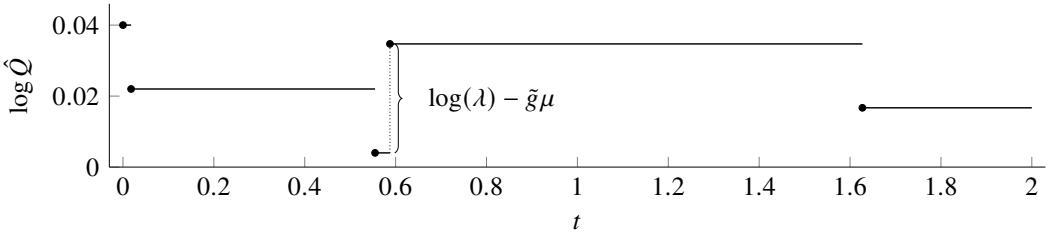
Panel A: The Jump Process for Industry Innovation Increments



Panel B: Sample Path for $\hat{q}$ Over $\Gamma(t)$

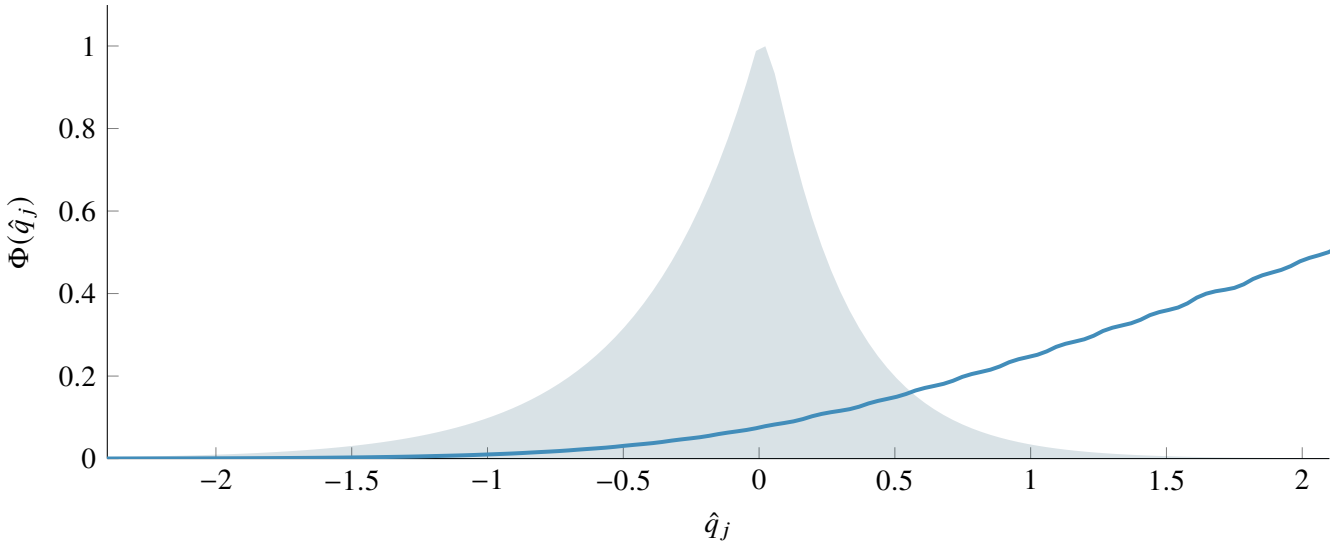


Panel C: Sample Path for $\hat{q}$ Over t



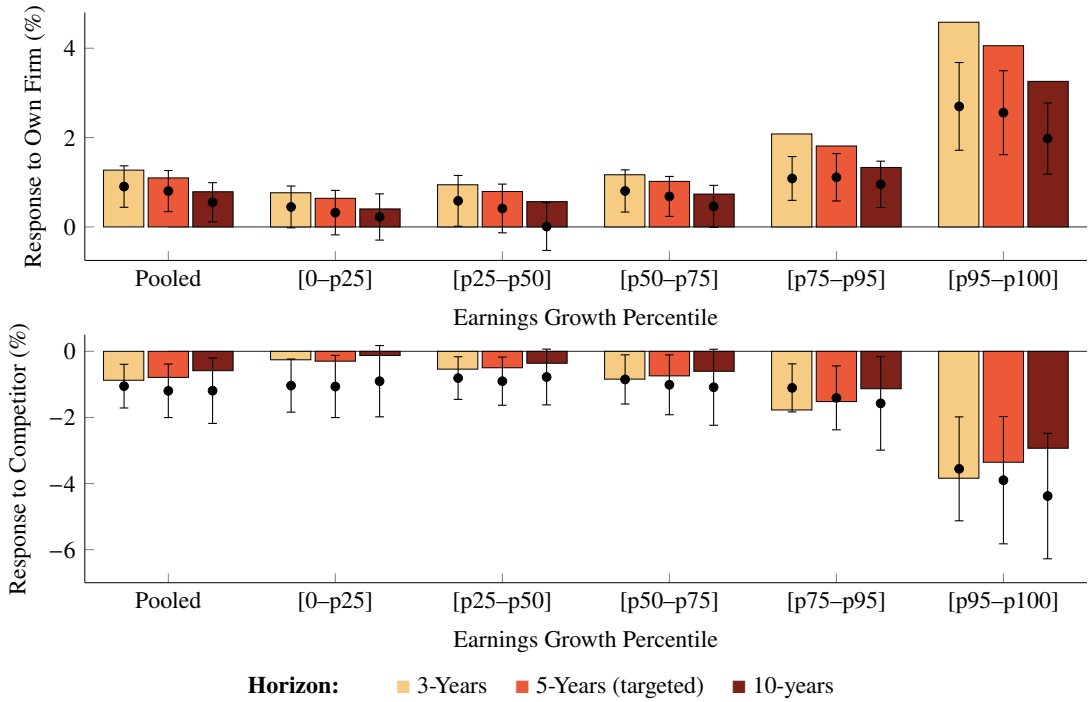
Note: Panel A relates business time, indexed by $\Gamma_j(t)$, to calendar time. Innovation occurs in bursts: a shock of size μ is instantaneous from the worker's perspective. Shocks to $d\Gamma_j$ arrive at the constant rate $1/\mu$, so the two time indexes align on average. Panel B shows a sample path of the state variable in business time; the vertical dotted lines mark the states seen by an external observer, while the continuous axis traces the entire process. Conditional on $d\Gamma_j(v,t) = \mu$, relative quality $\log\hat{Q}_j(v,t)$ sinks at rate $\tilde{g}\,d\Gamma_j(t)$ as competitors catch up, and the firm recovers through innovation jumps of size $\log\lambda$. Panel C translates the process back to calendar time and shows how the two Poisson processes interact; the intersections of the dotted lines with the sample paths in Panel B are the observed levels of $\log\hat{Q}$ in Panel C.

Figure SA.6: Implied Cost Function for Research and Development



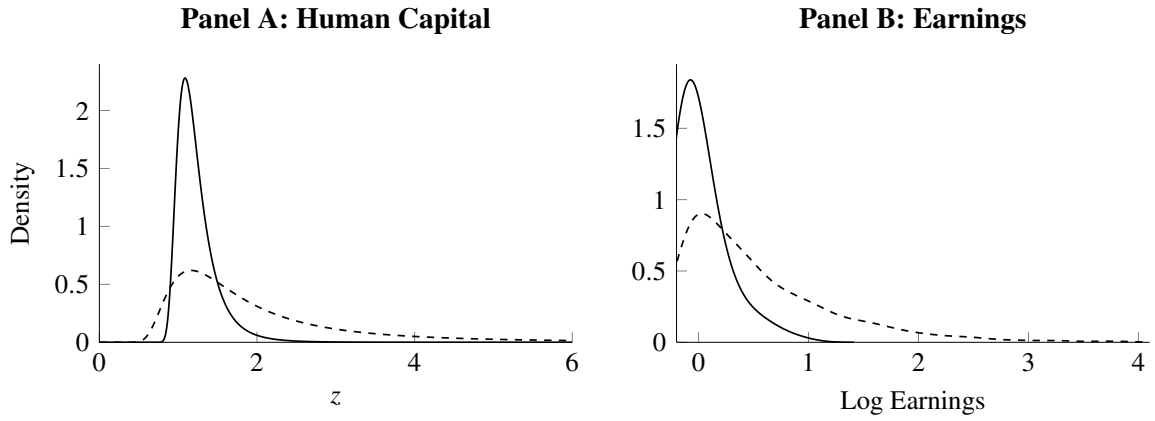
Note: The solid blue line plots the implied cost function for research and development that keeps the innovation rate stationary across the $\hat{Q}$ distribution. The shaded area shows the ergodic distribution of firm (relative) productivity $\hat{q}$.

Figure SA.7: Worker Wage Growth Responses Over 3, 5, and 10-Year Horizons: Model vs. Data



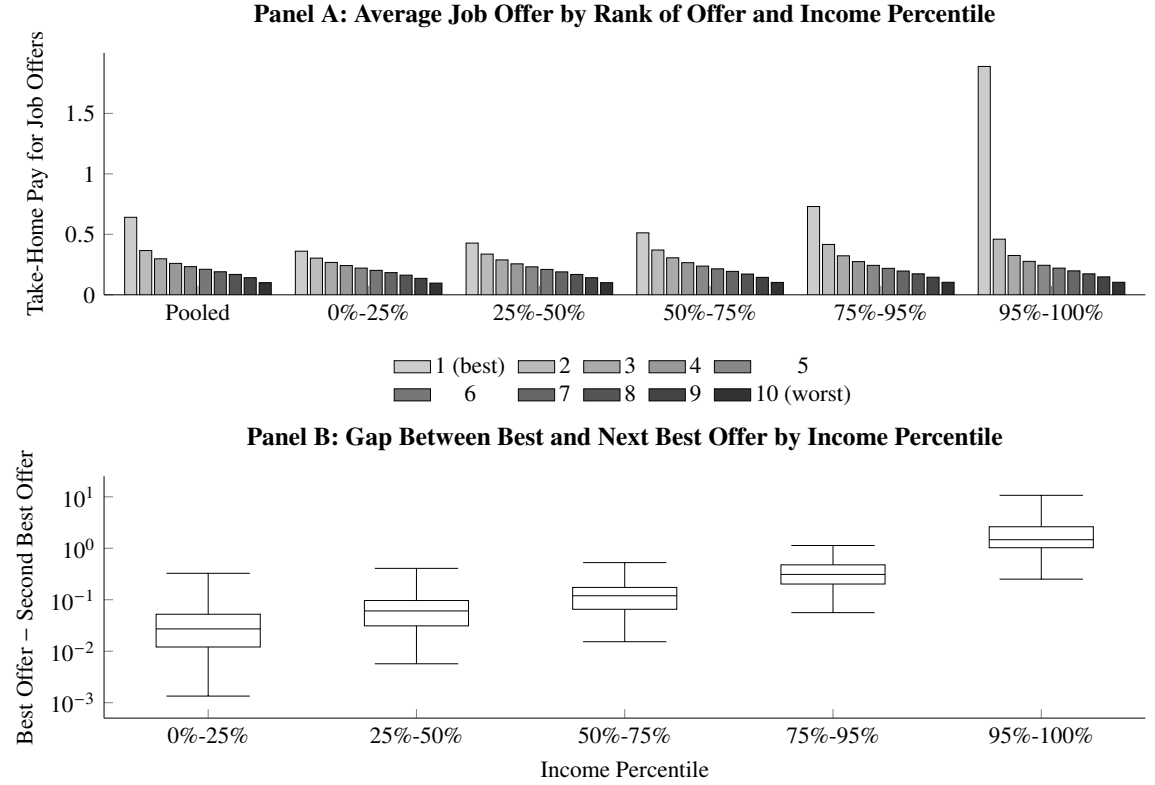
Note: The figure compares model-implied impulse responses of wage growth to product line shocks (columns) with their empirical counterparts (points with error bars), pooled and by income percentile, at 3, 5, and 10 year horizons. The competitor shock includes both declines in relative firm quality $\hat{Q}$ from the growth of competing firms and displacement of the product line by a competitor; own firm shocks are quality jumps within the worker's product line. Only the 5-year responses are targeted in the calibration.

Figure SA.8: Distributions of Worker Skill and Earnings in the Model for Managers (dashed) and Laborers (solid)



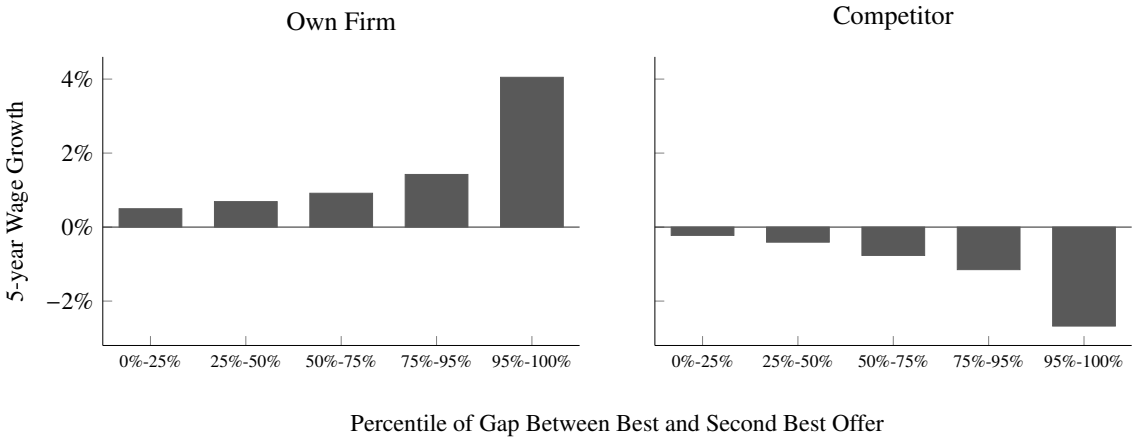
Note: Panel A of this figure shows the distribution of human capital draws, z_{ijt}, for both types. Both types have the same location parameter for the Fréchet distribution, $T = 2.03$, but different dispersion parameters. For managers, $\theta_m = 2.13$ and for laborers $\theta_l = 6.81$. Panel B of this figure shows the distribution of log earnings.

Figure SA.9: Distribution of Job Offers by Worker Income Group



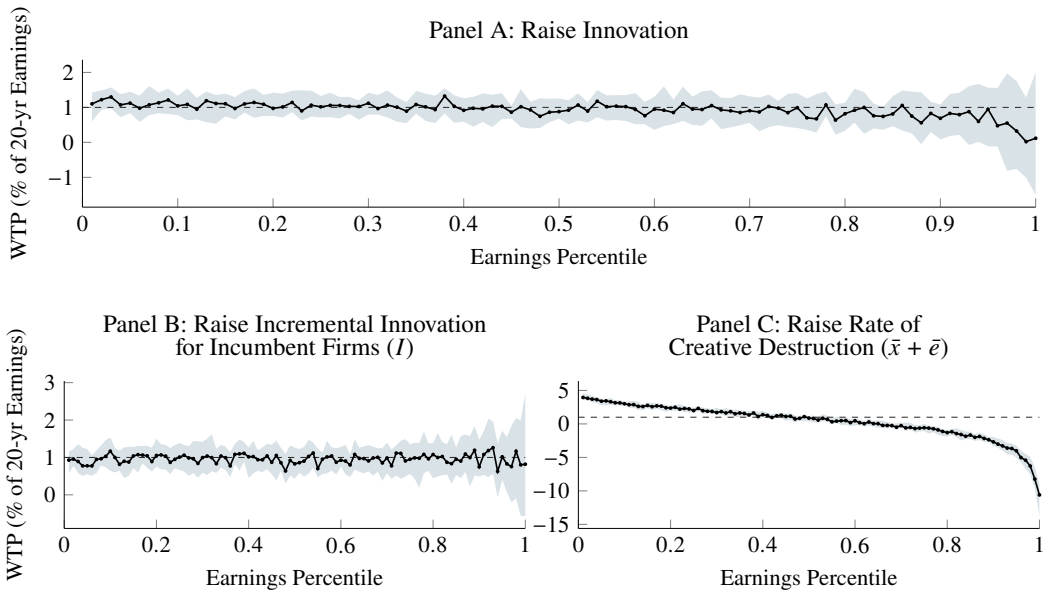
Note: The figure shows the distribution of the job offer set by income, with jobs ranked high (1) to low (5) and averaged within income groups. High-income workers have substantially worse outside options relative to their current employer, while lower-income workers face relatively similar offers. Panel A plots the average offer by income percentile and offer rank; Panel B shows boxplots of the gap by income percentile, with the y axis on a log scale.

Figure SA.10: Wage Growth Responses to Innovation by Outside-Option Gap



Note: This figure shows the difference in wage growth responses to innovation shocks by percentile of the gap between a worker's best option and their second best option. Workers who have substantially better first-best options see much larger impacts; this is especially concentrated in the very top of the distribution.

Figure SA.11: Willingness to Pay for a Temporary Increase in Innovation, $\gamma = 2$



Note: Panel A raises $\bar{x}$, $\bar{e}$ and I proportionally; Panel B raises I; Panel C raises $\bar{x}$ and $\bar{e}$ proportionally. Lines show medians and bands show the IQR; the dashed line is the complete-markets benchmark. For details see Section 4.3 and Appendix A.1.

Table SA.1: Robustness Checks for Pooled OLS Results

	Own Firm Innovation		Competitor Innovation	
	Estimate	SE	Estimate	SE
A. Firm Profits				
Baseline	5.664	0.571	-4.111	1.164
Enterprise Value Denominator	5.258	0.420	-2.781	0.930
Hoberg-Phillips Competitor	5.346	0.609	-4.528	0.777
Residualized Own-Firm Innovation	4.888	0.604	-4.404	1.265
B. Firm Total Wage Bill				
Baseline	4.939	0.461	-3.428	0.921
Enterprise Value Denominator	3.755	0.371	-3.17	0.950
Hoberg-Phillips Competitor	4.766	0.483	-4.833	0.688
Residualized Own-Firm Innovation	3.647	0.479	-4.056	1.059
C. Employment				
Baseline	2.979	0.400	-2.791	0.773
Enterprise Value Denominator	1.735	0.375	-2.967	0.919
Hoberg-Phillips Competitor	2.873	0.387	-4.756	0.750
Residualized Own-Firm Innovation	2.006	0.375	-3.480	0.872
D. Pay per worker				
Baseline	1.960	0.610	-0.637	1.202
Enterprise Value Denominator	2.020	0.528	-0.203	1.322
Hoberg-Phillips Competitor	1.893	0.619	-0.077	1.018
Residualized Own-Firm Innovation	1.641	0.608	-0.576	1.372
E. Individual Worker Earnings				
Baseline	0.803	0.235	-1.195	0.413
Enterprise Value Denominator	0.521	0.184	-0.843	0.302
Hoberg-Phillips Competitor	0.858	0.261	-1.011	0.256
Residualized Own-Firm Innovation	0.639	0.179	-1.171	0.412
Adjusted Gross Income	0.866	0.210	-1.491	0.274
Wage and Salary Income	0.683	0.216	-1.337	0.352

Note: This table reports robustness checks for the baseline results in Figures 1 and 2; all coefficients and standard errors are multiplied by 100. Residualized Innovation projects out predictable differences in the base rate of innovation, using a Poisson regression of the innovation measure on all of our controls together with industry and year fixed effects. The Hoberg-Phillips competitor definition builds industries from the 10-K textual similarity measure of Hoberg and Phillips (2016) (the TNIC3 definition) and recomputes competitor innovation using those industries in place of SIC3. The Enterprise value denominator normalizes the value of a patent filing by enterprise value rather than book assets.

Table SA.2: Wage Growth Responses to Innovation by Income Group: Robustness Checks

A. Own Firm Innovation	[0% - 25%)	[25% - 50%)	[50% - 75%)	[75% - 95%)	[95% - 100%]
Baseline	0.320	0.413	0.683	1.111	2.557
	(0.254)	(0.278)	(0.228)	(0.271)	(0.480)
Hoberg-Phillips Competitor	0.349	0.449	0.787	1.152	2.548
	(0.280)	(0.291)	(0.235)	(0.308)	(0.525)
Income Rank Firm Adjusted	0.698	0.494	0.710	1.239	2.185
	(0.304)	(0.235)	(0.207)	(0.278)	(0.483)
Adjusted Gross Income	0.197	0.375	0.520	1.011	2.424
	(0.215)	(0.136)	(0.213)	(0.240)	(0.497)
Wage and Salary Income	0.314	0.276	0.276	0.895	1.823
	(0.286)	(0.210)	(0.243)	(0.259)	(0.457)
B. Competitor Innovation					
Baseline	-1.065	-0.903	-1.012	-1.408	-3.898
	(0.480)	(0.372)	(0.462)	(0.493)	(0.982)
Hoberg-Phillips Competitor	-0.435	-1.021	-0.848	-1.256	-2.99
	(0.338)	(0.265)	(0.302)	(0.414)	(0.978)
Income Rank Firm Adjusted	-1.771	-0.801	-1.047	-0.936	-2.214
	(0.405)	(0.410)	(0.468)	(0.485)	(0.956)
Adjusted Gross Income	-0.875	-1.051	-1.587	-2.223	-4.301
	(0.220)	(0.254)	(0.345)	(0.461)	(0.897)
Wage and Salary Income	-0.798	-1.062	-1.386	-1.917	-3.808
	(0.345)	(0.364)	(0.434)	(0.459)	(0.839)

Note: This table shows a series of robustness checks for the results in figure 3. All coefficients and standard errors are multiplied by 100. "Income Rank Firm Adjusted" adjusts income by average firm pay before ranking individuals within industry when constructing the income rank percentiles for our interactions and fixed effects. The Hoberg-Phillips competitor definition uses the 10k textual similarity measure of Hoberg and Phillips to construct industries (using the TNIC3 definition) instead of SIC3.

Table SA.3: Targeted Moments in Model Calibration

Moment	Model	Data	Source
A. Firm Responses to Innovation (5-years, %)			
Profitability			Figure 1
Response to Firm Innovation	9.39	5.66	
Response to Competitor Innovation	-3.29	-4.11	
Employment			Figure 1
Response to Firm Innovation	7.54	2.98	
Response to Competitor Innovation	-2.65	-2.79	
Payroll			Figure 1
Response to Firm Innovation	8.11	4.94	
Response to Competitor Innovation	-2.85	-3.43	
B. Passthrough to Worker Earnings (Worker Earnings / Firm Profits, 5-years)			
Pooled			Figures 1 and 2
Response to Firm Innovation	0.11	0.14	
Response to Competitor Innovation	0.23	0.29	
[0–p25]			Figures 1 and 3
Response to Firm Innovation	0.06	0.06	
Response to Competitor Innovation	0.07	0.26	
[p25–p50]			Figures 1 and 3
Response to Firm Innovation	0.09	0.07	
Response to Competitor Innovation	0.16	0.22	
[p50–p75]			Figures 1 and 3
Response to Firm Innovation	0.11	0.12	
Response to Competitor Innovation	0.22	0.25	
[p75–p95]			Figures 1 and 3
Response to Firm Innovation	0.19	0.20	
Response to Competitor Innovation	0.43	0.34	
[p95–p100]			Figures 1 and 3
Response to Firm Innovation	0.43	0.45	
Response to Competitor Innovation	1.00	0.95	
C. Earnings Inequality (Earnings Share, %)			
[0–p25]	16.4	11.0	Detailed Earnings Record
[p25–p50]	19.4	17.0	
[p50–p75]	22.8	24.0	
[p75–p95]	23.3	29.0	
[p95–p100]	18.1	18.0	
D. Job Switching Probabilities (3-years)			
Pooled	51.8	41.0	Detailed Earnings Record
[0–p25]	65.1	50.3	
[p25–p50]	55.4	40.8	
[p50–p75]	47.9	37.1	
[p75–p95]	38.5	35.7	
[p95–p100]	31.5	37.6	
E. Labor Supply Elasticities by Firm Productivity Quartile			
[0–p25]	3.62	4.32	Seegmiller (2024)
[p25–p50]	2.86	2.71	
[p50–p75]	2.39	2.34	
[p75–p100]	1.68	1.17	
F. Other			
Labor Share	0.58	0.6	
Standard Deviation of 5-Year Wage Growth (%)	28	57	Detailed Earnings Record
KPSS Value share of Innovation coming from x / Radical Innovation (%)	34.6	34.9	Kogan et al. (2017) and equation (14)
90% - 10% spread in log revenue per product line	3.98	3.90	Berlingieri, Ridder, Lashkari, and Rigo (2025)

Note: Moments sourced from the Detailed Earnings Record are the authors' calculations on the regression sample used in this paper, with worker earnings measured as described in Section 2.1. Moments attributed to Kogan et al. (2017) use that paper's data on the dollar value of patents.

Table SA.4: Model Parameters

Description	Parameter	Value
A. Innovation Parameters		
Incremental Jump Rate	I	1.73
Displacive Innovation Rate	$\bar{x} + \bar{e}$	0.084
Incumbent External Innovation Rate	x	0.063
Innovation Jump Size	$\log \lambda$	0.038
Depreciation Rate	δ	0.050
Probability a τ shock resets industry lead	p_r	0.049
B. Production Parameters		
CES Markup Parameter	σ	6.08
Weight on Managers within CES Production Function	B	0.24
Elasticity parameter for CES Production Function	ψ	0.18
Skill bias of technology	α	0.58
Share of workers who are laborers	ω	0.81
C. Worker Parameters		
Dispersion Parameter for Manager Skill Distribution	θ_m	2.13
Dispersion Parameter for Laborer Skill Distribution	θ_l	6.81
Effort Cost Parameter	φ	2.38
Value of Outside Option	R	0.433
Human Capital Persistence	ρ	0.97
D. Fixed Parameters		
Number of Job Offers	K	10
Time-Step Given Shock	μ	0.5
Cost function parameter	ζ	2

Note: Table reports the parameter estimates used in our model solution. Table SA.3 reports the targeted moments along with model fit.

Table SA.5: Summary Statistics for Firm and Worker Samples

	Mean	5%	25%	Median	75%	95%
Panel A: Pooled Worker Sample Summary Stats						
Age	39.41	27	33	39	46	53
Wage	71,790	16,770	35,630	56,170	84,350	160,600
5-Year Log Wage Growth	-9.47%	-99.98%	-17.59%	-0.49%	13.85%	50.13%
Panel B: 75-95% Percentile Earners Sample Summary Stats						
Age	38.48	27	32	38	45	52
Wage	103,400	45,040	72,360	95,590	123,600	188,400
5-Year Log Wage Growth	-7.08%	-70.31%	-15.10%	-1.06%	10.98%	37.65%
Panel C: Top 5% of Earners Sample Summary Stats						
Age	40.87	27	35	41	47	53
Wage	263,400	80,470	127,800	177,100	268,400	636,700
5-Year Log Wage Growth	-4.44%	-81.68%	-18.17%	1.09%	19.17%	59.50%
Panel D: Firm Sample Summary Stats						
Payroll, Thousands	282,600	641	6,002	26,880	132,100	1,173,000
Employment, Thousands	8.203	0.024	0.195	0.93	4.2	35.7
Profits, Millions	532.2	1.264	10.55	45.25	214.7	1951

Note: The worker statistics are based on a linked dataset between the detailed earnings record (DER) and the current population survey (CPS). The firm data is the linked Compustat and Longitudinal Business Database (LBD).

Replication Appendix to
"Winners and Losers: Competition, Creative Destruction, and Labor Income Risk"

Brice Green Leonid Kogan Dimitris Papanikolaou Lawrence D. W. Schmidt

Submitted as supporting documentation under the Replication Policy.

A Numerical Implementation

A.1 Stationary distribution of number of product lines: $\mu \downarrow 0$ limit

In practice, we approximate the stationary distribution of product lines under the model with the $\mu \downarrow 0$ limit. Here, we solve for the measure of product lines that are associated with firms with different numbers of products under that assumption. In the next section, we will show that in practice this is nearly identical to the exact solution while being substantially easier to compute, which is the reason for the approximation. If we take the limit as $\mu \downarrow 0$, we can solve the KFE for this distribution in closed form. Since the probability of moving down in the distribution of number of lines exceeds the probability of moving up, so there will be a well-defined size distribution. Here are the key accounting identities which capture h_n the mass of *product lines* at firms which currently have n lines:

$$\dot{h}_1 = z - (x+\tau)h_1 + \tau h_2 = 0 \tag{94}$$

$$\dot{h}_n = x h_{n-1} - (x+\tau)h_n + \tau h_{n+1} = 0. \tag{95}$$

In a BGP in which the mass of firms is constant, entry must offset exit, which implies that $z = \tau h_1$, so $h_1 = \frac{z}{\tau} = \frac{\tau - x}{\tau}$. Next, let's guess and verify that $h_n = c h_{n-1}$ satisfies the recursion. Let's start with the first equation. If we know that $h_1 = \frac{\tau - x}{\tau}$, then we get $x h_1 = \tau h_2$, which implies that $c = \frac{x}{\tau} = \frac{x}{x+z}$. Next, we verify that $c = \frac{x}{\tau}$ satisfies the general second order difference equation in (95) for any $n \geq 2$:

$$x h_{n-1} - c(x+\tau)h_{n-1} + \tau c^2 h_{n-1} = h_{n-1}[x - \frac{(x+\tau)x}{\tau} + \frac{x^2}{\tau}] = h_{n-1} \cdot 0. \tag{96}$$

It is also easy to verify that the pmf $h_n = \frac{\tau - x}{\tau}\left[\frac{x}{\tau}\right]^{n-1}$ integrates to one.

Since all firms have at least 1 product, the measure of total firms will be less than the unit measure of total products; instead, it will equal $1 \,/\, \mathbb{E}[N_{f,t} | N_{f,t} > 0]$. Hence, if we want to compute the share of *firms* of each size, which we will denote by p_n, we need to do the

following transformation:

$$p_n = \frac{\frac{1}{n}h_n}{\sum_k \frac{1}{k}h_k} = \frac{-x}{\tau \log(1-\frac{x}{\tau})}\left[\frac{x}{\tau}\right]^{n-1}\frac{1}{n}. \tag{97}$$

A.2 Stationary distribution of number of product lines: Exact calculation with finite μ

Here, we instead derive the distribution of the number of product lines without taking the limit as $\mu \downarrow 0$. To do this, analogously with Klette and Kortum (2004), we will work with probability generating functions (PGFs). Suppose that we have a firm which currently has $N_{f,t}$ product lines. Then, the new number of product lines can be decomposed as

$$N_{f,t} = \begin{cases} \sum_{j=1}^{N_{f,t^-}} (S_{j,f,t} + I^x_{j,f,t}) & \text{if } \sum_{j=1}^{N_{f,t^-}} (S_{j,f,t} + I^x_{j,f,t}) > 0 \\ 1 & \text{otherwise} \end{cases}, \tag{98}$$

where $S_{j,f,t}$ captures product line survival and $I^x_{j,f,t}$ captures the successful arrival of new product lines. Let's define by $G(s) \equiv (p_\tau + (1-p_\tau)s)\exp(x\mu P_{x\text{ success}}(s-1))$ as the PGF of the change associated with an individual product line, where $p_\tau \equiv 1 - \exp(-\tau\mu)$ is the probability of displacement during a single industry innovation burst. The $P_{x\text{ success}}$ term is necessary to capture the fact that there is a very small probability that a successful external innovation may not result in winning the market.

Using properties of PGFs, we get that

$$\mathbb{E}[s^{N_{f,t}} | N_{f,t^-} = n, \mathrm{d}\Gamma_{j(f),t} = \mu] = G(s)^n + G(0)^n(s-1), \tag{99}$$

where the last term reflects that, in the stationary distribution, exiting firms are immediately replaced by new entrants with 1 product line. Now, let's find $H(s)$, the PGF of the stationary distribution.

$$\begin{aligned} H(s) \equiv \mathbb{E}[s^{N_{f,t}}] &= \mathbb{E}[E(s^{N_{f,t}} | N_{f,t^-}, \mathrm{d}\Gamma_{j(f),t} = \mu)] \\ &= \mathbb{E}[G(s)^{N_{f,t^-}}] + \mathbb{E}[G(0)^{N_{f,t^-}}](s-1) = H(G(s)) + H(G(0))(s-1). \end{aligned} \tag{100}$$

Let's denote by $E \equiv H(G(0))$, which captures the mass of entering firms. Let's next characterize $H(s)$ by recursive substitution:

$$H(s) = H(G(s)) + E(s-1) \tag{101}$$

$$H(G(s)) = H(G^{\circ 2}(s)) + E(G(s)-1) \tag{102}$$

$$H(s) = 1 + E\sum_{j=0}^{\infty}(G^{\circ j}(s) - 1), \tag{103}$$

where here we use the fact that $\lim_{N\to\infty} H(G^{\circ N}(s)) = 1$ for all $s \in [0,1]$, which will hold as long as all firms die out in the long run (ie each firm's number of products shrinks over time in expectation).

While $H(\cdot)$ does not satisfy a recursion which we can evaluate in closed form, we can use Fourier inversion techniques to recover the pmf of the stationary distribution. In particular, an application of Cauchy's integral formula implies that for an analytic function

$$p_n \;=\; \frac{H^{(n)}(0)}{n!} \;=\; \frac{1}{2\pi i}\oint_{|s|=r} \frac{H(s)}{s^{n+1}}\, ds, \tag{104}$$

where we take an integral around any circle $|s| = r \leq 1$. Next, if we write points on the unit circle as $s = \exp(i\theta)$, for $\theta \in [0, 2\pi]$, with $ds = i\exp(i\theta)d\theta$

$$p_n \;=\; \frac{1}{2\pi i}\int_0^{2\pi} H\big(e^{i\theta}\big)\, e^{-i(n+1)\theta}\, [i\exp(i\theta)d\theta] \;=\; \frac{1}{2\pi}\int_0^{2\pi} H\big(e^{i\theta}\big)\, e^{-in\theta}\, d\theta. \tag{105}$$

Now, if we approximate the integral using the trapezoid rule, we get the following discrete Fourier transform formula:

$$p_n \;=\; \frac{1}{M}\sum_{k=0}^{M-1} H\big(e^{2\pi ik/M}\big)\, e^{-2\pi ikn/M}. \tag{106}$$

Next, we need to pin down what E is. If we want the measure of firms to stay constant along the BGP, then entry needs to offset exit.

What is the rate of new entrants, E? Recall that mass e of entrants enter with probability $d\Gamma_{j,t}$. The probability they are successful is given by $P_{e\text{ success}}$ in equation (87) above, which will typically be very close to 1 since the probability of multiple successful business stealing innovations in the same $\mathrm{d}\Gamma_{j,t}$ burst is very unlikely. Hence, the total mass of successful entrants will be $E = (e\mu)P_{e\text{ success}}$.

If we define $E = (e\mu)P_{e\text{ success}}$, we solve for the measure of firms which have each number of product lines. This will integrate to a number equal to $1 \,/\, E[N_{f,t}|N_{f,t} > 0]$. An alternative way to pin down E is to solve for the level of E which sets $H(0) = 0$. This works because $H(0)$ is equal to p_0, the mass of firms with zero product lines. Therefore, setting E so that $H(0) = 0$ guarantees that each dead firm is replaced by a newborn one and that p_n captures the conditional probability distribution of firm sizes for all firms with positive numbers of product lines. Reassuringly, both methods yield the same conditional pmf for positive product lines, which closely coincides with what obtains from the $\mu \downarrow 0$ limit. Figure 12 illustrates this latter point in a numerical example.

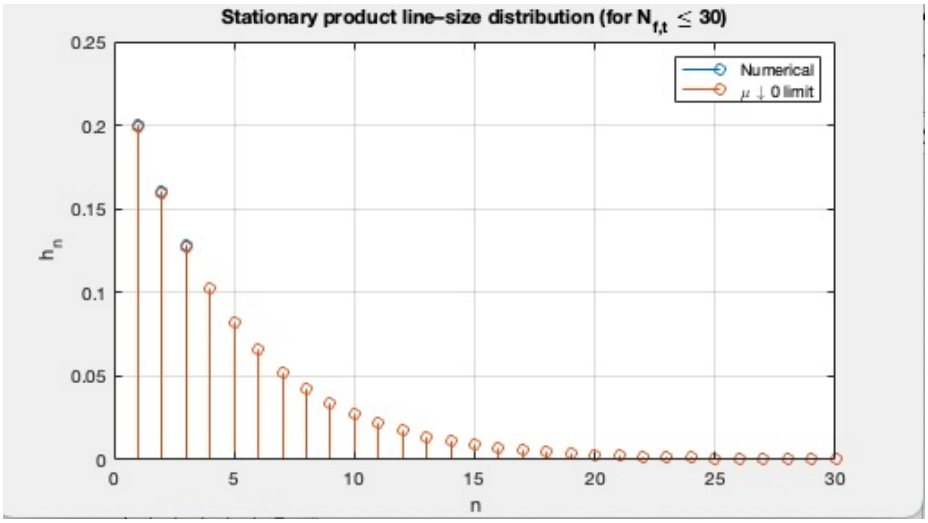


Figure 12: Comparison of h_n in exact calculation vs $\mu \downarrow 0$ analytical solution

A.3 Calibration Parameters

A.4 Model Mapping to Observable Measures of Innovation

The key explanatory variables in Section 2 consist of the value of innovation by the firm and its competitors. To interpret these objects through the lens of the model, consider the change in the market value of a product line as a result of innovation,

$$\Delta V(q_{f,l})(v,t^+) = \max\left(V\left(\hat{q}_{f,l}(v,t^+)\right) - V\left(\hat{q}_{f,l}(v,t^-)\right), 0\right). \tag{107}$$

This value can increase for two reasons: the firm has successfully innovated internally (raising the productivity advantage and hence the value of that product line), or it has innovated externally (adding a new product line to its portfolio).

The KPSS value of innovation for firm f in period t is the sum of this increase along the firm's product lines,

$$KPSS_{f,t} = \sum_{n=1}^{N_{f,t}} \Delta V(q_{f,l})(v,t^+), \tag{108}$$

which maps directly into the numerator of our empirical measure of firm innovation (6).

Innovation by competing firms is defined analogously as the sum of KPSS values across all other firms in industry j:

$$KPSS_j = \int_0^1 \Delta KPSS_{f,t}\, df \propto \mathrm{d}\Gamma_{j,t}. \tag{109}$$

Given the continuum of firms within each industry, the industry-level value of KPSS is simply

proportional to the realization of the industry innovation process defined in equation (23). This proportionality means that our empirical regressions of worker earnings on competitor innovation identify the effect of the industry-level subordinator.

A.5 Finding the stationary distribution

We approximate the product-line level quality on $\log\left(\hat{Q}\right)$ on an equally spaced grid with 201 points that has maximum of $\log(10)$ and a minimum of $\log(0.01)$. Let $\hat{Q}_j(v,t)$ be the quality of the product line and $\hat{q}_j(v,t)$ be the log-quality. Given a $d\Gamma$ shock, there are three possible things that can happen to a line: the firm can fail to innovate in that product line, successfully innovate in that product line (an I jump), or it can be displaced by a new firm.

Conditional on a $\mathrm{d}\Gamma_t$ shock, $\mathrm{d}\hat{q}_t$ has a finite support and probabilities given by the Poisson distribution. Specifically,

$$\mathrm{d}\hat{q}_t = \begin{cases} k\log\lambda - \tilde{g}\mathrm{d}\Gamma_t & \text{w/ prob. } \exp(-(I+\tau)\mathrm{d}\Gamma_t)\frac{[(I+\tau)\mathrm{d}\Gamma_t]^k}{k!}\left[\frac{I+\tau(1-p_r)}{I+\tau}\right]^k \\ k\log\lambda - \tilde{g}\mathrm{d}\Gamma_t - \hat{q}_{t^-} & \text{w/ prob. } \exp(-(I+\tau)\mathrm{d}\Gamma_t)\frac{[(I+\tau)\mathrm{d}\Gamma_t]^k}{k!}\left(1-\left[\frac{I+\tau(1-p_r)}{I+\tau}\right]^k\right) \end{cases}. \tag{110}$$

Notice that the process is constant whenever $\mathrm{d}\Gamma_t = 0$, so we can solve the stationary as if we are operating in discrete time, iterating on the stationary distribution across $\mathrm{d}\Gamma_t$ increments until convergence.

Since we operate on a discrete, bounded support, we impose clamps at the boundaries of the state space. In other words, if a jump moves $\hat{q}$ outside the state space, we replace the point in the support in (110) with the relevant boundary. Second, since the points in the support will typically not live exactly on the $\hat{q}$ over which we discretize, we construct a second stage lottery between the two adjacent grid points which has a mean equal to the desired point in the support. Since we work with a relatively fine grid in practice, this second stage lottery introduces very little additional randomness in the associated distribution.

Our grid is log-spaced while the quality per unit aggregate quality that is held constant along a balanced growth path is not in logs. Thus, when we approximate the ODEs along our grid, we linearly interpolate between adjacent points with weights that maintain the expected change in $\hat{Q}_t$.

In principle, there is no limit to the number of possible jumps that can occur for a single product line for a single firm over a single $d\Gamma$ shock because business time is no longer continuous but instead proceeds with a discrete step, μ. The probability of getting k jumps with rate I is

$$P(N = k) = \frac{(I\mu)^k}{k!} e^{-I\mu}$$

In practice, we choose a K_{max} such that $P(k > K_{max}) <$ tol where the tolerance is set to $1e - 10$. This allows us to work with a finite number of jumps and approximate them with a transition matrix along a discrete state-space.

To compute the stationary distribution, we solve the standard eigenvalue problem using Matlab's *eigs* command, normalizing the output to sum to one. We confirm the fixed point output by the solver by checking that

$$\pi_{\hat{q}} T_{\hat{q}} = \pi_{\hat{q}},$$

where $T_{\hat{q}}$ is the product line transition matrix and $\pi_{\hat{q}}$ is the steady state distribution of product line relative qualities. In practice, this is not sensitive to the number of gridpoints as long as the support is wide enough given the parameters, which we ensure by checking for excess density at the edges of the support.

A.6 Clearing Labor Markets

Recall that we do not have skill prices $w(\hat{Q})$ and $s(\hat{Q})$ in closed form, but instead have to solve for them by clearing markets. Given the distribution of $\hat{Q}$ which we find above, we can solve for wages that clear the supply and demand first order conditions discussed in the static equilibrium part of Section A.2.

We begin with two normalization restrictions. As long as T_τ for $\tau \in \{m, l\}$ is homogenous across jobs, we can't separately identify w from the number of efficiency units of labor. We choose $T_m = T_l$ so that the average number of efficiency units of each type equals one for type l, ie

$$T_l = \Gamma\left(1 - \tfrac{1}{\theta_l}\right)^{-1} \tag{111}$$

The functions $w(l)$ and $s(m)$ are only defined implicitly. To solve for them, we augment the system of equations to solve to include two labor market clearing conditions for each $\hat{Q}$ described above in equations 60 and 61. where $\mathcal{W}_i$ and $\mathcal{S}_i$ are defined as in the main text.

So, we can characterize the equilibrium in terms of the tuple $\{\mathbf{s}, \mathbf{w}, \hat{\mathbf{P}}, T_l, T_m, \bar{A}\}$ which satisfies profit maximization, labor supply functions (67), aggregate quality definitions(69), and satisfies the labor market clearing and normalization (111) equations for both workers and managers.

Given a guess of $\{\mathbf{w}, \mathbf{s}\}$, we can use the equations above to compute all other equilibrium objects. To compute the equilibrium conditions from a guess of $\{\mathbf{w}, \mathbf{s}\}$, we first compute the aggregate number of efficiency units (up to the scaling factors T_l and T_m) supplied of each type of labor.

This requires computing the expectation term which appears in (49), which we accomplish by simulating a large number of potential draws of peer firms' wages. We do this numerically by simulating one thousand possible draws of outside options where workers each face K of them total. To accomplish this, in each state of the discrete state-space, we assume at least one job the worker faces is from that space, and then draw $K-1$ jobs from the multinomial distribution implied by the steady state distribution of product lines, $\pi_{\hat{q}}$. This lets us compute the distribution of outside options, that we can then plug into expressions 49 and 67. We need these for each type in order to compute the vector of labor supply elasticities.

This gives us the ingredients we need to guess and check with an optimizer to find the appropriate skill price functions. Within the objective function we use (111) and its analogue for managers to compute the scaling factors T_l and T_m, (69) to compute $\bar{A}$, (63) to compute prices and finally we use the difference between the labor supply and demand conditions to check whether the labor markets clear at the proposed wages.

In practice, we rely on Matlab's *lsqnonlin* function with an initial guess of constant wages, an objective tolerance of 1e-6, and a maximum number of function evaluations of $1e5$. If the first order conditions do not clear, we add a very large constant to the loss function so that the objective function is pushed away from those parameters.

A.7 Simulating Firms and Workers

Given the schedule of wages, the transition process for product-line level states, and the stationary distribution for initial conditions, we can then directly simulate and then compute our target moments that are analogous to the data. Before we discuss estimating those regression coefficients, we will discuss the simulation process for firms and workers.

We draw initial conditions for N_w workers from the stationary distribution of product-line qualities $\pi_{\hat{q}}$. Each worker has K product lines that they track over time–this is the set of jobs they see available at each instant. At each point in time, workers maximize their income and they can switch costlessly between these K jobs.

Rather than simulating on the discrete grid, we simulate directly from the ODEs. While time is continuous, in practice we must simulate over a discrete time-step Δ_t. For convenience (and to avoid user error), we set that to μ, the time-step of the $d\Gamma$ shock. For each simulated

Δ_t, each worker i, and each job offer they have, k, we simulate the number $d\Gamma$ shocks they face in that product line as

$$P(d\Gamma = n) = \left(\frac{\Delta_t}{\mu}\right)^n \frac{e^{-\frac{\Delta_t}{\mu}}}{n!}$$

where we assume there can be no more than 20 jumps. This constructs a discrete CDF from which we can draw empirical jump numbers. We assume that the jumps across job offers are independent.

For each $d\Gamma$ they draw for each job over time step Δ_t, we simulate from the ODE given by 110 to get the next state. These simulated draws are no longer on the discrete grid, so to get exact per-efficiency-unit wages we interpolate over the log wage / log quality distribution for both skill types using Matlab's *pchip* function. We also enforce the reflecting barrier, as we have in our transition matrix, in order to make things tie out.

Each worker has by K job offers at each time which have wages per skill unit $w(\hat{Q}_{ik})$. They also have a number of skill units per job offer given by equations 18 and 30 in the main text. In the discrete time simulations, we approximate the OU process with an AR(1). At any given point in time, a worker chooses job k out of K offers to maximize take-home pay. This gives us two arrays of dimension $N_w \times T \times K$, where $T = \frac{8}{\Delta_t}$ is the number of periods required to compute our growth rates.

This gives us the path of employment and wages in the absence of additional shocks. As discussed in the main text, we use 5 year forward-looking income compared to the three years of past income. The extra shock we consider is an additional $d\Gamma$ shock that happens instantly at the end of the 3rd year of employment.

We consider each possible outcome from this the worker's income separately and weight them appropriately. This is to say we consider that conditional on a $d\Gamma$ shock, there is a discrete jump distribution from 0 up to a K_{max}, and for each of these outcomes we compute the expected counterfactual path of wages for each worker. For resets, we do the same.

As in the data, we winsorize our outcomes at the 1st and the 99th percentile. To compute these expected quantiles of outcomes, we compute the CDF of potential outcomes given a $d\Gamma$ shock. For a product line outcome, o_l, conditional on a shock, s_l, let $P(s_l)$ be the probability that that type of shock occurs. Let $P(\omega_l)$ be the probability of state ω_l in the discretized product line stationary distribution.

Then the PDF of an aggregate outcome for product line l, o_l is given by

$$P(o_l) = P(s_l)P(\omega_l)$$

which is a discrete distribution because of the discrete steady state distribution and the finite set of outcomes conditional on the shocks. The CDF, $P(o_{f,p} \leq O)$ is given by sorting and integrating over the discrete distribution. The inverse CDF immediately gives the quantiles of the product line outcomes.

Workers vary not only by initial states, but also by product line level human capital z_{il}. Let $P(z_{il}|\omega_l)$ be the pdf of human capital for worker i in product line j be the density of the human capital distribution given the stochastic process described in the main text for z_{ilt}. We approximate the distribution of $P(z_{il}|\omega_l)$ through standard simulation methods, drawing 15,000 simulated workers, and approximating the PDF with the empirical distribution of the simulations. Let $\hat{P}(z_{il}|\omega_l)$ be the realization of that empirical approximation, which gives PDF

$$P(o_{i,p}) \approx \hat{P}(z_{ij}|\omega_l)P(\omega_l)P(s_l).$$

The CDF follows immediately, as do the quantiles.

We take those winsorized product-line and worker-level outcomes, and use them to compute regression coefficients for both firms and workers. All variables are winsorized in the following sections. In the data, the outcomes are given by $dA_f = \frac{dV_f \mathbb{I}(dV_f > 0)}{V_{f,t}}$ where $dV_{f,t} = V_{f,t} - V_{f,t-}$ because we cannot identify impacts at the product-line level. However in the model, we can observe product lines directly, and

$$dA_{f,l,t} = \frac{\Delta V_{f,l,t} \mathbb{I}(\Delta V_{f,l,t} > 0)}{V_{f,l,t}}$$

where $\Delta V_{f,l,t} = V_{f,l,t} - V_{f,l,t-}$. To compute $dA_{f,t}$ we sum over product lines,

$$dA_{f,t} = \sum_{l=1}^{P} dA_{f,l,t}$$

Competitor innovation conditional on a $d\Gamma$ shock is simple in this model. Since firms are infinitesimal it collapses to expected firm-level $dA_{f,t}$:

$$dA_i = \int_0^1 dA_{k,t} dk = E(dA_{f,t}|d\Gamma)$$

In the absence of a $d\Gamma$ shock the firm state does not change so both measures are equal

to zero. This result is a key reason for using the machinery of the subordinated Γ process, because in continuous time if there were no innovation bursts, dAi would be a constant drift.

A.8 Computing Moments

For firms, we are interested in changes in five-year-forward expected growth paths of wage bill, profits, and employment. We have expressions that give instantaneous profit and employment given the current state, denote that vector y. To approximate a 5 year level of profits, we compute the forward-looking discounted value of y, but depreciate them at a 20% rate:

$$Y_{5yr} = (r - \Lambda)^{-1} y$$

where r is a diagonal matrix with 0.2 on the diagonal and Λ is the generator matrix for the stochastic process for the product line state. Instantaneous profits, employment, and pay follow from the numerical solution to the firm's problem described above.

For wage growth, we simulate forward a 5 year growth path for workers and compute log differences relative to a prior 3 years of employment. For the paths with an extra shock, described above, we do the same. The counterfactual wage growth path is the difference between these two outcomes.

The target moments of interest are largely regression coefficients of both firm and worker level outcomes on firm and competitor innovation. For some outcome, Y, we compute these directly as

$$\beta_o = E(X^T X)^{-1} E(X^T \tfrac{dY}{Y})$$

where all that varies is the outcome. We calculate $X^T X$ as

$$E(X^T X) = E\begin{pmatrix} dA_i^2 & dA_i dA_f \\ dA_i dA_f & dA_f^2 \end{pmatrix}$$

and

$$E(X^T Y_o) = E\begin{pmatrix} dA_i \frac{dY}{Y} \\ dA_f \frac{dY}{Y} \end{pmatrix}$$

Since $dA_i = E(dA_f)$ as described above is a constant, we can compute this once. However, it still remains to compute $E(dA_f)$ and $E(dA_f^2)$, and each $E(\frac{dY}{Y})$ and $E(dA_f \frac{dY}{Y})$.

Let $\hat{Q}_f = \{\hat{q}_{l,f}\}_{l=1}^{N_{lines}}$ be the set of product lines that the firm has. $dA_l = \frac{dV_l}{V_l}$, $dA_f = \frac{dV_f}{V_f} = \frac{V_{f,l}}{V_f} dA_l$, $A_f = \sum w_{f,l} a_{f,l}$; $w_l = \frac{V_{f,l}}{V_f}$ We need to compute:

$$E(dA_f|d\Gamma = \mu) = \frac{1}{N_{firms}} \sum_{f=1}^{N_{firms}} \sum_{l=1}^{N_f} w_{f,l} E(dA_{f,l}|\hat{q}_{f,l}, d\Gamma = \mu)$$

The expectation of the product line-level shock to the power k at the product line l for firm f given the quality $\hat{q}_{f,l}$, $E(dA^k_{f,l})$, is given by

$$\begin{aligned} E(dA^k_{f,l}|\hat{q}_{f,l}) &= p(\text{Lose Line})E(dA^k_{f,l}|\text{Lose Line}, \hat{q}_{f,l}) \\ &+ p(\Delta\hat{q}_{f,l} = -\mu\tilde{g})E(dA^k_{f,l}|\Delta\hat{q}_{f,l} = -\mu\tilde{g}, \hat{q}_{f,l}) \\ &+ p(\text{Gain Line})E(dA^k_{f,l}|\text{Gain Line}, \hat{q}_{f,l}) \\ &+ p(\text{Improve Line}) \sum_{j=1}^{N} p(n = j|\text{Improve Line})E(dA^k_{f,l}|n, \text{Improve Line}, \hat{q}_{f,l}) \end{aligned}$$

for $k = 1$.

For k=2 we also must include a cross term. Second moment computations become

$$E(dA^2_{f,l}|\hat{q}_{f,l}) = E(dA^2_{f,l,existing}|\hat{q}_{f,l}) + 2E(dA_{f,l,existing}|\hat{q}_{f,l})E(dA_{f,l,new}) + E(dA^2_{f,l,new})$$

Let Y_f be another firm-level outcome, e.g. profit, employment, or pay. dY_f is the instantaneous change in that outcome in real time. Again, for simplicity, this is all conditioning on the shock, $d\Gamma = \mu$. The expectation terms which appear in the regression are

$$E\begin{pmatrix} dA_i \frac{dY_f}{Y_f} \\ dA_f \frac{dY_f}{Y_f} \end{pmatrix} = \begin{pmatrix} E(dA_i)E(\frac{dY_f}{Y_f}) \\ E(dA_f \frac{dY_f}{Y_f}) \end{pmatrix}$$

are what is left to compute. Thus we need $E(\frac{dY_f}{Y_f})$ and $E(dA_f \frac{dY_f}{Y_f})$.

$$E(\frac{dY_f}{Y_f}|d\Gamma = \mu) = \sum_{s\in S} p(\hat{q}_{f,l} = \hat{q}(s)) \frac{1}{N_{firms}} \sum_{f=1}^{N_{firms}} \sum_{l=1}^{N_f} w^*_{f,l} E(\frac{dY_f}{Y_f}|\hat{q}_{f,l}, d\Gamma = \mu)$$

where $w^*_{f,l} = \frac{Y_{f,l}}{Y_f}$ for a given outcome. Similar to $dA_{f,l}$,

$$\begin{aligned} E(\frac{dY_{f,l}}{Y_{f,t}}|\hat{q}_{f,l}) &= p(\text{Lose Line})E\left(\frac{dY_{f,l}}{Y_{f,t}}|\text{Lose Line}, \hat{q}_{f,l}\right) \\ &+ p(\Delta\hat{q}_{f,l} = -\mu\tilde{g})E\left(\frac{dY_{f,l}}{Y_{f,t}}|\Delta\hat{q}_{f,l} = -\mu\tilde{g}, \hat{q}_{f,l}\right) \\ &+ p(\text{Gain Line})E\left(\frac{dY_{f,l}}{Y_{f,t}}|\text{Gain Line}, \hat{q}_{f,l}\right) \end{aligned}$$

$$+ \quad p(\text{Improve Line}) \sum_{j=1}^{N} p(n = j|\text{Improve Line}) E\left(\frac{dY_{f,l}}{Y_{f,t}}|n, \text{Improve Line}, \hat{q}_{f,l}\right)$$

$$E(dA_f dA_i) = E(dA_f)E(dA_i)$$

because firm f is atomistic.

$$E(dA_f^2|\hat{q}_{f,l}) = \frac{1}{N_{firms}} \sum_{f=1}^{N_{firms}} \left(\sum_{l=1}^{N_f} w_{f,l}^2 E(dA_{f,l}^2|\hat{q}_{f,l}) + \sum_{l=1}^{N_f} \sum_{m \neq l} w_{f,l} w_{f,m} E(dA_{f,l}|\hat{q}_{f,l}) E(dA_{f,m}|\hat{q}_{f,m}) \right)$$

$$E\left(dA_f \frac{dY_f}{Y_f}|\hat{q}_{f,l}\right) = \frac{1}{N_{firms}} \sum_{f=1}^{N_{firms}} \left(\sum_{l=1}^{N_f} w_{f,l} w_{f,l}^* E[dA_{f,l} \frac{dY_{f,l}}{Y_{f,l}}|\hat{q}_{f,l}] + \right.$$
$$\left. \sum_{l=1}^{N_f} \sum_{m \neq l} w_{f,l} w_{f,m}^* E(dA_{f,l}|\hat{q}_{f,l}) E(\frac{dY_{f,m}}{Y_{f,m}}|\hat{q}_{f,m}) \right)$$

To compute the unconditional expectations, we just integrate over the stationary distribution for the probability that product line l has quality $\hat{q}$.

$$E\left(dA_f \frac{dY_f}{Y_f}\right) = \sum_{s \in S} p(\hat{q}_{f,l} = \hat{q}(s)) E\left(dA_f \frac{dY_f}{Y_f}|\hat{q}_{f,l}\right),$$

which we can do analogously for any of these outcomes, following from the definition of conditional expectation.

For wage growth, we have an additional complication, which is that we are integrating over a different distribution of $\hat{q}_{f,l}$ when we condition on income. Rather than using the steady state product line distribution, we use the observed $\hat{q}_{f,i \in l}$ (where i indexes the worker) for a single product line in the firm, and the remainder are simulated from the product line stationary distribution. The firm's other product lines are drawn from the stationary distribution.

For workers, their wage growth only depends on the $\hat{q}_{i \in l}$, so

$$E(\frac{dW_i}{W_i} dA_{f,i \notin l}|\hat{q}_i, \hat{q}_l) = E(\frac{dW_i}{W_i}) E(dA_{f,l}).$$

While before we had a vector of product line outcomes, $\frac{dY_f}{Y_f}$ and a vector of product line dA_f shocks, in this case we have a scalar and a vector. Thus our above formula simplifies conditional on $\hat{q}_{f,i \in l}$,

$$E(\tfrac{dW_i}{W_i} dA_f | \hat{q}_{f,i\in l}) = \frac{1}{N_{firms}} \sum_{f=1}^{N_{firms}} \left[w_{f,l} E\left(\tfrac{dW_i}{W_i} dA_{f,i\in l} | \hat{q}_{f,l}\right) + E\left(\tfrac{dW_i}{W_i}\right) \sum_{m\neq l} w_{f,m} E(dA_{f,m} | \hat{q}_{f,m}) \right].$$

This integrates over the possible firms that a product line containing a worker, i working in product line l with quality $\hat{q}_{f,l}$ could have. We then need to integrate over the worker's $\hat{q}$ distribution, which we do by averaging across simulated workers,

$$E\left(\tfrac{dW_i}{W_i} dA_f\right) \approx \frac{1}{N_{workers}} \sum_{i=1}^{N_{workers}} E(\tfrac{dW_i}{W_i} dA_f | \hat{q}_{f,i\in l})$$

To condition on income, we just subset the worker distribution such that a worker meets a certain criterion for starting income.

The computation associated with looping over the quadratic set of cases, $l = m$ and $l \neq m$ is extremely costly, especially for firms a large number of product lines, which causes it to explode combinatorically. However, we can simplify the computation by boiling it down to a portfolio of two product lines: one which we take from the focal state we are conditioning on, and one which we take as an existing portfolio of product lines in the rest of the firm. Our conditioning information is only in a single one of the firms product lines, denoted $\hat{q}_{f,l}$, which we know is equal to $\hat{q}(s)$.

First, note that

$$E(dA_f^2 | \hat{q}_{f,l}) = E\left(w_{f,l}^2 dA_{f,l}^2 | \hat{q}_{f,l}\right) + \sum_{l\neq m} \left[E(w_{f,l} w_{f,m} dA_{f,l} dA_{f,m} | \hat{q}_{f,l}) + E(w_{f,m}^2 dA_{f,m}^2 | \hat{q}_{f,l}) \right]$$

Importantly, $dA_{f,m}$ and $dA_{f,l}$ are independent in this construction, so $dA_{f,m}^2$ does not depend on the value of $\hat{q}_{f,l}$ and the conditional expectation is equal to the unconditional. Consider the terms not involving the focal product line, l. They condition on $\hat{q}_{f,l}$ *only* through the denominator of $w_{f,l}$, V_f. Let $V_{f,-l} = \sum_{m\neq l} V_{f,m}$ be the leave-one-product-line out value of the firm. Identically, the change in firm value associated with an event in product line l is

$$dA_f = w_{f,l} dA_{f,l} = \frac{V_{f,l}}{V_f} \frac{dV_{f,l} \mathbb{I}(dV_{f,l} > 0)}{V_{f,l}}$$

For all other product lines, then, we can write

$$dA_f = w_{f,m} dA_{f,m} = \frac{V_{f,m}}{V_f} \frac{dV_{f,m} \mathbb{I}(dV_{f,m} > 0)}{V_{f,m}} = \frac{V_{f,-l}}{V_f} \frac{V_{f,m}}{V_{f,-l}} \frac{dV_{f,m} \mathbb{I}(dV_{f,m} > 0)}{V_{f,m}}$$

This means we can collapse all other product lines into a single $dA_{f,-l}$ and only consider a synthetic "two-product-line" firm for our simulations, speeding up computation considerably relative to looping over each product line and computing the product. All we need to do is track the total share of firm value associated with the non-focal product line, and use that as a normalizing factor. In other words, for each firm f we only need

$$dA_{f,-l} = \frac{\sum_{m \neq l} dV_{f,m} \mathbb{I}(dV_{f,l} > 0)}{V_{f,-l}}$$

and $V_{f,-l}$, which we can compute directly from our firm simulations. Second moment computations become

$$E(dA_f^2 | \hat{q}_{f,l}) = w_{f,l}^2 E(dA_{f,l}^2 | \hat{q}_{f,l}) + 2 w_{f,l} w_{f,-l} E(dA_{f,l} | \hat{q}_{f,l}) E(dA_{f,-l}) + w_{f,-l}^2 E(dA_{f,-l})^2$$

and for first moments we have

$$E(dA_f | \hat{q}_{f,l}) = w_f E(dA_{f,l} | \hat{q}_{f,l}) + w_{f,-l} E(dA_{f,-l})$$

for other outcomes it becomes

$$\begin{aligned} E\left(dA_f \tfrac{dY_f}{Y} | \hat{q}_{f,l}\right) = {} & w_{f,l}^y w_{f,l}^v E\left(dA_{f,l} \tfrac{dY_{f,l}}{Y} | \hat{q}_{f,l}\right) + w_{f,l}^v w_{f,-l}^y E(dA_{f,l} | \hat{q}_{f,l}) E\left(\tfrac{dY_{f,-l}}{Y_{f,-l}}\right) \\ & + w_{f,-l}^v w_{f,l}^y E(dA_{f,-l}) E\left(\tfrac{dY_{f,l}}{Y_{f,l}} | \hat{q}_{f,l}\right) + w_{f,-l}^v w_{f,-l}^y E\left(dA_{f,-l} \tfrac{dY_{f,-l}}{Y_{f,-l}}\right) \end{aligned}$$

All that we need to approximate, then, is the distribution of firm weights $w_{f,l}$ and $w_{f,-l}$ given the value of the focal product line, l. We do this by simulation, drawing 3,000 firms which we use to compute the distribution of weights. The expectations integrated over firms are computed numerically, such that (for example, in the case of a firm-level covariance with dA_f)

$$E\left(dA_f \tfrac{dY_f}{Y} | \hat{q}_{f,l}\right) \approx \frac{1}{N_{sims}} \sum_{\hat{f}=1}^{N_{sims}} E\left(dA_{\hat{f}} \tfrac{dY_{\hat{f}}}{Y} | \hat{q}_{\hat{f},l}\right)$$

where each firm $\hat{f}$ is drawn from the distribution of firms.

To simulate firms, we draw from the firm size distribution described in equation 97. For each product line we draw, we then simulate a quality from the discrete approximation to the stationary distribution. In order to match our Compustat sample, we mirror the censoring that happens in the data. A competitor shock type for a single product line firm eliminates that firm from the forward-looking sample. We account for this in the model by omitting that

possibility from the sample, and re-weighting the prevalence of single- and multiple-product line firms to account for the censoring.

Wages have one additional step, which comes from the fact that we have two skill types. Let $\frac{dW_i}{W_i}$ be the percent wage growth of worker i. That worker has a vector of human capital draws, z_i, of size K. In computing regression coefficients, we need to (optionally conditional on lagged income group) integrate over the distribution of human capital and available skill types.

The path of human capital is drawn from the same random normal process, but plugged into the inverse CDF of a Frechet distribution with parameters T_l and θ_l for skill type l and parameters T_m and θ_m for skill type m. This preserves the respective skill type distributions while saving simulation time. Workers are grouped into income bins based on the quantiles of their average three year income in the CDF of workers.

We need to be able to integrate over two distinct skill types. To do this, we compute the empirical share of the number of simulated workers, weighted by the share of the type in the economy, in each income bin. For each skill type, we compute

$$E\left(\frac{dW_i}{W_i}dA_f\right) \approx \frac{1}{N_{workers}} \sum_{i=1}^{N_{workers}} E(\frac{dW_i}{W_i}dA_f|\hat{q}_{f,l}(i))$$

where $\hat{q}_{f,l}(i)$ is the state associated with the worker at the time of the shock. We compute this separately for each type, and then average them based on the empirical shares. Let ω_l be the share of workers with skill type l and $\omega_m = 1 - \omega_l$ be the share associated with m. Then for a given income group, g, we first compute the total measure of workers. Let $s(i)$ be the skill type of worker i. Then the total measure of workers in an income group is $\sum_{i\in g} \omega_{s(i)}$ and the relative share of type l is given by $w_g(l) = \frac{\sum_{i\in g, s(i)=l} \omega_{s(i)}}{\sum_{i\in g} \omega_{s(i)}}$.

$$\beta_g = \left(w_g(l)E(X_g(l)^T X_g(l)) + w_g(m)E(X_g(m)^T X_g(m))\right)^{-1} \left(w_g(l)E(X_g(l)^T Y_g(l)) + w_g(m)E(X_g(m)^T Y_g(m))\right) \tag{112}$$

where the expectations are approximated by the empirical distribution of simulated workers, and calculated as described above.

The standard deviation of wage growth is equal to

$$\sigma(wg) = \sqrt{\omega_m \hat{\sigma}_m^2 + \omega_l \hat{\sigma}_l^2}$$

where $\hat{\sigma}^2$ is the empirical variance of wage growth rates for the simulated worker panel for each type.

For each income group we compute the share of total pay that goes to the group. For a group g the total income paid to that group is given by

$$I(g) = \omega_l \sum_{i \in g, s(i) \in l} z_{ij} w(\hat{q}_j) + \omega_m \sum_{i \in g, s(i) \in m} z_{ij} s(\hat{q}_j)$$

The share paid to each group is simply

$$\frac{I(g)}{\sum_{g' \in G} I(g')}.$$

Three year exit probabilities are computed based on the empirical frequency with which individuals switch jobs in the simulated panel, appropriately weighted by type.

A.9 Computing the objective function

The loss function is the traditional method of simulated moments quadratic form,

$$\mathcal{L}(\hat{\theta}) = g(\hat{\theta})' W g(\hat{\theta})$$

W is diagonal, and is fixed through the estimation in order to emphasize the fit of certain moments are more important than others. Because the moments are of different scales, we first scale them by their empirical counterparts before computing the loss.

We optimize the objective function using Matlab's surrogateopt, which performs radial basis function expansion for global optimization.

References

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